Physics

Class 11 (English Medium)

Solved Exercises — Sindh Textbook Board

14 chapters — 55 exercises — 433 questions

Unit 1: Physics and Measurements

Exercise — Multiple Choice Questions

The respective number of significant figures for the numbers 23.023, 0.0003 and 2.1 x 10^-3 are:

✓ (a) 5, 1, 2

(b) 5, 1, 5

(c) 5, 5, 2

(d) 4, 4, 2

Answer: (a) In 23.023 all five digits are significant, including the zero between non-zero digits. In 0.0003 the leading zeros are not significant, so it has only one significant figure, and in 2.1 x 10^-3 only the digits 2 and 1 are counted, so it has two.

Which among the following is the supplementary quantity?

(a) Mass

(b) Time

✓ (c) Solid angle

(d) Luminosity

Answer: (c) Mass, time and luminous intensity (luminosity) are fundamental quantities with their own base units. Solid angle, measured in steradian, is a supplementary quantity, like the plane angle.

The unit of solid angle is:

(a) Second

✓ (b) Steradian

(c) Kilogram

(d) Candela

Answer: (b) The solid angle is the three-dimensional angle subtended at a point, and its SI unit is the steradian (sr). The second, kilogram and candela are the units of time, mass and luminous intensity respectively.

The quantity having the same unit in all systems of units is:

(a) mass

✓ (b) time

(c) length

(d) temperature

Answer: (b) Length is measured in metre, centimetre or foot and mass in kilogram, gram or pound in different systems, so their units change. The second is used as the unit of time in the CGS, MKS and FPS systems alike.

Random errors can be eliminated by:

✓ (a) taking a number of observations and their mean

(b) measuring the quantity with more than one instrument

(c) eliminating the cause

(d) careful observations

Answer: (a) Random errors are positive in some readings and negative in others, so they partly cancel when many observations are averaged. Care and better instruments reduce them, but averaging repeated observations is the standard way to minimise their effect.

Systematic error can be:

✓ (a) either positive or negative

(b) negative only

(c) positive only

(d) zero error

Answer: (a) A systematic error always pushes the readings in one direction

for a given set-up, but that direction may be above or below the true value depending on its cause. For example, a zero error of a scale may add to every reading or subtract from every reading.

[MLT^-2] is the dimensional formula of:

(a) strain

✓ (b) force

(c) displacement

(d) pressure

Answer: (b) Force is mass x acceleration, so its dimensions are M x LT^-2 = [MLT^-2]. Strain is dimensionless, displacement is [L] and pressure is [ML^-1T^-2], so none of them matches.

Which of the following pairs has the same dimension?

(a) moment of inertia and torque

(b) surface tension and force

✓ (c) impulse and momentum

(d) specific heat and latent heat

Answer: (c) Impulse is force x time, [MLT^-2] x [T] = [MLT^-1], and momentum is mass x velocity, [M] x [LT^-1] = [MLT^-1]. The pairs in the other options differ in at least one power of M, L or T.

Dependent variable is:

(a) cause

✓ (b) effect

(c) cause and effect

(d) reason

Answer: (b) In an experiment the independent variable is the cause that the experimenter changes deliberately. The dependent variable is the effect, that is, the quantity that changes in response and is measured.

Dimensions of kinetic energy are the same as that of:

(a) Acceleration

(b) Velocity

(c) Work

(d) Force

Answer: (c) Kinetic energy is (1/2)mv^2, whose dimensions are [M] x [LT^-1]^2 = [ML^2T^-2]. Work is force x distance, [MLT^-2] x [L] = [ML^2T^-2], so work and all forms of energy share the same dimensions.

Exercise — Structured Questions (CRQs)

Question: Give an example of (I) a physical quantity which has a unit but no dimensions, (II) a physical quantity which has neither unit nor dimensions, (III) a constant which has a unit, (IV) a constant which has no unit.

Answer: (I) The plane angle has the unit radian but no dimensions, because it is a ratio of arc length to radius. (II) Strain has neither a unit nor dimensions, because it is a ratio of two lengths. (III) The universal gravitational constant G is a constant with a unit, N m^2/kg^2. (IV) The number pi is a constant with no unit, because it is a pure ratio of circumference to diameter.

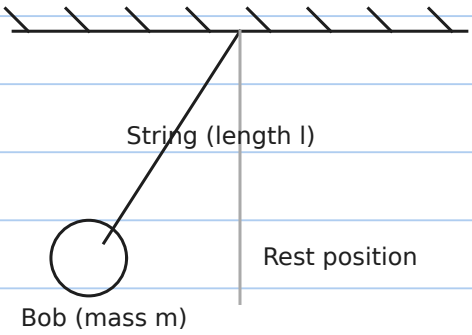
Question: When rounding the product or quotient of two measurements, is it necessary to consider significant digits?

Answer: Yes, it is necessary, because the result of a calculation cannot be more precise than the least precise measurement used in it. In multiplication and division the answer is rounded to the same number of significant figures as the measurement having the fewest significant figures. For example, 4.16 x 2.0 must be written as 8.3, not 8.32,

otherwise the answer would claim an accuracy that the data do not have.

Question: Derive the equation for the period of oscillations of a mass suspended on a vertical spring by dimensional analysis, i.e. T = 2pi sqrt(m/k).

Answer: Suppose the period T depends on the mass m and the spring constant k, so T is proportional to m^a k^b. The dimensions of k are force divided by length, that is [MT^-2], so writing [T] = [M]^a [MT^-2]^b gives a + b = 0 for mass and -2b = 1 for time, hence b = -1/2 and a = 1/2. Therefore T is proportional to sqrt(m/k), and experiment fixes the dimensionless constant as 2pi, giving T = 2pi sqrt(m/k).



Question: Find the dimensions of the following: (a) work (b) energy (c) power (d) momentum.

Answer: (a) Work is force x distance, so its dimensions are [MLT^-2] x [L] = [ML^2T^-2]. (b) Energy has the same dimensions as work, [ML^2T^-2], because energy is the capacity to do work. (c) Power is work divided by time, so its dimensions are [ML^2T^-3]. (d) Momentum is mass x velocity, so its dimensions are [M] x [LT^-1] = [MLT^-1].

Question: You measure the radius of a wheel to be 4.16 cm. If you multiply by 2 to get the diameter, should you write the result as 8 cm or as 8.32 cm? Justify your answer.

Answer: The diameter should be written as 8.32 cm. The number 2 here is an exact count, not a measurement, so it has unlimited significant figures and does not limit the answer. The measured radius has three significant

figures, so the diameter must also keep three significant figures; writing 8 cm would wrongly throw away measured precision.

Question: If y = a + bt + ct^2, where y is in metres and t in seconds, what is the unit of c?

Answer: Every term in the equation must have the same unit as y, namely the metre, so the term ct^2 must be measured in metres. Since t^2 has the unit s^2, c must have the unit metre divided by s^2, i.e. m/s^2. In other words, c has the unit of acceleration.

Question: Differentiate between accuracy and precision.

Answer: Accuracy tells how close a measured value is to the true value of the quantity, while precision tells how close repeated measurements are to one another. A result can be precise without being accurate, for example when a faulty instrument gives nearly the same wrong reading every time. Good measurement needs both: closeness to the true value and consistency between readings.

Question: Define dependent and independent variables.

Answer: The independent variable is the quantity that the experimenter changes deliberately, one step at a time, in order to study its effect. The dependent variable is the quantity that responds to that change and is measured during the experiment. For example, in studying how the length of a pendulum affects its period, length is independent and period is dependent.

Question: Differentiate between systematic error and random error.

Answer: A systematic error is caused by a fault in the instrument, a wrong zero, or a fixed flaw in the method, so it shifts every reading in the same direction by a similar amount. A random error is caused by unpredictable small changes in conditions or in the observer's judgement, so it makes readings scatter on both sides of the true value. Systematic errors are

removed by correcting the cause, while random errors are reduced by repeating readings and taking their mean.

Question: Enlist the limitations of dimensional analysis.

Answer: Dimensional analysis cannot determine dimensionless constants such as 2pi or 1/2 in a formula, because such numbers have no dimensions. It cannot distinguish between different quantities that have the same dimensions, such as work and torque, and it cannot deal with trigonometrical, exponential or logarithmic functions. It also cannot derive relations involving more than three fundamental quantities or quantities added or subtracted in a sum.

Question: Describe the least count of a Vernier caliper and a screw gauge micrometer.

Answer: The least count is the smallest measurement an instrument can make reliably. For a Vernier caliper it is one main scale division minus one Vernier scale division, commonly 1 mm - 0.9 mm = 0.1 mm = 0.01 cm. For a micrometer screw gauge it is the pitch of the screw divided by the number of divisions on the circular scale, commonly 0.5 mm / 50 = 0.01 mm = 0.001 cm, so the micrometer is the more precise instrument.

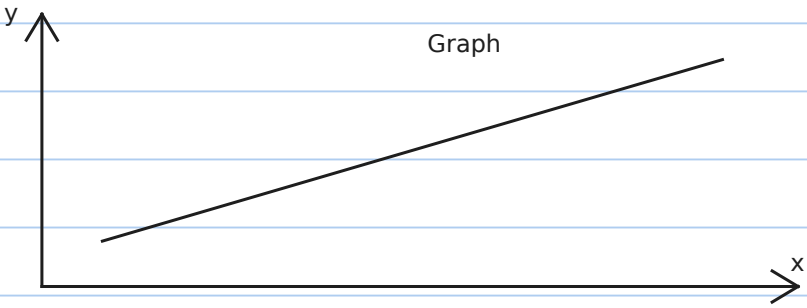
Question: Describe extrapolation methods.

Answer: Extrapolation means extending a graph or a trend beyond the range of the measured data in order to predict a value outside that range. In the graphical method the plotted curve is extended smoothly in the same trend, and the required value is read off it; in the formula method the equation fitted to the data is used with a value of the variable outside the measured range. Extrapolation assumes that the observed trend continues unchanged, so its predictions become less reliable the farther they go from the data.

Exercise — Structured Questions (ERQs)

Question: Discuss graphs and their various types, supported by an example.

Answer: A graph is a visual plot of the dependent variable against the independent variable, and it shows their relationship at a glance. Common types are the line graph, used for continuous change such as displacement against time; the bar graph, used for comparing separate categories such as the marks of different subjects; the pie chart, used for shares of a total; and the scatter graph, used to look for a correlation between two measured quantities. For example, a straight line on a velocity-time graph shows uniform acceleration, its slope gives the acceleration and the area under it gives the displacement.



Question: Elaborate the rules of significant figures. State the rules for determining the number of significant figures in a given measurement.

Answer: All non-zero digits are significant, and zeros between non-zero digits are also significant, as in 203 which has three significant figures. Leading zeros before the first non-zero digit are not significant, as in 0.003 which has one, while trailing zeros after a decimal point are significant, as in 2.50 which has three. In scientific notation only the digits before the power of ten are counted. For addition and subtraction the answer keeps the fewest decimal places of the data, and for multiplication and division it

keeps the fewest significant figures.

Question: What are the uses of dimensional analysis? Explain each of them.

Answer: First, it checks the correctness of an equation: if the dimensions of both sides are not equal, the equation must be wrong. Second, it helps derive a relation between quantities, as in finding that the period of a pendulum depends on sqrt(l/g). Third, it converts a quantity from one system of units to another by using the conversion factors of the fundamental units. Fourth, it checks the consistency of the units used in a calculation, which prevents mixing, for example, centimetres and metres in one formula.

Question: A Vernier and a micrometer are shown in the book (printed photographs under the headings Vernier and Micrometer). Observe their readings and write them correctly.

Answer: For the Vernier caliper, read the main scale just before the zero of the Vernier scale, then add the coinciding Vernier division multiplied by the least count of 0.01 cm, and finally apply any zero correction. For the micrometer, read the exposed sleeve (main scale) reading in millimetres, add the thimble division in line with the reference line multiplied by the least count of 0.01 mm, and apply any zero correction. Note: the photographs printed in the book are too blurred to read the exact coinciding divisions reliably, so the method above is the correct way to obtain the two readings from a clear figure.

Exercise — Numerical Problems

Question: What is the percent uncertainty in the measurement 3.67 +/-

0.25 m?

Answer: Given: measured value = 3.67 m, absolute uncertainty = 0.25 m

Step 1: Write the formula, percent uncertainty = (absolute uncertainty / measured value) x 100

Step 2: Substitute the values, percent uncertainty = (0.25 / 3.67) x 100

Step 3: Calculate, 0.25 / 3.67 = 0.0681, and 0.0681 x 100 = 6.81 percent

Final answer: The percent uncertainty is about 6.8 percent.

Question: What is the area, and its approximate uncertainty, of a circle with radius 3.7 x 10^4 cm?

Answer: Given: radius r = 3.7 x 10^4 cm, taking the uncertainty in r as half of the last quoted digit, 0.05 x 10^4 cm

Step 1: Find the area, A = pi r^2

Step 2: Substitute, A = 3.1416 x (3.7 x 10^4)^2 = 3.1416 x 1.369 x 10^9 = 4.30 x 10^9 cm^2

Step 3: Find the fractional uncertainty in r, 0.05 / 3.7 = 0.0135, and because A depends on r^2, the fractional uncertainty in A is doubled, 2 x 0.0135 = 0.027

Step 4: Find the absolute uncertainty, 0.027 x 4.30 x 10^9 = 0.12 x 10^9 cm^2, approximately 0.11 x 10^9 cm^2

Final answer: The area is (4.3 +/- 0.11) x 10^9 cm^2.

Question: An aeroplane travels at 850 km/h. How long does it take to travel 1.00 km?

Answer: Given: speed = 850 km/h, distance = 1.00 km

Step 1: Convert the speed to m/s, 850 x 1000 / 3600 = 236.1 m/s

Step 2: Convert the distance, 1.00 km = 1000 m

Step 3: Use time = distance / speed

Step 4: Substitute, time = 1000 / 236.1 = 4.24 s

Final answer: The aeroplane takes about 4.23 s to travel 1.00 km.

Question: A rectangular holding tank 25.0 m in length and 15.0 m in width is used to store water for a short period of time in an industrial plant. If 2980 m^3 of water is pumped into the tank, what is the depth of the water?

Answer: Given: length = 25.0 m, width = 15.0 m, volume of water = 2980 m^3

Step 1: Write the volume relation, volume = length x width x depth

Step 2: Rearrange for depth, depth = volume / (length x width)

Step 3: Substitute, depth = 2980 / (25.0 x 15.0) = 2980 / 375

Step 4: Calculate, depth = 7.95 m

Final answer: The depth of the water is 7.95 m.

Question: Find the volume of a rectangular underground water tank having dimensions 1.9 m by 1.2 m by 0.8 m.

Answer: Given: length = 1.9 m, width = 1.2 m, height = 0.8 m

Step 1: Write the formula, volume = length x width x height

Step 2: Substitute, volume = 1.9 x 1.2 x 0.8

Step 3: Multiply in stages, 1.9 x 1.2 = 2.28, and 2.28 x 0.8 = 1.824

Final answer: The volume of the tank is 1.824 m^3.

Question: Two students derive the following equations, in which x refers to

distance travelled, v to the speed, a to the acceleration, t to the time and the subscript (o) means a quantity at time t = 0: (a) x = vt^2 + 2at and (b) x = v_o t + 2at^2. Which of these could possibly be correct according to a dimensional check?

Answer: Given: [x] = L, [v] = LT^-1, [a] = LT^-2, [t] = T

Step 1: Check part (a), term by term: [v t^2] = LT^-1 x T^2 = LT, which is not L, and [2at] = LT^-2 x T = LT^-1, which is also not L, so part (a) is dimensionally incorrect

Step 2: Check part (b), term by term: [v_o t] = LT^-1 x T = L, which is correct, and [2at^2] = LT^-2 x T^2 = L, which is also correct

Step 3: Compare the two results: every term in (b) has the dimension of length, while no term in (a) does

Final answer: (a) is incorrect and (b) could possibly be correct.

Question: One hectare is defined as 10^4 m^2. One acre is 4 x 10^4 ft^2. How many acres are in one hectare? (Hint: 1 m = 3.28 ft.)

Answer: Given: 1 hectare = 10^4 m^2, 1 acre = 4 x 10^4 ft^2, 1 m = 3.28 ft

Step 1: Convert one square metre to square feet, 1 m^2 = (3.28)^2 ft^2 = 10.76 ft^2

Step 2: Convert one hectare to square feet, 10^4 x 10.76 = 1.076 x 10^5 ft^2

Step 3: Divide by the size of one acre, number of acres = 1.076 x 10^5 / (4 x 10^4) = 2.69

Final answer: One hectare is about 2.69 acres.

Question: A watch factory claims that its watches gain or lose not more

than 10 seconds in a year. How accurate is this watch, expressed as a percentage?

Answer: Given: maximum error = 10 s, time interval = 1 year

Step 1: Convert one year to seconds, 365 x 24 x 3600 = 3.15 x 10^7 s

Step 2: Write the formula, percentage accuracy error = (error / total time) x 100

Step 3: Substitute, (10 / 3.15 x 10^7) x 100 = 3.17 x 10^-5 percent

Final answer: The watch is accurate to about 3.16 x 10^-5 percent.

Question: The diameter of the Moon is 3480 km. What is the volume of the Moon? How many Moons would be needed to create a volume equal to that of Earth? (Hint: Radius of Earth = 6380 km)

Answer: Given: diameter of Moon = 3480 km so radius of Moon = 1740 km = 1.74 x 10^6 m, radius of Earth = 6380 km

Step 1: Find the volume of the Moon, V = (4/3) pi r^3

Step 2: Substitute, V = 1.333 x 3.1416 x (1.74 x 10^6)^3 = 4.189 x 5.27 x 10^18 = 2.2 x 10^19 m^3

Step 3: For the number of Moons, use the ratio of volumes, which is the cube of the ratio of radii, (6380 / 1740)^3

Step 4: Calculate the ratio, 6380 / 1740 = 3.667, and 3.667^3 = 49.3

Final answer: The volume of the Moon is about 2.2 x 10^19 m^3, and about 49.3 Moons would make up the volume of Earth.

Unit 2: Kinematics

Exercise — Multiple Choice Questions

To get a resultant displacement of 10 m, two displacement vectors of magnitude 6 m and 8 m should be combined:

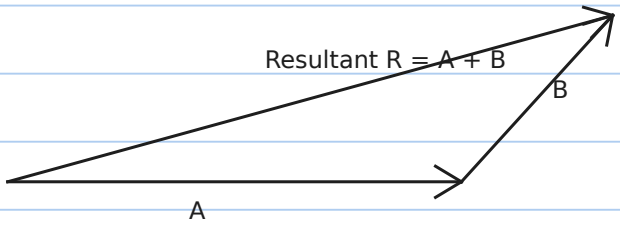
(a) Parallel

(b) Antiparallel

(c) At an angle of 45 deg

✓ (d) Perpendicular to each other

Answer: (d) Working: for perpendicular vectors the resultant is sqrt(6^2 + 8^2) = sqrt(36 + 64) = sqrt(100) = 10 m. Parallel vectors would give 14 m and antiparallel vectors only 2 m, so only the perpendicular combination gives 10 m.



The velocity of a particle at an instant is 10 m/s and after 5 sec the velocity of the particle is 20 m/s. The velocity at 3 sec before that instant is:

(a) 8

✓ (b) 4

(c) 6

(d) 7

Answer: (b) m/s. Working: the acceleration is (20 - 10) / 5 = 2 m/s^2, assuming uniform acceleration. Going 3 s back from the instant, the velocity was 10 - (2 x 3) = 4 m/s.

A ball is thrown upwards with a velocity of 100 m/s. It will reach the ground after:

(a) 10 sec

✓ (b) 20 sec

(c) 5 sec

(d) 40 sec

Answer: (b) Working: taking g = 10 m/s^2, the time to go up and come back is the total time of flight, 2u / g = (2 x 100) / 10 = 20 s. The ball spends 10 s rising and the same 10 s falling back to the ground.

Two projectiles are fired from the same point with the same speed at angles of projection 60 deg and 30 deg respectively. Which one of the following is true?

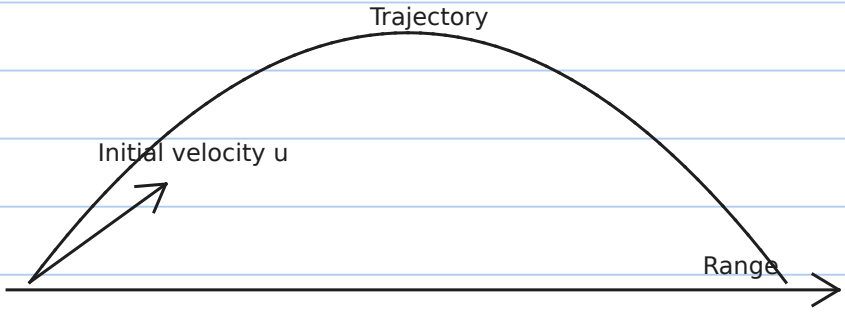
✓ (a) The range will be same

(b) Their maximum height will be same

(c) Their landing velocity will be same

(d) Their time of flight will be same

Answer: (a) The range depends on sin 2theta, and sin 120 deg = sin 60 deg, so the complementary angles 30 deg and 60 deg give equal ranges for the same speed. Their heights and times of flight differ, because those depend on sin theta and sin^2 theta, which are not equal for 30 deg and 60 deg.



The ratio of numerical values of average velocity and average speed of a body is always:

(a) Unity

✓ (b) Unity or less

(c) Unity or more

(d) Less than unity

Answer: (b) Average velocity is displacement divided by time and average speed is total distance divided by time. Displacement can never be greater than the distance travelled, so the ratio is 1 for straight motion in one direction and less than 1 whenever the path is not straight.

If the average velocity of a body becomes equal to its instantaneous velocity, the body is said to be moving with:

(a) Uniform acceleration

✓ (b) Uniform velocity

(c) Variable velocity

(d) Variable acceleration

Answer: (b) If the velocity does not change at any instant, then the velocity at every instant is the same as the average over any interval. In all other cases the instantaneous velocity keeps differing from the average value.

At the top of the trajectory of a projectile, the acceleration is:

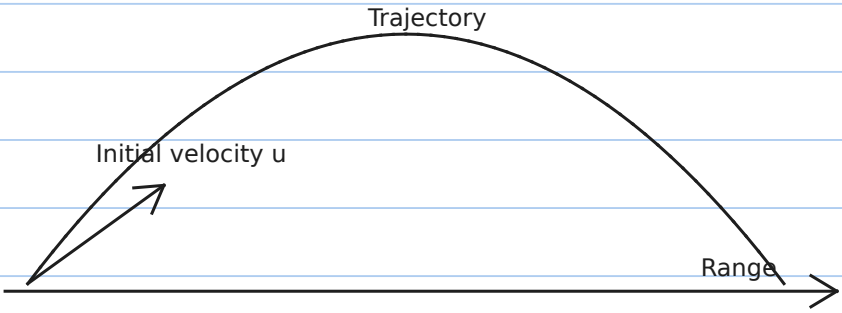
(a) maximum

(b) minimum

(c) zero

✓ (d) g

Answer: (d) Throughout the flight the only acceleration acting on a projectile is the acceleration due to gravity, acting vertically downward. At the top only the vertical component of velocity becomes zero for an instant; the acceleration remains g.



At what angle does the range of a projectile become equal to the height of the projectile?

(a) 65 deg

(b) 45 deg

✓ (c) 76 deg

(d) 30 deg

Answer: (c) Working: setting range equal to maximum height gives sin 2theta = (sin^2 theta) / 2, which reduces to tan theta = 4. The angle whose tangent is 4 is about 76 deg, so at that steep angle the two quantities are equal.

The angle at which the dot product becomes equal to the cross product is:

(a) 65 deg

✓ (b) 45 deg

(c) 76 deg

(d) 30 deg

Answer: (b) The magnitudes are AB cos theta for the dot product and AB sin theta for the cross product, and these are equal when sin theta = cos

theta, that is tan theta = 1. This happens at theta = 45 deg.

If the dot product of two non-zero vectors vanishes, the vectors will be:

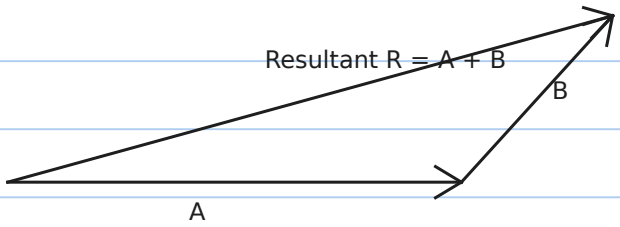
(a) in the same direction

(b) opposite direction to each other

(c) perpendicular to each other

(d) zero

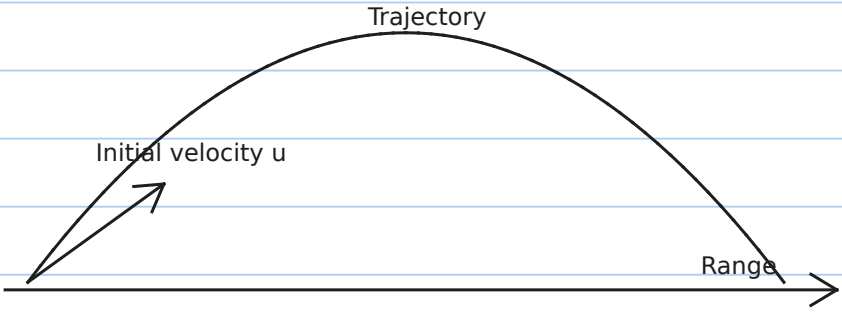
Answer: (c) The dot product is AB cos theta, and for non-zero vectors it can vanish only when cos theta = 0. That means theta = 90 deg, so the vectors are at right angles to each other.



Exercise — Structured Questions (CRQs)

Question: Is the range equivalent to the horizontal distance in projectile motion, or do they have distinct meanings? Explain.

Answer: They are closely related but not exactly the same idea. The range is the specific horizontal distance between the point of projection and the point where the projectile lands at the same level. The phrase horizontal distance is general and can mean the horizontal separation at any stage of the flight or between any two points. So range is one particular horizontal distance, the total one for the complete flight.

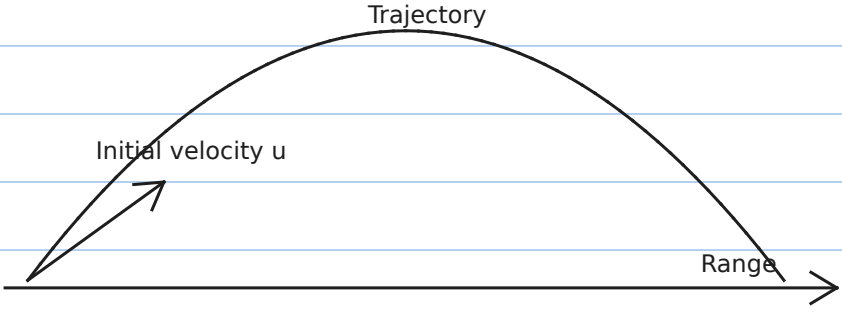


Question: If air resistance is taken into account in the case of projectile motion, then what parameters of the projectile are influenced?

Answer: Air resistance opposes the motion at every instant, so it reduces both the horizontal and the vertical component of velocity. As a result the maximum height, the range and the speed of landing all become smaller than their ideal values, and the trajectory is no longer a symmetric parabola, the falling part becoming steeper. The time of flight also changes, and strictly the horizontal velocity is no longer constant.

Question: What are some real-life examples of projectile motion? Can the motion of an aeroplane be considered as projectile motion? Explain.

Answer: A kicked football, a cricket ball hit by the bat, a basketball thrown towards the basket, water from a garden hose and a stone thrown from a cliff are all projectiles once they leave the hand or the bat and move under gravity alone. The motion of an aeroplane in normal flight is not projectile motion, because its engines provide thrust and its wings provide lift, so forces other than gravity act on it throughout. Only an object dropped or released from the aeroplane behaves as a projectile.



Question: Where do we use the concept of speed and velocity in our daily life? Can you give some examples?

Answer: Speed is used whenever only the rate of covering distance matters, for example the speedometer of a car, speed limits on roads, the speed of a ceiling fan or of an internet connection. Velocity is used when the direction also matters, for example in navigation of ships and aircraft, in weather reports giving wind movement, and in sports where a ball must be sent in a chosen direction. In short, speed tells how fast, while velocity tells how fast and in which direction.

Question: Can an object have an initial velocity of zero and still experience uniformly accelerated motion?

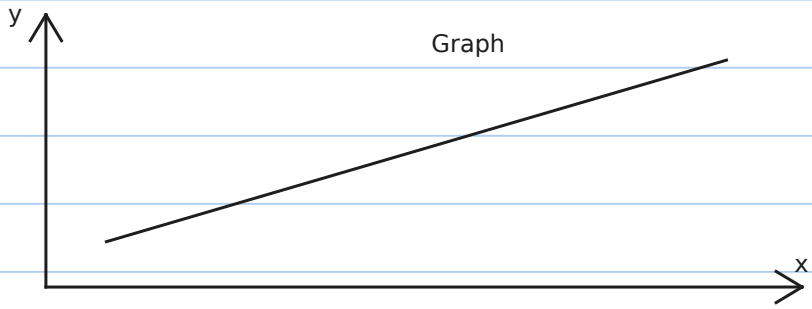
Answer: Yes, it can. Acceleration depends on the force acting on the body, not on its starting velocity, so a body dropped from rest starts with zero velocity but falls with the uniform acceleration g. Similarly, a car starting from rest with a steady engine force moves with roughly uniform acceleration in its first moments. Zero initial velocity only means the body begins from rest; it says nothing about how its velocity will change afterwards.

Exercise — Structured Questions (ERQs)

Question: What are the three equations of uniformly accelerated motion and how are they derived?

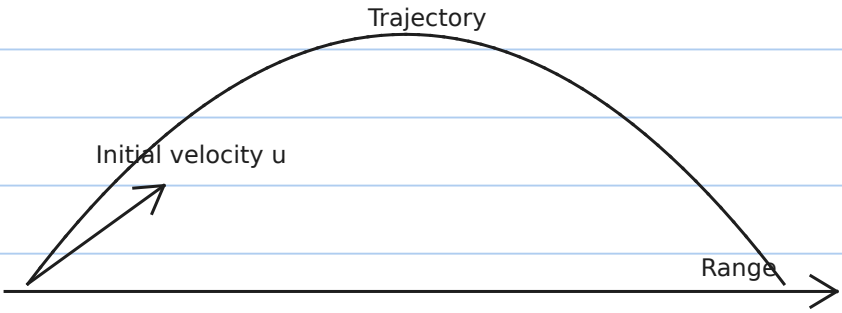
Answer: The first equation, v = u + at, comes directly from the definition of acceleration, a = (v - u) / t, rearranged for v. The second, s = ut + (1/2)at^2, comes from the area under the velocity-time graph, which is a rectangle of area ut plus a triangle of area (1/2)at^2. The third, v^2 =

u^2 + 2as, is obtained by eliminating t between the first two equations. Together they connect the five quantities u, v, a, t and s for motion with constant acceleration.



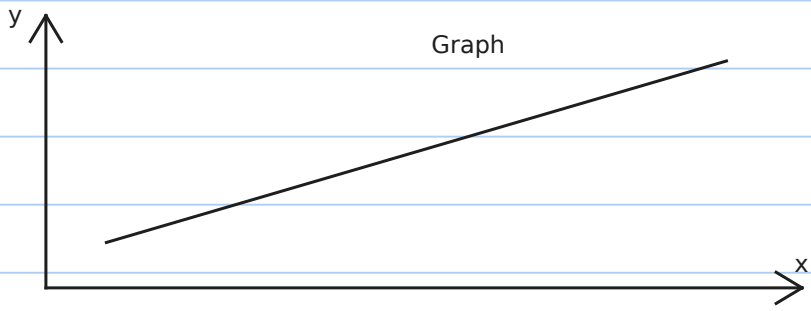
Question: How does the horizontal component of velocity in projectile motion behave?

Answer: The horizontal component, v_x = u cos theta, remains constant throughout the flight when air resistance is neglected. This is because no force acts in the horizontal direction, so there is no horizontal acceleration. The projectile therefore covers equal horizontal distances in equal times, and its horizontal motion is exactly like uniform motion in a straight line.



Question: What is the significance of the area under a velocity-time graph in the context of accelerated motion?

Answer: The area under a velocity-time graph gives the displacement of the body during that time interval. This is because a thin strip of the area has height v and width dt, and v x dt is the small displacement in that short interval. The rule holds for uniform as well as accelerated motion, and in addition the slope of the same graph gives the acceleration.



Question: How do the initial launch angle and velocity affect projectile motion?

Answer: Increasing the launch velocity increases everything: the range and maximum height grow with u^2 and the time of flight grows with u. The launch angle divides the velocity into components: a larger angle gives more height and time of flight but less horizontal speed, while a smaller angle gives a flatter, faster flight. The range is greatest at 45 deg, and two complementary angles give the same range with different heights and times.

Question: Explain the concept of independence between horizontal and vertical motion in projectile motion.

Answer: The horizontal and vertical motions of a projectile take place independently of each other, each obeying its own equations. Horizontally there is no acceleration, so the velocity stays u cos theta, while vertically gravity produces the acceleration g and the velocity changes as in free fall. The two motions are linked only by the common time t, which is why the combination of steady horizontal motion and accelerated vertical motion produces the parabolic path.

Exercise — Numerical Problems

Question: A helicopter is ascending at the rate of 12 m/s. At a height of 80

m above the ground, a package is dropped. How long does the package take to reach the ground? (Ans: 5.4 sec)

Answer: Given: initial velocity of package u = +12 m/s (upward, the same as the helicopter), height = 80 m, g = 10 m/s^2

Step 1: Take upward as positive, so the displacement to the ground is s = -80 m and the acceleration is a = -10 m/s^2

Step 2: Use s = ut + (1/2)at^2, giving -80 = 12t - 5t^2

Step 3: Rearrange into a quadratic, 5t^2 - 12t - 80 = 0

Step 4: Solve, t = [12 + sqrt(144 + 1600)] / 10 = [12 + sqrt(1744)] / 10 = [12 + 41.76] / 10 = 5.38 s, rejecting the negative root

Final answer: The package takes about 5.4 s to reach the ground.

Question: Two tug boats are towing a ship, each exerts a force of 6000 N, and the angle between the two ropes is 60 deg. Calculate the resultant force on the ship. (Ans: 10392 N)

Answer: Given: F1 = 6000 N, F2 = 6000 N, angle between the forces theta = 60 deg

Step 1: Write the parallelogram law, R = sqrt(F1^2 + F2^2 + 2 F1 F2 cos theta)

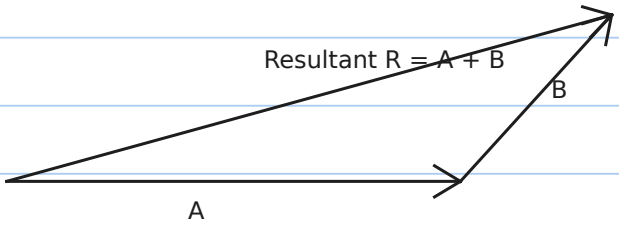
Step 2: Substitute, R = sqrt(6000^2 + 6000^2 + 2 x 6000 x 6000 x cos 60 deg)

Step 3: Calculate inside the root, 3.6 x 10^7 + 3.6 x 10^7 + (7.2 x 10^7 x 0.5) = 1.08 x 10^8

Step 4: Take the square root, R = sqrt(1.08 x 10^8) = 10392 N

Final answer: The resultant force on the ship is 10392 N, acting along the

bisector of the angle between the ropes.



Question: A car starts from rest and moves with a constant acceleration. During the 5th second of its motion, it covers a distance of 36 metres. Calculate: (a) the acceleration of the car, (b) the total distance covered by the car during this time. (Ans: 8 m/s^2, 100 m)

Answer: Given: initial velocity u = 0, distance covered in the 5th second = 36 m

Step 1: Use the formula for distance in the nth second, s_n = u + (a/2)(2n - 1), with n = 5

Step 2: Substitute, 36 = 0 + (a/2)(9), so 36 = 4.5a

Step 3: Solve part (a), a = 36 / 4.5 = 8 m/s^2

Step 4: For part (b), use s = ut + (1/2)at^2 with t = 5 s, giving s = 0 + (1/2) x 8 x 25 = 100 m

Final answer: The acceleration is 8 m/s^2 and the total distance covered in 5 s is 100 m.

Question: Show that the ranges of a projectile at complementary angles are the same, with examples.

Answer: Given: range formula R = (u^2 sin 2theta) / g for projection angle theta

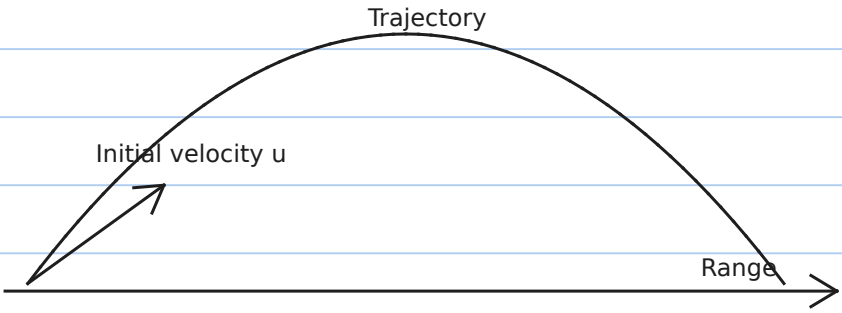
Step 1: Write the range for the complementary angle (90 deg - theta), R' =

(u^2 sin 2(90 deg - theta)) / g = (u^2 sin(180 deg - 2theta)) / g

Step 2: Use the identity sin(180 deg - 2theta) = sin 2theta, so R' = (u^2 sin 2theta) / g = R

Step 3: Give an example: for the same speed u, angles 30 deg and 60 deg give sin 60 deg and sin 120 deg, and both sines equal 0.866, so both ranges are equal

Final answer: Projectiles projected at complementary angles, such as 30 deg and 60 deg, have the same range.



Question: At what angle does the range of a projectile become equal to the height of the projectile? (Ans: 76 deg)

Answer: Given: range R = (u^2 sin 2theta) / g and maximum height H = (u^2 sin^2 theta) / (2g), and R = H

Step 1: Equate the two expressions, (u^2 sin 2theta) / g = (u^2 sin^2 theta) / (2g), and cancel u^2 / g

Step 2: Write sin 2theta = 2 sin theta cos theta, so 2 sin theta cos theta = (sin^2 theta) / 2

Step 3: Divide both sides by sin theta cos theta, giving tan theta = 4

Step 4: Find the angle, theta = arctan(4) = 76 deg

Final answer: The range equals the maximum height when the angle of projection is about 76 deg.

Question: A mortar shell is fired at a ground level target 500 m distance with an initial velocity of 90 m/s. What is the launch angle? (Ans: 71.4 deg)

Answer: Given: range R = 500 m, initial velocity u = 90 m/s, g = 9.8 m/s^2

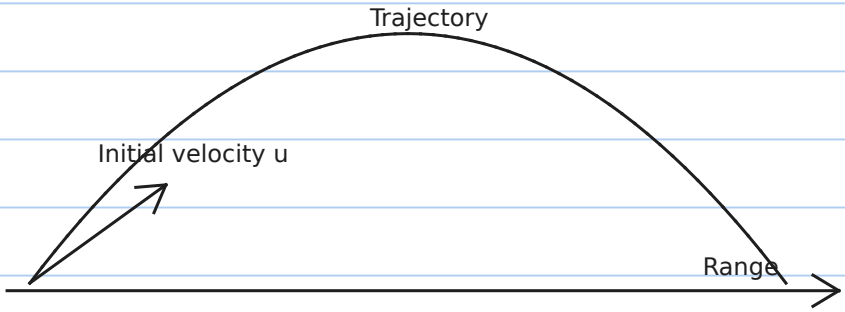
Step 1: Write the range formula, R = (u^2 sin 2theta) / g

Step 2: Rearrange, sin 2theta = Rg / u^2

Step 3: Substitute, sin 2theta = (500 x 9.8) / (90)^2 = 4900 / 8100 = 0.605

Step 4: Find 2theta, 2theta = sin^-1(0.605) = 37.2 deg or 180 - 37.2 = 142.8 deg, so theta = 18.6 deg or 71.4 deg

Final answer: The launch angle is 71.4 deg for the high trajectory (the shell can also reach the target at the complementary angle of 18.6 deg).



Unit 3: Dynamics

Exercise — Multiple Choice Questions

The rate of change of linear momentum of a body is called a:

✓ (a) Linear force

(b) Angular force

(c) Power

(d) Impulse

Answer: (a) By Newton's second law, the resultant force acting on a body is equal to the rate of change of its linear momentum, F = delta p / delta t. Impulse is force multiplied by time, and power is the rate of doing work, so neither of them is a rate of change of momentum.

The term mass refers to the same physical concept as:

(a) weight

✓ (b) inertia

(c) force

(d) acceleration

Answer: (b) Mass is the measure of the inertia of a body, that is, of its resistance to any change in its state of rest or motion. A body of greater mass is harder to start, stop or turn, while weight is a force that depends on gravity and is a different quantity.

Which one of the following forces is also called a self adjusting force?

✓ (a) frictional force

(b) tension

(c) Weight

(d) Thrust

Answer: (a) Static friction adjusts its own magnitude to match the applied force, up to its limiting value, so that the body stays at rest. Tension, weight and thrust do not change themselves in this self adjusting way.

The laws of motion show the relationship between:

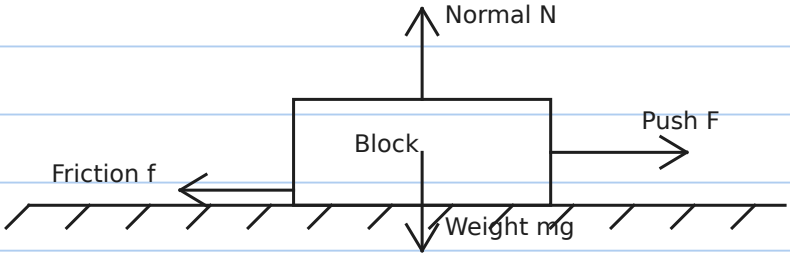
(a) velocity and acceleration

(b) mass and velocity

(c) mass and acceleration

✓ (d) force and acceleration

Answer: (d) Newton's second law, F = ma, states that the acceleration produced in a body is directly proportional to the resultant force and inversely proportional to its mass. The laws of motion are therefore fundamentally a statement of how force and acceleration are related for a given mass.



The motion of a rocket in space is according to the law of conservation of:

(a) Energy

✓ (b) linear momentum

(c) mass

(d) angular momentum

Answer: (b) The rocket throws exhaust gases backwards with momentum in one direction, so the rocket gains equal momentum in the forward direction. The total momentum of the rocket and gases stays constant,

which is why a rocket works even in empty space.

A bomb of mass 12 kg, initially at rest, explodes into two pieces of masses 4 kg and 8 kg. The speed of the 8 kg mass is 6 m/s. The kinetic energy of the 4 kg mass is:

(a) 32 J

(b) 48 J

(c) 114 J

✓ (d) 288 J

Answer: (d) Working: by conservation of momentum, 4 x v = 8 x 6, so the 4 kg piece moves at v = 12 m/s. Its kinetic energy is (1/2) x 4 x 12^2 = 2 x 144 = 288 J.

If momentum is increased by 20 percent, then kinetic energy increases by:

✓ (a) 44 percent

(b) 55 percent

(c) 66 percent

(d) 77 percent

Answer: (a) Kinetic energy is p^2 / (2m), so it grows with the square of the momentum. A 20 percent increase makes the momentum 1.20 times, and 1.20^2 = 1.44, which is a 44 percent increase in kinetic energy.

The kinetic energy of a body of mass 2 kg and momentum of 2 N s is:

✓ (a) 1 J

(b) 2 J

(c) 3 J

(d) 4 J

Answer: (a) Working: kinetic energy = p^2 / (2m) = 2^2 / (2 x 2) = 4 / 4 = 1 J. This relation is useful whenever the momentum and mass are known instead of the velocity.

For the same kinetic energy, the momentum is maximum for:

(a) an electron

(b) a proton

(c) a deuteron

✓ (d) alpha particle

Answer: (d) For a fixed kinetic energy, momentum is p = sqrt(2m x K.E.), so the momentum is larger for the larger mass. The alpha particle is the heaviest of the four, being about four times as heavy as a proton, so it carries the greatest momentum.

A 3 kg bowling ball experiences a net force of 15 N. What will be its acceleration?

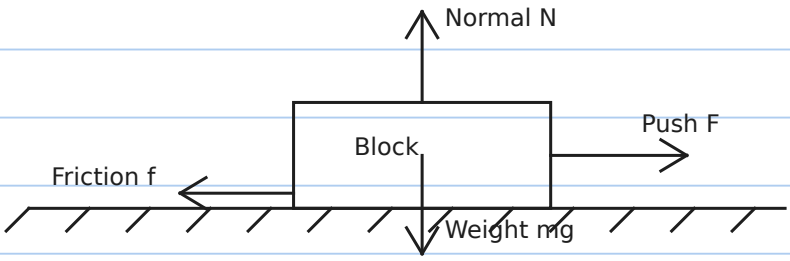
(a) 35 m/s^2

(b) 7 m/s^2

✓ (c) 5 m/s^2

(d) 35 m/s^2

Answer: (c) Working: from F = ma, the acceleration is a = F / m = 15 / 3 = 5 m/s^2. The acceleration acts in the direction of the net force on the ball.



Exercise — Structured Questions (CRQs)

Question: State Newton's second law of motion.

Answer: Newton's second law states that the acceleration of a body is directly proportional to the resultant force acting on it and inversely

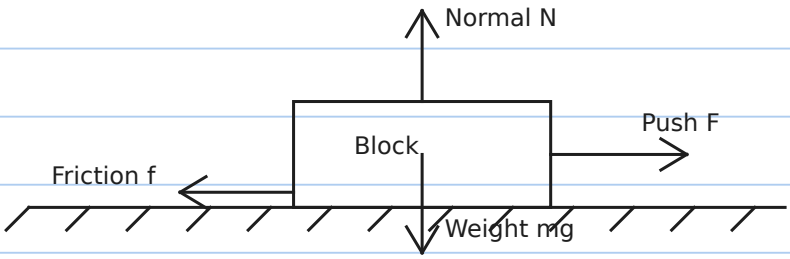
proportional to its mass, and it takes place in the direction of the force. In symbols, F = ma, where F is the net force in newtons, m the mass in kilograms and a the acceleration in m/s^2. It also has the more general form that force is the rate of change of momentum.

Question: How does mass affect an object's acceleration?

Answer: For the same applied force, a larger mass gives a smaller acceleration, because a = F / m and acceleration is inversely proportional to mass. Doubling the mass halves the acceleration if the force is unchanged. This is why a loaded truck speeds up much more slowly than an empty one driven with the same engine force.

Question: What are the different types of forces that can act on an object?

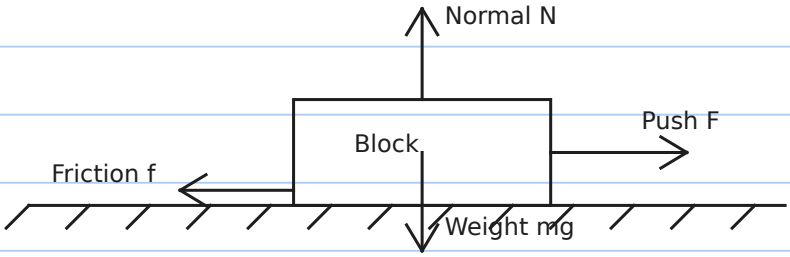
Answer: Forces may be contact forces or field forces acting at a distance. Contact forces include the push or pull of a hand, friction, tension in a string, the normal reaction of a surface and air resistance. Field forces include the gravitational force (weight), the electric force and the magnetic force, which act without touching the body. In every case a force can change the speed, the direction or the shape of the object.



Question: How do you calculate net force?

Answer: First list all the forces acting on the body and show them on a free body diagram, with their directions. Then add the forces as vectors, taking forces in one chosen direction as positive and those in the opposite direction as negative, or resolving them into x and y components when they are not in a line. The vector sum so obtained is the net force, and it

alone decides the acceleration through F_net = ma.



Question: State the law of conservation of momentum.

Answer: The law states that when no external force acts on a system, the total linear momentum of the system remains constant. In other words, the total momentum before a collision or an explosion is equal to the total momentum after it, even though individual bodies may exchange momentum. This law follows from Newton's second and third laws.

Question: What is the difference between elastic and inelastic collisions?

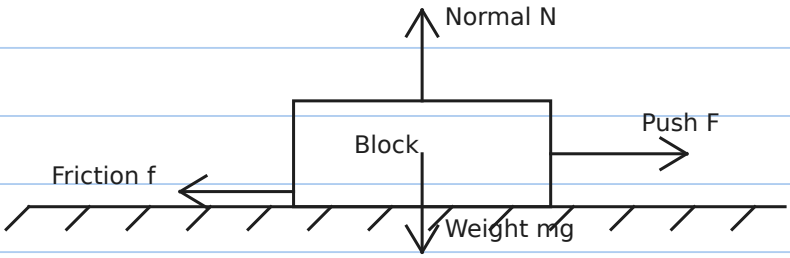
Answer: In an elastic collision both momentum and kinetic energy are conserved, and the bodies bounce apart without any permanent change of shape, like two billiard balls. In an inelastic collision momentum is conserved but kinetic energy is not, because part of it changes into heat, sound and deformation. If the bodies stick together after impact, the collision is called perfectly inelastic and the loss of kinetic energy is the maximum possible.

Question: How does impulse relate to force and time?

Answer: Impulse is the product of a force and the time for which it acts, J = F x delta t, and it is equal to the change in momentum of the body. A large force acting for a short time can therefore produce the same impulse as a small force acting for a long time. This is why extending the time of impact, as with a fielder pulling his hands back while catching a ball, reduces the force experienced.

Question: How does friction influence the motion of an object?

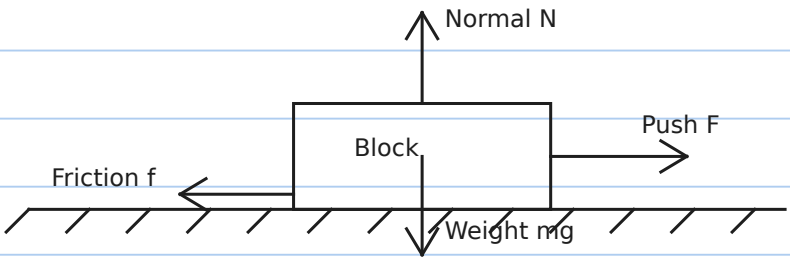
Answer: Friction opposes the relative motion, or the tendency of motion, between surfaces in contact, so it slows a moving object and can bring it to rest. It wastes part of the kinetic energy as heat, which is why extra force is needed to keep an object moving at constant speed on a rough surface. At the same time friction is useful, because walking, braking of vehicles and gripping of objects are all impossible without it.



Exercise — Structured Questions (ERQs)

Question: Explain Newton's Second Law of Motion and how it relates force, mass and acceleration. Provide an example to illustrate the concept.

Answer: Newton's second law says that a resultant force produces an acceleration in its own direction, the acceleration being directly proportional to the force and inversely proportional to the mass, F = ma. It means a bigger push speeds a body up faster, while a heavier body responds more slowly to the same push. For example, a force of 20 N gives a 2 kg trolley an acceleration of 10 m/s^2, but the same force gives a 5 kg trolley only 4 m/s^2. The law also explains everyday facts such as why a loaded vehicle needs a stronger engine force to accelerate like an empty one.



Question: How does the conservation of momentum apply in each type of collision? Give real-life examples of both types of collision.

Answer: Momentum is conserved in every type of collision provided no external force acts during the short impact, because the internal action and reaction forces cancel. In an elastic collision kinetic energy is also conserved, as in the collision of billiard balls or of hard steel balls in Newton's cradle, where the balls separate with almost the same total kinetic energy. In an inelastic collision kinetic energy is partly lost as heat, sound and deformation, as when two cars collide and crumple, or when a bullet embeds itself in a wooden block and the two move together.

Question: Explain how the concept of impulse is related to the change in momentum of an object. Provide an example of an everyday life where impulse plays a significant role.

Answer: Impulse is defined as force multiplied by the time of its action, and the impulse-momentum theorem states that this impulse is exactly equal to the change in momentum, F x delta t = m v - m u. So whenever a force acts for some time, it changes the momentum of the body by a definite amount. A common example is a cricketer catching a fast ball: by pulling the hands backwards he increases the stopping time, so the same change of momentum is produced by a smaller, safer force on his hands.

Question: State the law of conservation of momentum. Provide an example from everyday life that demonstrates the principle of momentum conservation.

Answer: The law states that in a closed system with no external force, the total momentum before an interaction equals the total momentum after it. A clear everyday example is the recoil of a gun: before firing, the gun and bullet together have zero momentum, and after firing the forward momentum of the bullet is balanced by the equal backward momentum of the gun. The same principle is seen when a person jumps from a small boat and the boat moves backwards in the opposite direction.

Exercise — Numerical Problems

Question: A car weighing 9800 N is moving with a speed of 40 km/h. On the application of brakes it comes to rest after travelling a distance of 50 metres. Calculate the average retarding force. (Ans: 1234.57 N)

Answer: Given: weight W = 9800 N, initial speed = 40 km/h, final speed = 0, distance = 50 m, g = 9.8 m/s^2

Step 1: Find the mass, m = W / g = 9800 / 9.8 = 1000 kg

Step 2: Convert the speed, 40 km/h = 40 x 1000 / 3600 = 11.11 m/s

Step 3: Use v^2 = u^2 + 2as, so 0 = (11.11)^2 + 2a(50), giving 0 = 123.4 + 100a

Step 4: Solve for the retardation, a = -123.4 / 100 = -1.235 m/s^2

Step 5: Find the force, F = ma = 1000 x 1.235 = 1235 N in magnitude (using more exact figures, 1234.57 N)

Final answer: The average retarding force is about 1234.57 N, acting opposite to the motion.

Question: A helicopter weighs 3920 N. Calculate the force on it if it is

ascending up at the rate of 2 m/s^2. What will be the force on the helicopter if it is moving up with a constant speed of 4 m/s? (Ans: 4720 N, 3920 N)

Answer: Given: weight W = 3920 N, upward acceleration a = 2 m/s^2, g = 9.8 m/s^2

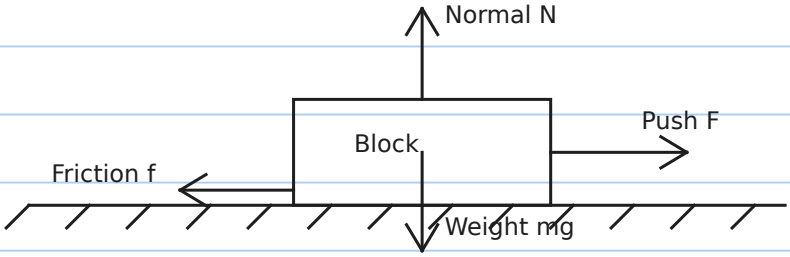
Step 1: Find the mass, m = W / g = 3920 / 9.8 = 400 kg

Step 2: For accelerated ascent, the lifting force must overcome weight and also provide ma, so F = m(g + a)

Step 3: Substitute, F = 400 x (9.8 + 2) = 400 x 11.8 = 4720 N

Step 4: At a constant speed of 4 m/s the acceleration is zero, so the lifting force only balances the weight, F = W = 3920 N

Final answer: The force is 4720 N while accelerating upward, and 3920 N while rising at constant speed.



Question: A 100 grams bullet is fired from a 10 kg gun with a speed of 1000 m/s. What is the speed of recoil of the gun? (Ans: 10 m/s)

Answer: Given: mass of bullet m = 100 g = 0.1 kg, speed of bullet v = 1000 m/s, mass of gun M = 10 kg, initial momentum = 0

Step 1: Write the conservation of momentum, total momentum after firing = total momentum before firing, so mv + MV = 0 in signed form, i.e. mv = MV in magnitude

Step 2: Substitute, 0.1 x 1000 = 10 x V

Step 3: Solve, 100 = 10V, so V = 10 m/s

Final answer: The gun recoils with a speed of 10 m/s in the direction opposite to the bullet.

Question: A machine gun fires 10 bullets per second into a target. Each bullet weighs 20 gram and had a speed of 1500 m/s. Find the necessary force to hold the gun in position. (Ans: 300 N)

Answer: Given: number of bullets per second n = 10, mass of each bullet m = 20 g = 0.02 kg, speed of each bullet v = 1500 m/s

Step 1: Find the momentum carried away by one bullet, p = mv = 0.02 x 1500 = 30 kg m/s

Step 2: Find the momentum carried away per second, 10 x 30 = 300 kg m/s per second

Step 3: Force equals rate of change of momentum, so the gun experiences a backward force of 300 N, and an equal forward force must be applied to hold it

Final answer: A force of 300 N is necessary to hold the gun in position.

Question: A 50 grams bullet is fired into a 10 kg block that is suspended by a long cord so that it can swing as a pendulum. If the block is displaced so that its centre of gravity rises by 10 cm, what was the speed of the bullet? (Ans: 281.4 m/s)

Answer: Given: mass of bullet m = 50 g = 0.05 kg, mass of block M = 10 kg, rise of centre of gravity h = 10 cm = 0.10 m, g = 9.8 m/s^2

Step 1: Find the speed V of the block plus bullet just after impact from the pendulum rise, V = sqrt(2gh)

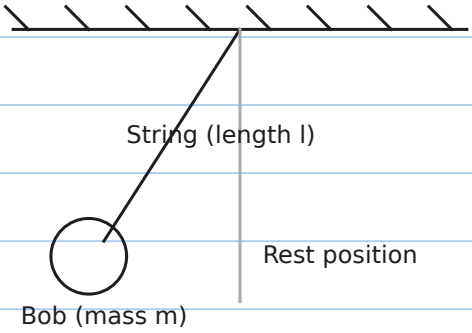
Step 2: Substitute, V = sqrt(2 x 9.8 x 0.10) = sqrt(1.96) = 1.4 m/s

Step 3: Apply conservation of momentum for the impact, m x v = (M + m) x V

Step 4: Substitute, 0.05 x v = (10 + 0.05) x 1.4 = 10.05 x 1.4 = 14.07

Step 5: Solve, v = 14.07 / 0.05 = 281.4 m/s

Final answer: The speed of the bullet was 281.4 m/s.



Question: A 70-gram ball collides with another ball of mass 140 gram. The initial velocity of the first ball is 9 m/s to the right while the second ball is at rest. If the collision is perfectly elastic, what would be the velocity of the two balls after the collision? (Ans: -3 m/s, 6 m/s)

Answer: Given: m1 = 70 g, m2 = 140 g, u1 = 9 m/s, u2 = 0, perfectly elastic collision

Step 1: Write the elastic collision formula for the first ball, v1 = [(m1 - m2) / (m1 + m2)] x u1

Step 2: Substitute, v1 = [(70 - 140) / (70 + 140)] x 9 = (-70 / 210) x 9 = -3 m/s

Step 3: Write the formula for the second ball, v2 = [2m1 / (m1 + m2)] x u1

Step 4: Substitute, v2 = [140 / 210] x 9 = 6 m/s

Step 5: Interpret the signs: the minus sign means the first ball rebounds to

the left, while the second ball moves to the right

Final answer: The first ball moves at -3 m/s (3 m/s to the left) and the second ball at 6 m/s to the right.

Question: A truck weighing 2500 kg and moving with a velocity of 21 m/s collides with a stationary car weighing 1000 kg. The truck and car move together after the impact. Calculate their common velocity. (Ans: 15 m/s)

Answer: Given: mass of truck m1 = 2500 kg, velocity of truck u1 = 21 m/s, mass of car m2 = 1000 kg, velocity of car u2 = 0

Step 1: Write the conservation of momentum, m1 u1 + m2 u2 = (m1 + m2) V

Step 2: Substitute, 2500 x 21 + 1000 x 0 = (2500 + 1000) V

Step 3: Calculate the left side, 52500 = 3500 V

Step 4: Solve, V = 52500 / 3500 = 15 m/s

Final answer: The common velocity of the truck and car after the impact is 15 m/s in the original direction of the truck.

Unit 4: Rotational and Circular Motion

Exercise - Multiple Choice Questions (MCQs)

One radian is about

(a) 25 deg

(b) 37 deg

(c) 45 deg

✓ (d) 57 deg

Answer: (d) Working: one full circle is 2 pi radians = 360 deg, so 1 radian = 360 / (2 pi) = 57.3 deg, which is about 57 deg. The radian is the angle for which the arc length equals the radius of the circle.

A wheel turns with constant angular speed, then:

(a) each point on its rim moves with constant velocity

(b) each point on its rim moves with constant acceleration

✓ (c) the wheel turns through equal angles in equal times

Answer: (c) Working: constant angular speed means the angular displacement per second never changes, so equal angles are swept in equal times. A rim point still changes the direction of its velocity, so its velocity is not constant even though its speed is.

The rotational inertia of a wheel about its axle does not depend upon its

(a) diameter

(b) mass

(c) distribution of mass

✓ (d) speed of rotation

Answer: (d) Working: moment of inertia I = sum of m r^2 depends only on the total mass, the size of the body and how the mass is distributed about the axis. It is a property of the body and the axis, not of how fast the body happens to be spinning.

A force of given magnitude is to be applied to a wheel. The torque can be maximized by

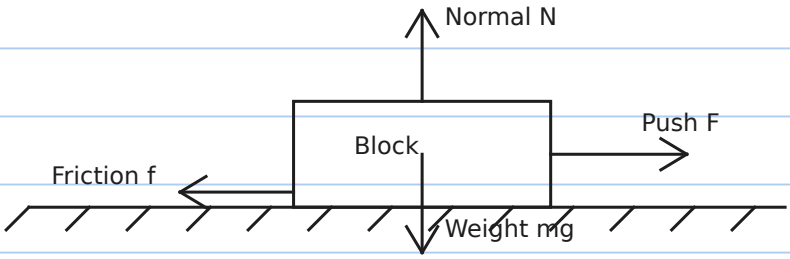
(a) applying the force near the axle, radially outward from the axle

(b) applying the force near the rim, radially outward

(c) applying the force near the axle, parallel to a tangent to the wheel

✓ (d) applying the force at the rim, tangent to the rim

Answer: (d) Working: torque = r F sin theta, so it is largest when both the moment arm r and sin theta are largest. Applying the force at the rim gives the greatest r, and applying it tangentially makes theta = 90 deg so sin theta = 1.



For an object rotating about a fixed axis, I is its rotational inertia and alpha is its angular acceleration. The relation tau = I alpha

(a) is the definition of torque

(b) is the definition of rotational inertia

(c) is the definition of angular acceleration

✓ (d) follows directly from Newton's second law

Answer: (d) Working: applying F = m a to each particle of the rotating body and summing the moments gives sum of tau = I alpha. It is the rotational form of Newton's second law, not a definition of any single

quantity.

The angular momentum vector of Earth about its rotation axis, due to its daily rotation, is directed

(a) tangent to the equator towards east

(b) tangent to the equator towards the west

✓ (c) north

(d) towards the Sun

Answer: (c) Working: by the right-hand rule, curling the fingers in the direction of rotation (west to east) makes the thumb point along the axis towards the north pole. Angular momentum L = I omega is a vector along the rotation axis, so it points north.

A stone of 2 kg is tied to a 0.50 m long string and swung around a circle at angular velocity of 12 rad/s. The net torque on the stone about the center of the circle is

✓ (a) 0 N.m

(b) 6 N.m

(c) 12 N.m

(d) 72 N.m

Answer: (a) Working: the stone moves with constant angular velocity, so its angular acceleration alpha is zero and tau = I alpha = 0. Also, the tension acts along the string towards the center, so its moment arm about the center is zero and it can produce no torque about that point.

A man, with his arms at his sides, is spinning on a light frictionless turntable. When he extends his arms

(a) his angular velocity increases

(b) his angular velocity remains same

(c) his rotational inertia decreases

✓ (d) his angular momentum remains the same

Answer: (d) Working: there is no external torque about the vertical axis, so L = I omega is conserved. Extending the arms increases I, so his angular velocity actually decreases, but the product I omega stays constant.

A space station revolves around the Earth as a satellite, 100 km above the Earth's surface. What is the net force on an astronaut at rest inside the space station?

(a) equal to her weight on earth

✓ (b) a little less than her weight on earth

(c) less than half her weight on earth

Answer: (b) Working: gravity still acts on the astronaut and provides the centripetal force that keeps her in orbit, and at only 100 km height g is only slightly smaller than at the surface. Her apparent weight is zero because she and the station fall together, but the real net (gravitational) force on her is not zero.

If the external torque acting on a body is zero, then its

(a) angular momentum is zero

✓ (b) angular momentum is conserved

(c) angular acceleration is maximum

(d) rotational motion is maximum

Answer: (b) Working: torque is the rate of change of angular momentum, tau = dL/dt, so zero net torque means dL/dt = 0 and L stays constant. The angular momentum itself need not be zero; the body may keep spinning at a steady rate.

Exercise - Structured Questions (CRQs)

Question: For an isolated rotating body, what is the relation between

angular velocity and radius?

Answer: For an isolated body no external torque acts, so angular momentum L = I omega stays constant. Since I grows with the square of the size (for a particle I = m r^2), a larger radius means a larger I and therefore a smaller omega; in simple form omega is inversely related to r^2 for a given mass. So when the body spreads out it spins slower, and when it contracts it spins faster.

Question: When the moment of inertia of a rotating body is halved, then what will be the effect on angular velocity?

Answer: With no external torque, angular momentum L = I omega is conserved, so I1 omega1 = I2 omega2. If I2 = I1/2, then omega2 = 2 omega1, so the angular velocity doubles. This is why a spinning skater rotates much faster when the arms are pulled in, reducing the moment of inertia.

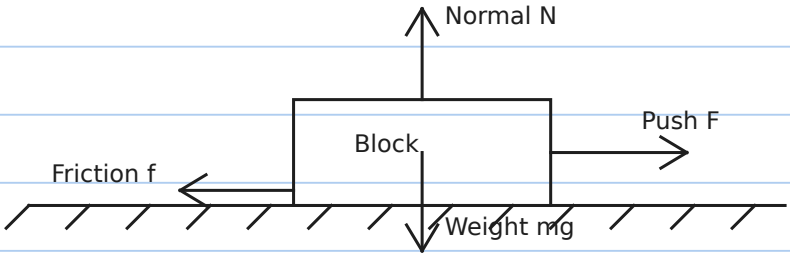
Question: Compare the kinematics equations of linear motion and circular motion.

Answer: The equations have exactly the same form, with angular quantities replacing linear ones: theta replaces s, omega replaces v, and alpha replaces a. Thus omega = omega_o + alpha t matches v = u + a t, theta = omega_o t + (1/2) alpha t^2 matches s = u t + (1/2) a t^2, and omega^2 = omega_o^2 + 2 alpha theta matches v^2 = u^2 + 2 a s. The similarity exists because rotation about a fixed axis is the direct analogue of straight-line motion.

Question: Can a small force ever exert a greater torque than a larger force? Give reason.

Answer: Yes, it can, because torque depends on the moment arm as well as the force: tau = r F sin theta. A small force applied far from the axis and at right angles to the arm can give a larger r F product than a big force

applied close to the axis or nearly along the arm. This is why a long spanner loosens a tight nut more easily than a short one.



Question: Give two real world applications of angular momentum.

Answer: First, ice skaters and divers use conservation of angular momentum: pulling the arms and legs in reduces I and increases the spin rate, while stretching out slows the spin for a safe landing. Second, gyroscopes and spinning satellites use the fixed direction of angular momentum to stay stable and hold their orientation in space. Bicycle and motorcycle wheels likewise resist tipping because of the angular momentum of the spinning wheels.

Question: Derive the relationship between torque and angular acceleration.

Answer: Consider a particle of mass m moving in a circle of radius r under a tangential force F. Newton's second law gives F = m a_tan = m r alpha, since a_tan = r alpha. Multiplying both sides by r gives r F = m r^2 alpha, and r F is the torque tau while m r^2 is the moment of inertia I of the particle, so tau = I alpha. Summing over all particles of a rigid body gives the general result: sum of tau = I alpha.

Question: List the factors on which moment of inertia depends.

Answer: Moment of inertia depends on the total mass of the body, on the distribution of that mass about the axis (how far the mass lies from the axis), and on the position and direction of the axis of rotation. It does not depend on angular velocity, torque or angular acceleration. The same

body therefore has different moments of inertia about different axes.

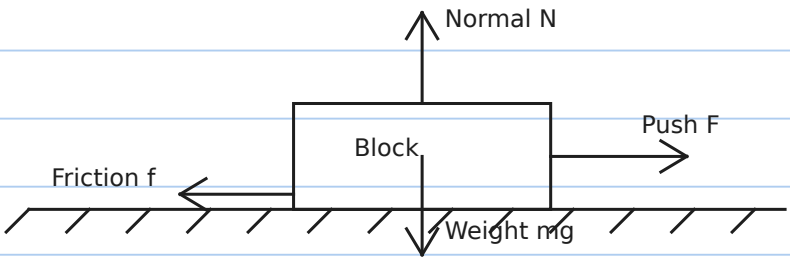
Exercise - Structured Questions (ERQs)

Question: State and explain the law of conservation of angular momentum. Give two examples to illustrate it.

Answer: The law states that when no resultant external torque acts on a body or system, its total angular momentum remains constant, that is I omega = constant. It follows from tau = dL/dt: if tau = 0 then L cannot change. For example, a diver who leaves the board rotating tucks into a ball, reducing I, so omega increases and more somersaults are completed before entering the water. Similarly, the planets keep orbiting the Sun with constant angular momentum because the gravitational force acts along the line joining them to the Sun and exerts no torque about the Sun.

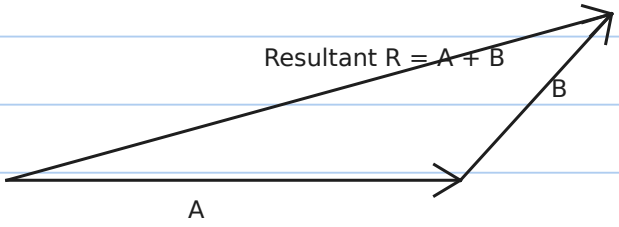
Question: Discuss the forces acting on a banked curve and derive the relation between curve angle and velocity.

Answer: On a road banked at angle theta, the weight m g acts vertically down and the normal reaction N acts perpendicular to the road surface. Resolving N: the vertical component N cos theta balances the weight, N cos theta = m g, and the horizontal component N sin theta points towards the center of the curve and supplies the centripetal force, N sin theta = m v^2 / r. Dividing the two equations gives tan theta = v^2 / (r g), so the ideal banking angle satisfies v = sqrt(r g tan theta). At this speed the vehicle can take the curve without relying on friction.



Question: Derive the formula for centripetal acceleration using fundamental principles and equations, illustrating each step of the derivation.

Answer: Consider a body moving with constant speed v in a circle of radius r; in a small time delta t it moves through a small angle delta theta, and its velocity vector turns by the same angle delta theta. The change in velocity delta v forms a small triangle similar to the position triangle, giving delta v / v = delta s / r = v delta t / r, so delta v / delta t = v^2 / r. Taking the limit as delta t goes to zero gives the instantaneous acceleration a_c = v^2 / r, directed towards the center of the circle. Using v = r omega, it can also be written as a_c = r omega^2.



Question: Derive the formula for the moment of inertia of a uniform rod or a solid sphere. Clearly illustrate each step of the derivation.

Answer: For a uniform rod of mass M and length L rotating about an axis through its center and perpendicular to it, divide the rod into small elements of mass dm = (M/L) dx at distance x from the center. The moment of inertia is I = integral of x^2 dm from x = -L/2 to +L/2, which

gives I = (M/L) x (L^3/12) evaluated over the length, that is I = (1/12) M L^2. For a solid sphere, the sphere is sliced into thin discs perpendicular to the axis and integrating their moments of inertia over the sphere gives I = (2/5) M R^2. In both cases the mass distribution about the axis, through the sum or integral of m r^2, fixes the result.

Question: Define torque and explain how it differs from force in linear motion.

Answer: Torque is the turning effect of a force about an axis, defined as tau = r x F with magnitude tau = r F sin theta, where r is the moment arm from the axis to the point of application. In linear motion a force alone changes the linear momentum and produces linear acceleration through F = m a, regardless of where it is applied. Torque, by contrast, depends on both the force and where and at what angle it is applied, and it produces angular acceleration through tau = I alpha. A force through the axis causes translation effects only, while an off-axis force also causes rotation.

Exercise - Numerical Problems

Question: A car mechanic applies a force of 800 N to a wrench for the purpose of loosening a bolt. He applies the force which is perpendicular to the arm of the wrench. The distance from the bolt to the mechanic's hand is 0.40 m. Find out the magnitude of the torque applied.

Answer: Given: F = 800 N, r = 0.40 m, force perpendicular to the arm so theta = 90 deg and sin theta = 1

Step 1: Write the torque formula, tau = r F sin theta

Step 2: Substitute the values, tau = 0.40 x 800 x 1

Step 3: Multiply, tau = 320 N.m

Final answer: 320 N.m

Question: A car accelerates uniformly from rest and reaches a speed of 22 m/s in 9 s. If the diameter of a tire is 58 cm, find (a) the number of revolutions the tire makes during this motion, assuming no slipping, and (b) the final rotational speed of the tire in revolutions per second.

Answer: Given: u = 0, v = 22 m/s, t = 9 s, diameter = 0.58 m so radius r = 0.29 m

Step 1: Find the distance travelled using average speed, s = ((u + v)/2) x t = (22/2) x 9 = 99 m

Step 2: Find the circumference of the tire, C = 2 pi r = 2 x 3.14 x 0.29 = 1.82 m

Step 3: For (a), number of revolutions = s / C = 99 / 1.82 = 54.3, about 55 rev

Step 4: For (b), final rotational speed = v / C = 22 / 1.82 = 12.1 rev/s, about 12 rev/s

Final answer: (a) about 55 rev, (b) about 12 rev/s

Question: An ordinary workshop grindstone has a radius of 7.5 cm and rotates 6500 rev/min. (a) Calculate the magnitude of centripetal acceleration at its edge in m/s^2 and convert it into multiples of g. (b) What is the linear speed of a point on its edge?

Answer: Given: r = 7.5 cm = 0.075 m, rotation rate = 6500 rev/min = 6500/60 = 108.3 rev/s

Step 1: Find the angular velocity, omega = 2 pi x 108.3 = 681 rad/s

Step 2: For (a), a_c = r omega^2 = 0.075 x (681)^2 = 0.075 x 4.64x10^5

= 3.47x10^4 m/s^2

Step 3: Express in multiples of g, 3.47x10^4 / 9.8 = 3.55x10^3, so a_c = 3.55x10^3 g

Step 4: For (b), v = r omega = 0.075 x 681 = 51 m/s

Final answer: (a) 3.47x10^4 m/s^2, equal to 3.55x10^3 g, (b) 51 m/s

Question: A satellite is orbiting the Earth with an orbital velocity of 3200 m/s. What is the orbital radius?

Answer: Given: v = 3200 m/s, G = 6.67x10^-11 N.m^2/kg^2, M_earth = 5.98x10^24 kg

Step 1: For a circular orbit, v = sqrt(G M / r), so r = G M / v^2

Step 2: Calculate G M = 6.67x10^-11 x 5.98x10^24 = 3.99x10^14 m^3/s^2

Step 3: Calculate v^2 = (3200)^2 = 1.024x10^7 m^2/s^2

Step 4: Divide, r = 3.99x10^14 / 1.024x10^7 = 3.90x10^7 m

Final answer: 3.897x10^7 m (about 3.9x10^7 m)

Question: A satellite wishes to orbit the earth at a height of 100 km (approximately 60 miles) above the surface of the earth. Determine the speed, acceleration and orbital period of the satellite. (Given: M_earth = 5.98 x 10^24 kg, R_earth = 6.37 x 10^6 m)

Answer: Given: h = 100 km = 1.0x10^5 m, R = 6.37x10^6 m, M = 5.98x10^24 kg, G = 6.67x10^-11 N.m^2/kg^2

Step 1: Find the orbital radius, r = R + h = 6.37x10^6 + 0.10x10^6 = 6.47x10^6 m

Step 2: Find the speed, v = sqrt(G M / r) = sqrt(3.99x10^14 / 6.47x10^6)

= sqrt(6.17x10^7) = 7.85x10^3 m/s

Step 3: Find the acceleration, a = v^2 / r = (7.85x10^3)^2 / 6.47x10^6 = 6.17x10^7 / 6.47x10^6 = 9.53 m/s^2

Step 4: Find the period, T = 2 pi r / v = 2 x 3.14 x 6.47x10^6 / 7.85x10^3 = 5177 s

Step 5: Convert the period to hours, T = 5177 / 3600 = 1.44 hours

Final answer: speed 7.85x10^3 m/s, acceleration 9.53 m/s^2, period 1.44 hours

Question: A thin disk with a 0.3 m diameter and a total moment of inertia of 0.45 kg.m^2 is rotating about its center of mass. There are three rocks with masses of 0.2 kg on the outer part of the disk. Find the total moment of inertia of the system.

Answer: Given: I_disk = 0.45 kg.m^2, diameter = 0.3 m so r = 0.15 m, three rocks each m = 0.2 kg at the rim

Step 1: Moment of inertia of one rock about the center, I_rock = m r^2 = 0.2 x (0.15)^2 = 0.2 x 0.0225 = 0.0045 kg.m^2

Step 2: For three rocks, I_rocks = 3 x 0.0045 = 0.0135 kg.m^2

Step 3: Add the disk and the rocks, I_total = 0.45 + 0.0135 = 0.4635 kg.m^2

Final answer: 0.464 kg.m^2

Question: What is the ideal banking angle for a gentle turn of 1.20 km radius on a highway with a 105 km/h speed limit (about 65 mi/h), assuming everyone travels at the limit?

Answer: Given: r = 1.20 km = 1200 m, v = 105 km/h = 105 x 1000 / 3600 = 29.2 m/s, g = 9.8 m/s^2

Step 1: Write the banking relation, tan theta = v^2 / (r g)

Step 2: Calculate v^2 = (29.2)^2 = 851

Step 3: Calculate r g = 1200 x 9.8 = 11760

Step 4: tan theta = 851 / 11760 = 0.0724

Step 5: theta = arctan(0.0724) = 4.14 deg

Final answer: 4.14 deg

Question: A 1500-kg car moving on a flat, horizontal road negotiates a curve. If the radius of the curve is 35 m and the coefficient of static friction between the tires and dry pavement is 0.523, find the maximum speed the car can have and still make the turn successfully.

Answer: Given: m = 1500 kg, r = 35 m, mu_s = 0.523, g = 9.8 m/s^2

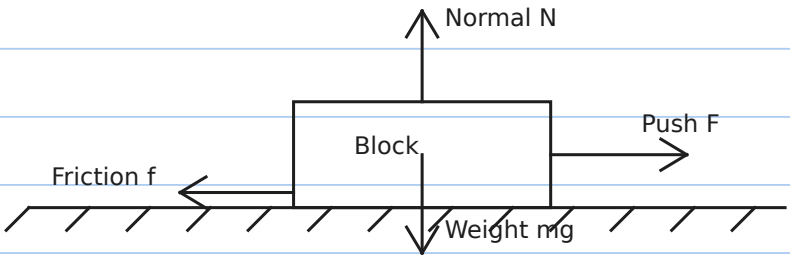
Step 1: On a flat curve friction alone provides the centripetal force, m v^2 / r = mu_s m g

Step 2: Cancel m and rearrange, v_max = sqrt(mu_s r g)

Step 3: Substitute, v_max = sqrt(0.523 x 35 x 9.8) = sqrt(179.4)

Step 4: Take the square root, v_max = 13.4 m/s, about 13.5 m/s

Final answer: 13.5 m/s



Question: A system of points is shown in figure 4.22, with three particles each of mass 0.3 kg lying in the same plane at distances of 40 cm, 20 cm

and 50 cm from a central particle on the given axis. What is the moment of inertia of the system about the given axis?

Answer: Given: each particle m = 0.3 kg, distances r1 = 0.40 m, r2 = 0.20 m, r3 = 0.50 m, the central particle lies on the axis so its distance is zero

Step 1: Write I = sum of m r^2 for all particles

Step 2: Calculate the squares, (0.40)^2 = 0.16, (0.20)^2 = 0.04, (0.50)^2 = 0.25, sum = 0.45 m^2

Step 3: Multiply by the mass, I = 0.3 x 0.45 = 0.135 kg.m^2

Final answer: 0.135 kg.m^2

Question: (a) What is the angular momentum of a 2.9 kg uniform cylindrical grinding wheel of radius 20 cm when rotating 1550 rpm? (b) How much torque is required to stop it in 6 s?

Answer: Given: M = 2.9 kg, R = 20 cm = 0.20 m, rotation rate = 1550 rpm, stopping time t = 6 s

Step 1: Moment of inertia of a solid cylinder, I = (1/2) M R^2 = 0.5 x 2.9 x (0.20)^2 = 0.5 x 2.9 x 0.04 = 0.058 kg.m^2

Step 2: Angular velocity, omega = 1550 x 2 pi / 60 = 162 rad/s

Step 3: For (a), L = I omega = 0.058 x 162 = 9.4 kg.m^2/s, about 9.5 kg.m^2/s

Step 4: For (b), torque = rate of change of angular momentum, tau = delta L / t = 9.4 / 6 = 1.6 N.m, applied opposite to the rotation

Final answer: (a) 9.5 kg.m^2/s, (b) 1.6 N.m opposing the rotation

Question: Determine the angular momentum of the Earth (a) about its rotation axis (assume the Earth as a uniform sphere), and (b) in its orbit

around the Sun (take Earth as a particle orbiting the Sun). The Earth has mass 6 x 10^24 kg and radius 6.4 x 10^6 m, and is 1.5 x 10^8 km from the Sun.

Answer: Given: M = 6x10^24 kg, R = 6.4x10^6 m, orbital radius r = 1.5x10^8 km = 1.5x10^11 m, one day = 86400 s, one year = 3.16x10^7 s

Step 1: For (a), I of a uniform sphere = (2/5) M R^2 = 0.4 x 6x10^24 x (6.4x10^6)^2 = 0.4 x 6x10^24 x 4.10x10^13 = 9.8x10^37 kg.m^2

Step 2: Spin angular velocity, omega = 2 pi / 86400 = 7.27x10^-5 rad/s

Step 3: Spin angular momentum, L = I omega = 9.8x10^37 x 7.27x10^-5 = 7.1x10^33 kg.m^2/s, about 7.3x10^33 kg.m^2/s

Step 4: For (b), orbital speed, v = 2 pi r / T = 2 x 3.14 x 1.5x10^11 / 3.16x10^7 = 2.98x10^4 m/s

Step 5: Orbital angular momentum, L = M v r = 6x10^24 x 2.98x10^4 x 1.5x10^11 = 2.7x10^40 kg.m^2/s

Final answer: (a) 7.3x10^33 kg.m^2/s, (b) 2.7x10^40 kg.m^2/s

Unit 5: Work, Energy and Power

Exercise - Multiple Choice Questions (MCQs)

Work done by centripetal force is always:

(a) Maximum

(b) Minimum

✓ (c) Zero

(d) None of these

Answer: (c) Working: the centripetal force always acts at right angles to the instantaneous displacement of the body, so theta = 90 deg and W = F s cos 90 deg = 0. It changes only the direction of motion, never the speed or kinetic energy.

A body of mass 5 kg is moving with a momentum of 10 kg m/s. A force of 0.2 N acts on it in the direction of motion of the body for 10 seconds. The increase in kinetic energy is:

(a) 2.8 J

(b) 3.2 J

(c) 3.8 J

✓ (d) 4.4 J

Answer: (d) Working: initial velocity v1 = p/m = 10/5 = 2 m/s, so initial KE = (1/2) x 5 x 4 = 10 J. The impulse F t = 0.2 x 10 = 2 kg m/s raises the momentum to 12 kg m/s, giving final KE = p^2/(2m) = 144/10 = 14.4 J, an increase of 4.4 J.

The kinetic energy of a light and a heavy object is same. Which object has

maximum momentum?

(a) Light object

✓ (b) Heavy object

(c) Both have same momentum

(d) N.O.T

Answer: (b) Working: momentum and kinetic energy are related by p = sqrt(2 m KE). For the same KE, the momentum is larger for the larger mass, so the heavy object has the greater momentum even though it moves more slowly.

Two bodies of mass 1 kg and 2 kg have equal momentum. Then the ratio of their kinetic energies is:

✓ (a) 2:1

(b) 3:1

(c) 1:3

(d) 1:1

Answer: (a) Working: for equal momentum, KE = p^2/(2m) is inversely proportional to mass. So KE of the 1 kg body : KE of the 2 kg body = 2 : 1, the lighter body having twice the kinetic energy.

A body fallen from height h. After it has fallen a height h/2, it will possess:

(a) only potential energy

(b) only kinetic energy

✓ (c) half potential half kinetic energy

(d) more kinetic and less potential

Answer: (c) Working: at height h/2 the body still holds PE = m g (h/2), exactly half of its initial m g h. By conservation of energy the other half has been converted into kinetic energy, so the two forms are equal at the halfway point.

Which of the following quantity can be calculated by multiplying force and

velocity?

(a) acceleration

✓ (b) power

(c) torque

(d) work

Answer: (b) Working: instantaneous power is P = F . v, the product of the force and the velocity of its point of application along the force direction. Work is force times displacement, and torque is force times moment arm, so neither of those fits.

The minimum velocity given to an object so that it emerges out from the gravitational field of earth is about _____:

✓ (a) 11.2 km/s

(b) 15.3 km/s

(c) 5 km/s

(d) 9.8 km/s

Answer: (a) Working: the escape velocity v = sqrt(2 g R) for the Earth works out to about 11.2 km/s. Given this speed at the surface, the object just reaches infinity with zero speed and never falls back.

When one joule of work is done on a body in one second, power of body is said to be:

✓ (a) One watt

(b) 0.5 watt

(c) zero

(d) 100 watt

Answer: (a) Working: power is the rate of doing work, P = W/t = 1 J / 1 s = 1 watt. The watt is defined exactly as one joule per second.

The absolute potential energy of an object depends on:

✓ (a) The object's mass and height

(b) The object's mass and speed

(c) The object's shape and size

(d) The object's color and temperature

Answer: (a) Working: gravitational potential energy is U = m g h measured from a chosen reference level such as the Earth's surface (more exactly it depends on mass and distance from the Earth's center). Speed belongs to kinetic energy, while shape, color and temperature do not enter the formula.

The escape velocity of a planet depends on which of the following factors?

(a) The mass of the planet only

(b) The radius of the planet only



(c) Both the mass and the radius of the planet

(d) The density of the planet

Answer: (c) Working: escape velocity is v = sqrt(2 G M / R), so it increases with the planet's mass and decreases with its radius. Neither mass alone nor radius alone is enough to fix its value.

Exercise - Structured Questions (CRQs)

Question: How does work relate to the transfer of energy?

Answer: Work is the process by which energy is transferred from one body or system to another: when a force does work on a body, energy equal to that work is transferred to the body. The work-energy theorem states that the net work done equals the change in kinetic energy of the body. If the work is positive the body gains energy, and if negative (for example by friction) it loses energy.

Question: How is power related to work and time?

Answer: Power is the rate at which work is done, P = W / t, so it tells how quickly energy is transferred. Two machines may do the same total work, but the one that does it in less time has the greater power. Its SI unit is the watt, where 1 watt = 1 joule per second.

Question: What is the difference between average power and instantaneous power?

Answer: Average power is the total work divided by the total time taken, P_avg = W / t, and describes performance over a whole interval. Instantaneous power is the rate of doing work at one particular instant, P = F . v, and it can change from moment to moment. For a car accelerating, for example, the instantaneous power keeps changing while the average is a single value for the whole journey.

Question: How does gravitational potential energy change with height and mass?

Answer: Gravitational potential energy is given by U = m g h, so it is directly proportional to both the mass and the height above the reference level. Doubling the mass doubles the stored energy, and lifting a body twice as high doubles it as well. The energy is stored because work was done against gravity in raising the body.

Question: What is the law of conservation of energy?

Answer: The law states that energy can neither be created nor destroyed; it can only be changed from one form into another, so the total energy of an isolated system remains constant. In mechanical problems without friction, the sum of kinetic and potential energy stays the same at every point. When friction acts, the missing mechanical energy appears as heat, and the total including heat is still conserved.

Question: How does energy efficiency play a role in various energy transformations?

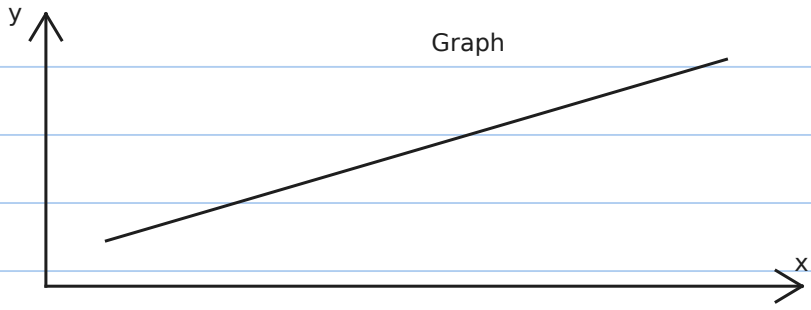
Answer: Efficiency is the ratio of useful energy output to total energy input, usually expressed as a percentage, and it is always less than 100 percent in real devices. In every transformation some energy is wasted, mostly as heat due to friction, electrical resistance or sound. Knowing the efficiency helps us compare machines and fuels and guides designs that waste less energy, for example in engines, bulbs and power stations.

Question: What is the work-energy theorem and how is it expressed mathematically?

Answer: The work-energy theorem states that the net work done on a body by all the forces acting on it equals the change in its kinetic energy. Mathematically, W = delta KE = (1/2) m v^2 - (1/2) m u^2, where u is the initial and v the final velocity. It follows directly from Newton's second law combined with the equations of motion, and it holds even when the force is not constant.

Question: How is the work done by a variable force calculated?

Answer: For a variable force the work is found by dividing the displacement into very small parts, over each of which the force is nearly constant, and adding up all the small amounts F delta x; in calculus form W = integral of F dx. Graphically it is the area under the force-displacement curve between the starting and finishing positions. This method works for forces such as a spring force that change with position.



Exercise - Structured Questions (ERQs)

Question: How the work done in gravitational field is independent of path

Answer: Gravity is a conservative force, so the work it does depends only on the vertical height difference between the start and finish, W = m g h. Along any sloping or curved path, only the vertical component of each small displacement contributes to the work, and these vertical parts always add up to the same total height change. Horizontal parts contribute nothing because gravity is vertical. Hence a body gains or loses the same potential energy whether it is lifted straight up or moved along any other route between the same two levels.

Question: Calculate gravitational potential energy at a certain height due to work against gravity.

Answer: To raise a mass m slowly through a height h, an upward force equal to its weight m g must be applied over the distance h. The work done against gravity is therefore W = F x s = m g h. This work is not lost; it is stored in the body as gravitational potential energy, so U = m g h. For example, a 2 kg body raised 5 m gains U = 2 x 9.8 x 5 = 98 J of potential energy.

Question: Describe that the gravitational PE is measured from a reference level and can be positive or negative, to denote the orientation from the reference level.

Answer: Potential energy has no absolute zero of its own, so it is always measured from an agreed reference level, commonly the ground or a tabletop, where U is taken as zero. A body above the reference level has positive PE, U = +m g h, because work must be done to lift it there. A body below the reference level (for example in a basement or a well) has

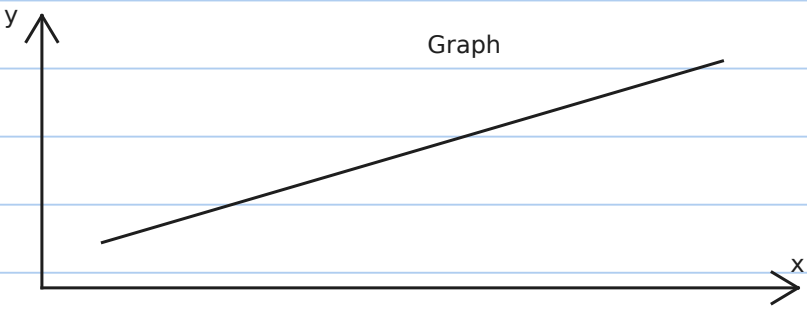
negative PE, U = -m g h, relative to that level. Only differences in potential energy between two positions have physical meaning, and they are the same whatever reference level is chosen.

Question: Show that the work done in gravitational field is independent of path

Answer: Consider a body moved from point A to a point B that is h lower, first by sliding down a smooth incline and then by a stepped or curved path. On the incline of angle theta and length L, gravity's component along the path is m g sin theta and the work is m g sin theta x L = m g h, since L sin theta = h. On any other path, breaking the path into small vertical and horizontal pieces, the horizontal pieces contribute no work and the vertical pieces add to h, again giving W = m g h. Since the result is the same for every path, work in a gravitational field depends only on the endpoints, not on the path.

Question: Define work by variable force. Calculate the work done from the force-displacement graph.

Answer: Work by a variable force is the work done when the magnitude or direction of the force changes during the displacement, calculated as W = integral of F dx over the path. On a force-displacement graph this work equals the area enclosed under the curve between the initial and final positions. For example, if the force rises steadily from 0 to 10 N over 4 m the graph is a triangle, and the work is (1/2) x 4 x 10 = 20 J. The same area method is used for a stretching spring, whose force grows with extension.



Exercise - Numerical Problems

Question: A man pulls a trolley through a distance of 10 m by applying a force of 50 N which makes an angle of 60 deg with the horizontal. Calculate the work done by the man.

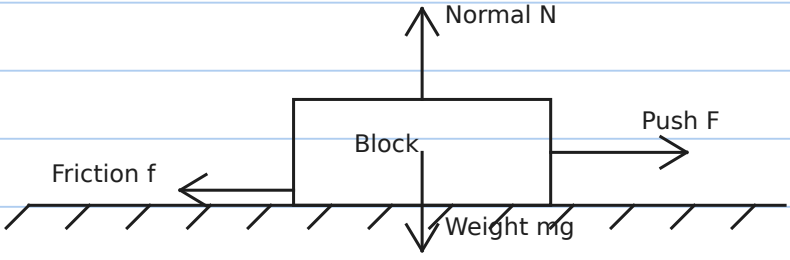
Answer: Given: F = 50 N, s = 10 m, theta = 60 deg, cos 60 deg = 0.5

Step 1: Write the work formula, W = F s cos theta

Step 2: Substitute the values, W = 50 x 10 x 0.5

Step 3: Multiply, W = 250 J

Final answer: 250 J



Question: A 100 kg man runs up a long flight of stairs in 9.8 second. The vertical height of the stair is 10 m. Calculate its power.

Answer: Given: m = 100 kg, h = 10 m, t = 9.8 s, g = 9.8 m/s^2

Step 1: Work done against gravity, W = m g h = 100 x 9.8 x 10 = 9800 J

Step 2: Write the power formula, P = W / t

Step 3: Substitute, P = 9800 / 9.8 = 1000 W

Final answer: 1000 Watts

Question: When an object is thrown upwards, it rises to a height 'h'. How high is the object in terms of h, when it has lost one third of its original kinetic energy?

Answer: Given: maximum height = h, fraction of original KE lost = 1/3

Step 1: At the maximum height all the original KE has become PE, so original KE = m g h

Step 2: Losing one third of the KE means the PE gained is (1/3) m g h

Step 3: If the body is now at height h1, then m g h1 = (1/3) m g h

Step 4: Cancel m g, giving h1 = h/3

Final answer: h/3

Question: A 70 kg man runs up a hill through a height of 3 m in 2 seconds. (a) How much work does he do against gravitational field? (b) What is the average power output?

Answer: Given: m = 70 kg, h = 3 m, t = 2 s, g = 9.8 m/s^2

Step 1: For (a), work against gravity, W = m g h = 70 x 9.8 x 3 = 2058 J, about 2060 J

Step 2: For (b), average power, P = W / t = 2058 / 2 = 1029 W, about 1030 W

Final answer: (a) 2060 J, (b) 1030 Watt

Question: A neutron travels a distance of 12 m in a time interval of 3.6 x 10^-4 sec. Assuming its speed was constant, find its kinetic energy. Take

the mass of neutron 1.7 x 10^-27 kg.

Answer: Given: s = 12 m, t = 3.6x10^-4 s, m = 1.7x10^-27 kg, 1 eV = 1.6x10^-19 J

Step 1: Find the speed, v = s / t = 12 / 3.6x10^-4 = 3.33x10^4 m/s

Step 2: Square the speed, v^2 = (3.33x10^4)^2 = 1.11x10^9 m^2/s^2

Step 3: Calculate KE = (1/2) m v^2 = 0.5 x 1.7x10^-27 x 1.11x10^9 = 9.4x10^-19 J

Step 4: Convert to electron volts, KE = 9.4x10^-19 / 1.6x10^-19 = 5.9 eV, about 5.78 eV

Final answer: 5.78 eV (about 5.9 eV)

Question: A stone is thrown vertically upwards and can reach to a height of 10 m, find the speed of stone, when it is just 2 m above the ground.

Answer: Given: maximum height H = 10 m, height of the point = 2 m, g = 9.8 m/s^2

Step 1: By energy conservation, the KE at 2 m equals the PE difference between 10 m and 2 m, (1/2) m v^2 = m g (10 - 2)

Step 2: Cancel m and rearrange, v^2 = 2 g x 8 = 2 x 9.8 x 8 = 156.8

Step 3: Take the square root, v = sqrt(156.8) = 12.5 m/s

Final answer: 12.5 m/s

Question: The potential energy of a body at the top of a building is 200 Joules when it is dropped its kinetic energy just before striking the ground is 160 Joules, find the work done against the air resistance.

Answer: Given: PE at top = 200 J, KE just before ground = 160 J, PE at ground = 0

Step 1: Without air resistance the KE at the ground would equal the full 200 J of PE lost

Step 2: The shortfall is the energy used against air resistance, W = 200 - 160

Step 3: Subtract, W = 40 J

Final answer: 40 J

Question: Find the energy equivalent of 1 gram.

Answer: Given: m = 1 gram = 1x10^-3 kg, c = 3x10^8 m/s

Step 1: Write Einstein's relation, E = m c^2

Step 2: Calculate c^2 = (3x10^8)^2 = 9x10^16 m^2/s^2

Step 3: Substitute, E = 1x10^-3 x 9x10^16 = 9x10^13 J

Final answer: 9 x 10^13 J

Question: A 1 Kilowatt motor pump, pumps the water from the ground to a height of 10 m. Find, how much litres of water it can pump in one hour.

Answer: Given: P = 1 kW = 1000 W, t = 1 hour = 3600 s, h = 10 m, g = 10 m/s^2, density of water = 1000 kg/m^3 so 1 litre has mass 1 kg

Step 1: Total energy supplied, E = P t = 1000 x 3600 = 3.6x10^6 J

Step 2: This energy becomes PE of the water, m g h = E, so m = E / (g h) = 3.6x10^6 / (10 x 10) = 3.6x10^4 kg

Step 3: Convert mass of water to litres, volume = 3.6x10^4 litres

Final answer: 3.6 x 10^4 litre

Question: A rocket of mass 2 kg is launched in air, when it attains height of 15 m the 400 Joules of its chemical fuel burns. Find the speed of rocket at

that height.

Answer: Given: m = 2 kg, fuel energy E = 400 J, h = 15 m, g = 10 m/s^2

Step 1: Potential energy at 15 m, PE = m g h = 2 x 10 x 15 = 300 J

Step 2: The rest of the fuel energy is kinetic, KE = E - PE = 400 - 300 = 100 J

Step 3: From KE = (1/2) m v^2, v^2 = 2 x 100 / 2 = 100

Step 4: Take the square root, v = 10 m/s

Final answer: 10 m/s

Question: A motor pumps the water at the rate 500 gram/minute to the height of 120 m. If the motor is 50% efficient then how much input electric power is needed?

Answer: Given: mass rate = 500 gram/minute = 0.5 kg per 60 s, h = 120 m, g = 10 m/s^2, efficiency = 50% = 0.5

Step 1: Useful work per minute, W = m g h = 0.5 x 10 x 120 = 600 J

Step 2: Useful output power, P_out = 600 / 60 = 10 W

Step 3: Input power = P_out / efficiency = 10 / 0.5 = 20 W

Final answer: 20 watt

Unit 6: Fluid Statics

Exercise - Multiple Choice Questions (MCQs)

A completely submerged object always displaces its own _____

(a) weight of fluid

✓ (b) volume of fluid

(c) density of a fluid

(d) Area of fluid

Answer: (b) Working: a completely submerged object takes the place of a volume of fluid exactly equal to its own volume, whatever its shape or weight. This displaced volume is what fixes the buoyant force through Archimedes' principle.

The pressure exerted on the ground by a man is greatest when:

(a) he stands with both feet flat on the ground

(b) he stands flat on one foot

✓ (c) he stands on the toes of one foot

(d) he lies down on the ground

Answer: (c) Working: pressure = force / area, and the force is the man's weight in every case. Standing on the toes of one foot gives the smallest contact area, so the pressure on the ground is greatest; lying down gives the largest area and the least pressure.

In a stationary homogeneous liquid:

(a) pressure is the same at all points

(b) pressure depends on the direction

(c) pressure is independent of any atmospheric pressure on the upper surface of the liquid

✓ (d) pressure is the same at all points at the same level

Answer: (d) Working: in a still liquid the pressure depends only on depth, P = P_o + rho g h, so all points at the same depth (same level) have the same pressure. Points at different depths have different pressures, so pressure is not the same everywhere.

One piston in a hydraulic lift has an area that is twice the area of the other. When the pressure at the smaller piston is increased by delta p, the pressure at the larger piston:

(a) increases by 2 delta p

(b) increases by delta p/2

✓ (c) increases by delta p

(d) increases by 4 delta p

Answer: (c) Working: by Pascal's principle a change of pressure applied to an enclosed liquid is transmitted undiminished to every part of the liquid. The larger piston therefore experiences the same pressure increase; it is the force that grows, because its area is larger.

In a vacuum, an object has

✓ (a) No buoyant force

(b) no mass

(c) no weight

(d) none of these

Answer: (a) Working: buoyant force is the upthrust from the surrounding fluid, so in a vacuum, with no fluid around it, an object experiences no buoyant force. The object still has its mass, and near the Earth it still has weight; only the upthrust is absent.

The pressure at the bottom of a pond does NOT depend on

(a) Water density

(b) the depth of the pond

✓ (c) the surface area of the pond

(d) None of these

Answer: (c) Working: pressure at depth is P = rho g h, set by the density of the water and the depth below the surface only. A wide pond and a narrow tank of the same depth have the same pressure at the bottom.

Question: A rock suspended by a weighing scale weighs 5 N out of water and 3 N when submerged in water. What is the buoyant force on the rock? (a) 3 N (b) 5 N (c) 8 N (d) 15 N

Answer: (none of the listed options) 2 N. Working: the buoyant force equals the apparent loss of weight, which is 5 N - 3 N = 2 N. The correct value 2 N is not among the printed options (a) to (d), so none of the listed options is correct.

"An object completely submerged in a fluid displaces its own volume of fluid." This is:

(a) Pascal's paradox

✓ (b) Archimedes' principle

(c) Pascal's principle

(d) true, but none of the above

Answer: (b) Working: Archimedes' principle concerns a body immersed in a fluid: it displaces its own volume of fluid and experiences an upthrust equal to the weight of that displaced fluid. Pascal's principle is about transmission of pressure, which is a different result.

Salt water has greater density than freshwater. A boat floats in both freshwater and saltwater. The buoyant force on the boat in salt water is _____ that in freshwater.

✓ (a) equal to

(b) smaller than

(c) larger than

(d) same as

Answer: (a) Working: a floating boat is in equilibrium, so the buoyant force equals its weight in either liquid, and the weight does not change. In denser salt water the boat simply displaces a smaller volume, floating slightly higher, while the force stays the same.

You fill a tall glass with ice and then add water to the level of the glass's rim, so some fraction of the ice floats above rim. When the ice melts, what's happens to the water level?

(a) the water overflows the rim

(b) the water level drops below the rim

✓ (c) the water level stays at the rim

(d) it depends on the difference in density between water and ice

Answer: (c) Working: floating ice displaces a volume of water whose weight equals the weight of the ice. When it melts it produces exactly that same volume of water, so it just refills the space it had displaced and the level at the rim is unchanged.

Exercise - Short Questions (SRQs)

Question: State Pascal's principle. Describe its two applications.

Answer: Pascal's principle states that pressure applied to an enclosed fluid is transmitted equally and undiminished to every part of the fluid and to the walls of the container. One application is the hydraulic press or car lift, where a small force on a small piston produces a large force on a large piston. A second application is the hydraulic brake system of vehicles,

where pressure applied at the master cylinder is transmitted through brake oil to operate all the wheel brakes together.

Question: (a) Why must a liquid and not a gas be used as the 'fluid' in a hydraulic machine? (b) On what other important property of a liquid do hydraulic machines depend?

Answer: (a) A liquid is practically incompressible, so the pressure applied to it is transmitted at once and the piston movement is passed on without being wasted in compressing the fluid; a gas would simply compress and absorb the push. (b) Hydraulic machines also depend on the liquid being able to flow and take the shape of the connecting pipes, carrying the pressure to every piston. In practice the liquid must also not corrode the machine parts, which is why special hydraulic oils are used.

Question: Why don't ships made of iron sink?

Answer: Iron itself is denser than water, but a ship is not a solid block of iron: its hull encloses a large volume of air, so the average density of the whole ship is less than the density of water. The hollow hull displaces a weight of water equal to the ship's total weight before it is fully submerged. By the law of flotation it therefore floats, with only part of the hull below the waterline.

Question: Why do you float higher in salt water than in fresh water?

Answer: Salt water is denser than fresh water, so for the same submerged volume it provides a greater buoyant force. Your body needs to displace a weight of water equal to your own weight in order to float. In salt water that weight is displaced with a smaller submerged volume, so you sit higher in the water; in fresh water you must sink deeper to displace the same weight.

Question: What is the difference between being immersed and being submerged in water?

Answer: Immersed means the body is placed in the water and may be only partly below the surface; part of it can remain above the water. Submerged means the body is completely under the water, entirely surrounded by it. A floating ship is immersed but not submerged, while a stone resting on the bottom is fully submerged.

Question: Define surface tension and give its any two applications.

Answer: Surface tension is the property of a liquid surface that makes it behave like a stretched elastic membrane; it is defined as the force per unit length acting along the surface, with unit N/m. Applications include the use of detergents, whose lowered surface tension helps water spread into fabrics for cleaning, and the working of paints, inks and sprays that must spread evenly over a surface. It also explains why small insects can walk on water and why falling drops are nearly spherical.

Question: A swimmer dives off a raft in a pool. Does the raft rise or sink in the water? What happens to the water level in the pool? Give reasons for your answer.

Answer: While the swimmer stands on the raft, the raft displaces water equal to the combined weight of the raft and the swimmer, so it sits low. When the swimmer dives off, the raft only has to support its own weight, displaces less water, and rises. The swimmer, now floating or swimming in the pool, displaces his own weight of water, so the total displaced volume is the same as before and the water level of the pool stays unchanged.

Question: Distinguish between flotation and upthrust.

Answer: Upthrust (buoyant force) is the upward force that a fluid exerts on any body partly or wholly immersed in it, equal to the weight of the fluid displaced. Flotation is the special condition in which a body rests in equilibrium on or in the fluid because the upthrust exactly balances its weight. So upthrust is a force, while flotation is the resulting state of a

body when that force equals its weight; a sinking body also experiences upthrust but does not float.

Exercise - Extended Response Questions (ERQs)

Question: Discuss buoyancy in liquids and gases.

Answer: Buoyancy is the upward force exerted by a fluid on a body immersed in it, caused by the pressure of the fluid increasing with depth, so the upward push on the lower surface exceeds the downward push on the upper surface. Its magnitude equals the weight of fluid displaced, as stated by Archimedes' principle. In liquids the effect is strong because liquids are dense: ships float, swimmers are supported, and submerged objects feel lighter. In gases the same effect exists but is much weaker because air is about 800 times less dense; still, it lifts hot-air balloons and helium balloons, and precision weighing must allow for the small upthrust of air on the object being weighed.

Question: Describe law of flotation in liquids and gases.

Answer: The law of flotation states that a floating body displaces its own weight of the fluid in which it floats; that is, the weight of displaced fluid equals the weight of the body. In liquids this explains why a ship floats at a depth where the displaced water weighs as much as the ship, and why the submerged fraction of a floating body equals the ratio of its density to the liquid's density. In gases the same law applies to balloons: a balloon rises until the weight of displaced air equals the total weight of the balloon, its gas and its load, and then it floats at that level. Hydrometers and lactometers are practical instruments based on this law.

Question: Discuss surface tension with at least three experiments.

Answer: Surface tension is the tangential force per unit length with which a liquid surface contracts, making the surface act like a stretched membrane. In the first experiment, a greased needle or a razor blade placed gently on water floats although the metal is denser than water, held up by the stretched surface film. In the second experiment, a loop of thread on a soap film stays circular once the film inside it is pricked, showing that the outer film pulls it equally in all directions. In the third experiment, water rises in a fine capillary tube and forms a concave surface, showing the surface pulling the liquid up the walls. A fourth everyday observation is that rain drops and falling water droplets are nearly spherical, the shape of minimum surface area.

Question: Explain Archimedes principle and find gold purity by using density.

Answer: Archimedes' principle states that when a body is immersed in a fluid it experiences an upthrust equal to the weight of the fluid displaced, so the apparent loss of weight of the body equals the weight of the displaced fluid. To test the purity of gold, the ornament is weighed in air (W1) and then weighed fully immersed in water (W2); the loss W1 - W2 is the weight of water displaced. The relative density is then W1 / (W1 - W2), and multiplying by 1000 kg/m^3 gives the density of the sample. Pure gold has a density of about 19300 kg/m^3; if the measured density matches this value the gold is pure, while a lower value shows it is mixed with a lighter metal such as copper or silver.

Exercise - Numerical Problems

Question: In a hydraulic press a force of 20 N is applied to a piston of area

0.20 m^2. The area of the other piston is 2.0 m^2. What is (a) the pressure transmitted through the fluid; (b) the force on the piston?

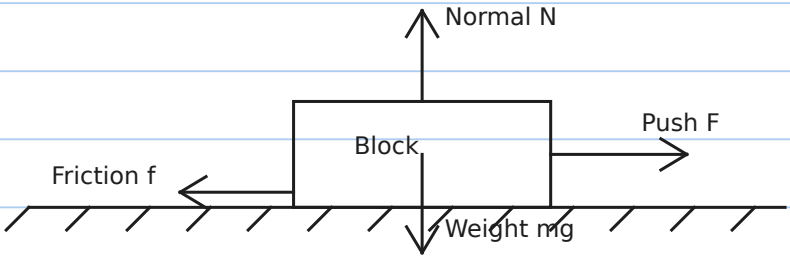
Answer: Given: F1 = 20 N, A1 = 0.20 m^2, A2 = 2.0 m^2

Step 1: For (a), pressure at the small piston, p = F1 / A1 = 20 / 0.20 = 100 Pa

Step 2: By Pascal's principle this same pressure is transmitted through the fluid

Step 3: For (b), force on the large piston, F2 = p x A2 = 100 x 2.0 = 200 N

Final answer: (a) 100 Pa, (b) 200 N



Question: The pressure in a water pipe in the ground floor of a building is 4x10^5 Pa but three floors up it is only 2x10^5 Pa. What is the height between the ground floor and the third floor? The water in the pipe may be assumed to be stationary; density of water = 1x10^3 Kg/m^3

Answer: Given: p_ground = 4x10^5 Pa, p_third = 2x10^5 Pa, rho = 1x10^3 kg/m^3, g = 10 m/s^2

Step 1: Pressure difference, delta p = 4x10^5 - 2x10^5 = 2x10^5 Pa

Step 2: This difference is due to the water column of height h, delta p = rho g h

Step 3: Rearrange, h = delta p / (rho g) = 2x10^5 / (1x10^3 x 10)

Step 4: Divide, h = 2x10^5 / 1x10^4 = 20 m

Final answer: 20 m

Question: The small piston of hydraulic press has an area of 10.0 cm^2. If the applied force is 50.0 N, what must the area of the large piston to exert a pressing force of 4800 N?

Answer: Given: A1 = 10.0 cm^2, F1 = 50.0 N, F2 = 4800 N

Step 1: Pressure at the small piston, p = F1 / A1 = 50.0 / 10.0 = 5.0 N/cm^2

Step 2: The same pressure acts on the large piston, so A2 = F2 / p

Step 3: Substitute, A2 = 4800 / 5.0 = 960 cm^2

Final answer: 960 cm^2

Question: Mechanical advantage of a hydraulic jack is 420. Find the weight of the heaviest automobile that can be lifted by an applied force of 55 N.

Answer: Given: mechanical advantage = 420, applied force = 55 N

Step 1: Mechanical advantage = load lifted / applied force

Step 2: Rearrange, load = mechanical advantage x applied force = 420 x 55

Step 3: Multiply, load = 23100 N, about 23000 N

Final answer: 23000 N (23100 N)

Question: A flat-bottom river barge is 30 ft wide, 85 ft long and 15 ft. deep, (a) how many m^3 of water will displace while the top stays 1 m above the water? (b) What load in tons will the barge contain under these conditions if the empty barge weighs 160 tons in dry dock?

Answer: Given: width = 30 ft = 9.14 m, length = 85 ft = 25.9 m, depth = 15 ft = 4.57 m, top stays 1 m above water, empty barge = 160 tons,

density of water = 1000 kg/m^3

Step 1: Submerged depth = 4.57 - 1 = 3.57 m

Step 2: For (a), displaced volume V = length x width x submerged depth = 25.9 x 9.14 x 3.57 = 845 m^3 (about 3.0x10^4 ft^3)

Step 3: Weight of displaced water = 845 m^3 x 1000 kg/m^3 = 8.45x10^5 kg = 845 tons, which by flotation equals barge plus load

Step 4: For (b), load = 845 - 160 = 685 tons, roughly 7x10^2 tons (the book's rough ft-based estimate is about 800 tons)

Final answer: (a) about 8.5x10^2 m^3, (b) about 685 tons

Question: A canal lock gate is 20 m wide and 10 m deep. Calculate the thrust acting on it assuming that the water in the canal is in level with the top of the gate. Density of water is 1000 kg/m^3

Answer: Given: width = 20 m, depth h = 10 m, rho = 1000 kg/m^3, g = 9.8 m/s^2

Step 1: Pressure grows with depth, so use the average pressure at the mean depth h/2 = 5 m

Step 2: Average pressure, p_avg = rho g (h/2) = 1000 x 9.8 x 5 = 49000 Pa

Step 3: Area of the gate, A = 20 x 10 = 200 m^2

Step 4: Thrust = p_avg x A = 49000 x 200 = 9.8x10^6 N

Final answer: 9.8x10^6 N

Question: A tank 4 m long, 3 m wide and 2 m deep is filled to the brim with paraffin (density 800 kg/m^3). Calculate the pressure on the base. What is the thrust on the base?

Answer: Given: length = 4 m, width = 3 m, depth h = 2 m, rho = 800 kg/m^3, g = 10 m/s^2

Step 1: Pressure on the base, p = rho g h = 800 x 10 x 2 = 16000 Pa

Step 2: Area of the base, A = 4 x 3 = 12 m^2

Step 3: Thrust = p x A = 16000 x 12 = 192000 N = 1.92x10^5 N

Final answer: 16000 Pa, 192000 N (the book's printed 19200 N misses a zero)

Question: A rectangular boat is 4.0 m wide, 8.0 m long, and 3.0 m deep. (a) How much water will it displace if the top stays 1.0 m above the water? (b) What load will the boat contain under these conditions if the empty boat weighs 8.60x10^4 N in dry dock? (Weight density of water = 9800 N/m^3)

Answer: Given: width = 4.0 m, length = 8.0 m, depth = 3.0 m, top stays 1.0 m above water, empty boat weight = 8.60x10^4 N, weight density of water = 9800 N/m^3

Step 1: Submerged depth = 3.0 - 1.0 = 2.0 m

Step 2: For (a), displaced volume V = 8.0 x 4.0 x 2.0 = 64.0 m^3

Step 3: Weight of displaced water = 64.0 x 9800 = 627200 N, which equals boat weight plus load

Step 4: For (b), load = 627200 - 8.60x10^4 = 541200 N = 5.41x10^5 N

Final answer: (a) 64.0 m^3, (b) 5.41x10^5 N

Question: A hot air balloon has a volume of 2200 m^3. The density of air at temperature of 20 deg C is 1.205 kg/m^3. The density of the hot air inside the balloon at a temperature of 100 deg C is 0.946 kg/m^3. How

much weight can the hot air can lift?

Answer: Given: V = 2200 m^3, rho_out = 1.205 kg/m^3, rho_in = 0.946 kg/m^3, g = 9.8 m/s^2

Step 1: Density difference, delta rho = 1.205 - 0.946 = 0.259 kg/m^3

Step 2: Net lifting mass = delta rho x V = 0.259 x 2200 = 569.8 kg

Step 3: Liftable weight = 569.8 x 9.8 = 5584 N, about 5590 N

Final answer: 5590 N

Question: A spherical balloon has a radius of 7.15 m and is filled with helium. How large a cargo can it lift, assuming that the skin and structure of the balloon have a mass of 930 kg? Neglect the buoyant force on the cargo volume itself.

Answer: Given: r = 7.15 m, skin and structure mass = 930 kg, density of air = 1.29 kg/m^3, density of helium = 0.179 kg/m^3, g = 9.8 m/s^2

Step 1: Volume of the sphere, V = (4/3) pi r^3 = (4/3) x 3.14 x (7.15)^3 = 4.19 x 365.5 = 1531 m^3

Step 2: Mass of displaced air = 1.29 x 1531 = 1975 kg

Step 3: Mass of helium = 0.179 x 1531 = 274 kg

Step 4: Net lift in mass = 1975 - 274 - 930 = 771 kg, about 770 kg

Step 5: As a weight this is 771 x 9.8 = 7.6x10^3 N

Final answer: about 770 kg, weight about 7.6x10^3 N (the book's printed 920 kg does not follow from these standard densities)

Unit 7: Fluid Dynamics

Exercise - Multiple Choice Questions (MCQs)

For an incompressible fluid, the flow rate is

(a) equal for all surfaces

✓ (b) constant throughout the pipe

(c) greater for the larger parts of the pipe

(d) none of the above

Answer: (b) Working: by the equation of continuity A v = constant, the volume flow rate is the same at every section of the pipe. Where the pipe is narrow the speed is high and where it is wide the speed is low, but their product stays constant.

Bernoulli's principle states that for horizontal flow of a fluid through a tube, the sum of the pressure and energy of motion per unit volume is

(a) increasing with time

(b) decreasing with time

✓ (c) constant

(d) varying with time

Answer: (c) Working: for horizontal flow Bernoulli's equation reduces to P + (1/2) rho v^2 = constant along a streamline. If the speed rises at some section, the pressure there falls by exactly the amount needed to keep the sum unchanged.

Which of the following is associated with the law of conservation of energy in fluids?

(a) Archimedes' principle

✓ (b) Bernoulli's principle

(c) Pascal's principle

(d) equation of continuity

Answer: (b) Working: Bernoulli's equation is derived by applying the work-energy theorem to a flowing fluid, so it is a statement of conservation of energy per unit volume. The equation of continuity instead expresses conservation of mass, and Pascal's and Archimedes' principles belong to fluid statics.

As the speed of a moving fluid increases, the pressure in the fluid

(a) increases

(b) remains constant

✓ (c) decreases

(d) may increase or decrease, depending on the viscosity

Answer: (c) Working: from Bernoulli's equation P + (1/2) rho v^2 = constant for horizontal flow, a greater speed means a greater kinetic energy per unit volume. The pressure must therefore fall so that the total remains constant.

If the cross-sectional area of a pipe decreases, what happens to the fluid velocity?

✓ (a) Increases

(b) Decreases

(c) Remains the same

(d) Depends on the fluid density

Answer: (a) Working: the equation of continuity, A1 v1 = A2 v2, requires the same volume to pass every section each second. A smaller area must therefore be compensated by a larger velocity, which is why water speeds up at the narrow nozzle of a hose.

A sky diver falls through the air at terminal velocity. The force of air resistance on him is

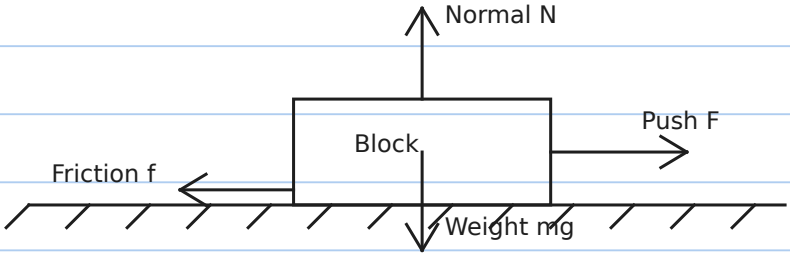
(a) half his weight

✓ (b) equal to his weight

(c) twice his weight

(d) Cannot be determined from the information given

Answer: (b) Working: at terminal velocity the diver falls at constant speed, so the net force on him is zero. The upward air resistance must therefore balance his downward weight exactly, which is the defining condition of terminal velocity.



Wind speeding up as it blows over the top of a hill

(a) Increases atmospheric pressure there

✓ (b) decreases atmospheric pressure there

(c) doesn't affect atmospheric pressure there

(d) equal's atmospheric pressure

Answer: (b) Working: by Bernoulli's principle, faster-moving air has lower pressure. The wind moves faster over the hilltop than in the still air around it, so the pressure over the top is reduced.

Question: A fluid is undergoing "incompressible" flow. This means that: (a) the pressure at a given point cannot change with time (b) the velocity at a given point cannot change with time (c) the velocity must be the same everywhere (d) the pressure must be the same everywhere (e) the density cannot change with time or location

Answer: (e) the density cannot change with time or location. Working: incompressible flow is defined by the density of the fluid remaining constant throughout the flow, as is nearly true for liquids. Pressure and velocity may still vary from point to point; steady flow, not incompressibility, is the condition about velocity at a fixed point.

A fluid is undergoing steady flow. Therefore:

(a) the velocity of any given molecule of fluid does not change

(b) the pressure does not vary from point to point

✓ (c) the velocity at any given point does not vary with time

(d) the density does not vary from point to point

Answer: (c) Working: in steady flow every particle arriving at a fixed point has the same velocity as the particles that passed that point earlier. The velocity may still differ from one point of the flow to another, for example between a wide and a narrow section.

The equation of continuity for fluid flow can be derived from the conservation of:

(a) energy

✓ (b) mass

(c) volume

(d) pressure

Answer: (b) Working: the equation A1 v1 = A2 v2 (for an incompressible fluid) follows because the mass of fluid entering a pipe section must equal the mass leaving it, since fluid is neither created nor stored inside. Conservation of energy, by contrast, leads to Bernoulli's equation.

Exercise - Structured Questions (CRQs)

Question: What is difference between streamline and turbulent flow?

Answer: In streamline (laminar) flow every particle follows a smooth path, the streamlines never cross, and the layers of fluid slide over one another in an orderly way, as at low speeds. In turbulent flow the motion is irregular and disorderly, with eddies and whirls, and streamlines change from moment to moment. Turbulence sets in at high speeds or past obstacles and wastes more energy in friction.

Question: Would a drinking straw work in space where there is no gravity? Explain.

Answer: A drinking straw works because atmospheric pressure pushes the drink up the straw when the air inside it is sucked out, creating a pressure difference. In open space there is no surrounding air pressure, so even in a spaceship cabin a straw would work only as long as the cabin air pressure exists; in a true vacuum outside, there would be no pressure to push the liquid up. Gravity itself is not what drives the drink upward, but on Earth gravity keeps the liquid settled at the bottom of the glass.

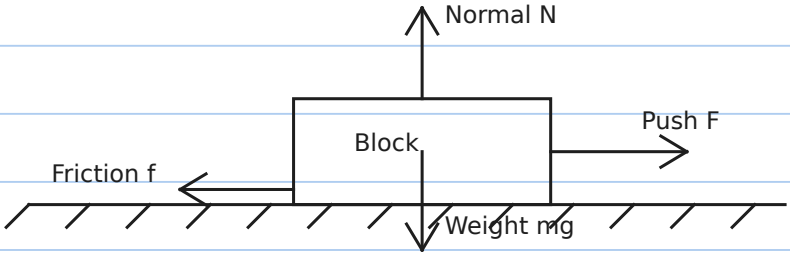
Question: Why do airplanes take off into wind?

Answer: Taking off into the wind means the air flows over the wings faster than the plane's ground speed alone would give. Since lift increases with the speed of air over the wing, the plane reaches the lift needed for take-off at a lower ground speed and over a shorter runway distance. It also gives the pilot better control during the slowest, most critical part of the flight.

Question: Describe terminal velocity in liquids.

Answer: When a small sphere falls through a liquid, three forces act on it: its weight pulling it down, the upthrust of the liquid, and the viscous drag acting upward, which grows with speed according to Stokes' law, F = 6 pi eta r v. At first the sphere accelerates, but as the drag grows the net force

falls to zero and the sphere then falls at a constant maximum speed called its terminal velocity. For a sphere it is given by v_t = (2 g r^2 (rho - sigma)) / (9 eta), where rho is the density of the sphere and sigma that of the liquid.



Question: Discuss the significance of Reynolds number.

Answer: The Reynolds number, Re = rho v L / eta, is a dimensionless quantity that predicts whether a flow will be laminar or turbulent. For flow in a pipe, values below about 2000 indicate smooth laminar flow, while values above about 4000 indicate turbulent flow, with a transition region in between. Engineers use it to design pipelines, aircraft and ships, because turbulent flow causes much greater drag and energy loss than laminar flow.

Question: State Bernoulli's principle.

Answer: Bernoulli's principle states that for the steady flow of an ideal (incompressible, non-viscous) fluid, the sum of the pressure energy, kinetic energy and potential energy per unit volume remains constant along a streamline. Mathematically, P + (1/2) rho v^2 + rho g h = constant. A direct consequence is that where the speed of the fluid is greater, its pressure is smaller.

Question: Give two applications of Bernoulli's principle.

Answer: One application is the aeroplane wing: air moves faster over the curved upper surface, so the pressure above the wing is lower than below it and the difference provides lift. A second application is the venturi meter

and the carburetor/atomizer, where fluid speeding through a narrow throat has reduced pressure, which is used to measure flow speed or to draw up fuel or perfume spray.

Question: 'Fluid flow is turbulent rather than laminar', support this statement.

Answer: In most practical situations speeds are high and passages are large, so the Reynolds number far exceeds the critical value of about 2000, and the flow becomes turbulent. Smoke rising from a cigarette flows smoothly for a short distance and then breaks into swirls, showing how easily laminar flow breaks down in real conditions. Rivers, winds, blood in arteries during exertion and water in household pipes all show eddies and mixing, which confirms that everyday fluid flow is mostly turbulent rather than laminar.

Question: Discuss importance of Stokes law.

Answer: Stokes' law, F = 6 pi eta r v, gives the viscous drag on a small sphere moving slowly through a fluid, and it is the basis for finding terminal velocity. It is used to measure the viscosity of liquids in the falling-sphere viscometer and to find the size of tiny particles such as droplets in the Millikan oil-drop experiment. It also explains why fine dust and mist settle very slowly through air while larger drops fall quickly.

Question: Justify spin of ball in Bernoulli's principle.

Answer: A spinning ball drags a layer of air around with it, so on one side the surface moves in the same direction as the oncoming air and the air speed increases, while on the other side it decreases. By Bernoulli's principle the side with faster air has lower pressure, and this pressure difference pushes the ball sideways, curving its path. This is the Magnus effect, used by cricket and football players to swing or curve the ball.

Exercise - Structured Questions (ERQs)

Question: Derive equation of continuity. Also show its physical significance.

Answer: Consider an incompressible fluid flowing through a pipe of varying cross-section, with area A1 and speed v1 at one section and A2 and v2 at another. In time delta t the mass entering is rho A1 v1 delta t and the mass leaving is rho A2 v2 delta t; since no fluid accumulates inside, these masses are equal, giving A1 v1 = A2 v2, the equation of continuity. Its physical significance is that the volume flow rate is the same everywhere: the fluid speeds up in narrow parts and slows down in wide parts. It is the fluid form of the law of conservation of mass.

Question: Derive Bernoulli's equation.

Answer: Take an ideal fluid flowing through a pipe from point 1 (pressure P1, height h1, speed v1) to point 2 (P2, h2, v2). The pressure forces do net work (P1 - P2) times the volume V of fluid that passes. This work equals the gain in kinetic energy (1/2) rho V (v2^2 - v1^2) plus the gain in potential energy rho V g (h2 - h1), by the work-energy theorem. Dividing by V and rearranging gives P1 + (1/2) rho v1^2 + rho g h1 = P2 + (1/2) rho v2^2 + rho g h2, that is P + (1/2) rho v^2 + rho g h = constant along a streamline. The equation is simply conservation of energy per unit volume of the fluid.

Question: Discuss viscous force in fluids.

Answer: When layers of a fluid move relative to each other, internal friction opposes the motion; the tangential force responsible is called the viscous force, and the property is viscosity. For a layer of area A with velocity gradient dv/dx, the viscous force is F = eta A (dv/dx), where eta is

the coefficient of viscosity, measured in Pa.s. Viscosity is why honey pours slowly while water pours quickly, why lubricating oil must have a suitable viscosity, and why a falling object in a fluid reaches a terminal velocity instead of accelerating forever.

Question: Define fluid dynamics and explain its significance in the study of fluids. How does it differ from fluid statics?

Answer: Fluid dynamics is the branch of physics that deals with fluids in motion: their speeds, pressures, flow rates and the forces involved in flow. It is significant because almost every practical use of fluids, from water supply and blood circulation to flight and weather, involves moving fluids, and laws such as the continuity equation and Bernoulli's equation let us predict and design for these flows. Fluid statics, in contrast, deals with fluids at rest and concerns pressure at a depth, Pascal's principle and buoyancy. The two together give the complete description of fluid behavior.

Question: Discuss the concept of Reynolds number and its significance in fluid dynamics. Explain how Reynolds number relates to the transition between laminar and turbulent flow.

Answer: The Reynolds number is the ratio of inertial effects to viscous effects in a flow, Re = rho v L / eta, where L is a characteristic length such as the pipe diameter; it has no units. Its significance is that flows with the same Reynolds number behave similarly, so it is used in model testing of aircraft, ships and pipelines. For pipe flow, when Re is below about 2000 viscous forces dominate and the flow stays laminar and orderly. When Re rises above about 4000 inertia dominates, the streamlines break up into eddies, and the flow becomes turbulent; between these values the flow is in an unstable transition state.

Exercise - Numerical Problems

Question: Two spherical raindrops of equal size are falling through air at a velocity of 0.08 m/s. If the drops join together forming a large spherical drop, what will be the new terminal velocity?

Answer: Given: terminal velocity of each small drop v1 = 0.08 m/s, two drops of equal radius r join to form one drop of radius R

Step 1: Volume doubles, (4/3) pi R^3 = 2 x (4/3) pi r^3, so R^3 = 2 r^3 and R = 2^(1/3) r = 1.26 r

Step 2: Terminal velocity is proportional to the square of the radius, v proportional to r^2

Step 3: v2 / v1 = (R / r)^2 = (2^(1/3))^2 = 2^(2/3) = 1.587

Step 4: v2 = 1.587 x 0.08 = 0.127 m/s, about 0.13 m/s

Final answer: 0.13 m/s

Question: Calculate the viscous drag on a drop of oil of 0.1 mm radius falling through air at its terminal velocity. (Viscosity of air = 1.8 x 10^-5 Pa. s; density of oil = 850 kg/m^3)

Answer: Given: r = 0.1 mm = 1.0x10^-4 m, rho_oil = 850 kg/m^3, g = 9.8 m/s^2, air density about 1.29 kg/m^3

Step 1: At terminal velocity the viscous drag balances the effective weight of the drop, F = (weight - upthrust) = (4/3) pi r^3 (rho_oil - rho_air) g

Step 2: Volume factor, (4/3) pi r^3 = 4.19 x (1.0x10^-4)^3 = 4.19x10^-12 m^3

Step 3: Density difference = 850 - 1.29 = 848.7 kg/m^3

Step 4: F = 4.19x10^-12 x 848.7 x 9.8 = 3.48x10^-8 N

Final answer: 3.48x10^-8 N

Question: What area must a heating duct have if air moving 3.0 m/s along it can replenish the air every 15 minutes in a room of volume 300 m^3. Assume air density remains constant.

Answer: Given: room volume V = 300 m^3, time = 15 minutes = 900 s, air speed v = 3.0 m/s

Step 1: Required volume flow rate, Q = V / t = 300 / 900 = 0.333 m^3/s

Step 2: From Q = A v, area A = Q / v = 0.333 / 3.0 = 0.111 m^2

Step 3: If the duct is square, its side is sqrt(0.111) = 0.33 m

Final answer: 0.11 m^2 (a square duct of side 0.33 m)

Question: Water circulates throughout a house in a hot-water heating system. If the water is pumped at a speed of 0.50 m/s through a 4.0 cm diameter pipe in the basement under a pressure of 3.0 atm, what will be the flow speed and pressure in a 2.6 cm diameter pipe on the second floor 5.0 m above? Assume the pipes do not divide into branches.

Answer: Given: v1 = 0.50 m/s, d1 = 4.0 cm, d2 = 2.6 cm, P1 = 3.0 atm = 3.04x10^5 Pa, h = 5.0 m, rho = 1000 kg/m^3, g = 9.8 m/s^2, 1 atm = 1.01x10^5 Pa

Step 1: By continuity A1 v1 = A2 v2, so v2 = v1 (d1/d2)^2 = 0.50 x (4.0/2.6)^2 = 0.50 x 2.37 = 1.18 m/s, about 1.2 m/s

Step 2: Apply Bernoulli, P2 = P1 + (1/2) rho (v1^2 - v2^2) - rho g h

Step 3: (1/2) rho (v1^2 - v2^2) = 500 x (0.25 - 1.40) = -575 Pa, and rho g h = 1000 x 9.8 x 5.0 = 49000 Pa

Step 4: P2 = 304000 - 575 - 49000 = 254425 Pa = 2.5 atm

Final answer: flow speed 1.2 m/s, pressure 2.5 atm

Question: What is the volume rate of flow of water from a 1.85 cm diameter faucet if the pressure head is 12 m?

Answer: Given: diameter d = 1.85 cm = 0.0185 m, pressure head h = 12 m, g = 9.8 m/s^2

Step 1: By Torricelli's theorem the speed of efflux is v = sqrt(2 g h) = sqrt(2 x 9.8 x 12) = sqrt(235.2) = 15.3 m/s

Step 2: Cross-sectional area, A = pi d^2 / 4 = 3.14 x (0.0185)^2 / 4 = 2.69x10^-4 m^2

Step 3: Volume flow rate, Q = A v = 2.69x10^-4 x 15.3 = 4.1x10^-3 m^3/s

Final answer: 4.1x10^-3 m^3/s (the book prints 4.6x10^-3 m^3/s)

Question: The stream of water emerging from a faucet 'neck down' as it falls. The cross-sectional area is 1.2 cm^2 and 0.35 cm^2. The two levels are separated by a vertical distance of 45 mm as shown in figure. At what rate does water flow from the tap?

Answer: Given: A1 = 1.2 cm^2, A2 = 0.35 cm^2, vertical separation h = 45 mm = 4.5 cm, g = 980 cm/s^2

Step 1: By continuity the flow rate Q is the same at both levels, v1 = Q / A1 and v2 = Q / A2

Step 2: By Bernoulli (or free fall through h), v2^2 - v1^2 = 2 g h = 2 x 980 x 4.5 = 8820 cm^2/s^2

Step 3: Substitute the speeds, Q^2 (1/A2^2 - 1/A1^2) = 8820, so Q^2 (1/0.1225 - 1/1.44) = Q^2 (8.163 - 0.694) = 7.47 Q^2 = 8820

Step 4: Q^2 = 1181, so Q = 34 cm^3/s

Final answer: 34 cm^3/s

Question: Water leaves the jet of a horizontal hose at 10 m/s. If the velocity of water within the hose is 0.40 m/s, calculate the pressure within the hose. Density of water is 1000 kg/m^3 and atmospheric pressure is 100000 Pa.

Answer: Given: jet speed v1 = 10 m/s, speed inside hose v2 = 0.40 m/s, rho = 1000 kg/m^3, atmospheric pressure = 100000 Pa, hose is horizontal so heights are equal

Step 1: Apply Bernoulli between the jet (at atmospheric pressure) and the inside of the hose, P_atm + (1/2) rho v1^2 = P_hose + (1/2) rho v2^2

Step 2: Rearrange, P_hose = 100000 + (1/2) x 1000 x (10^2 - 0.40^2)

Step 3: Calculate, (1/2) x 1000 x (100 - 0.16) = 500 x 99.84 = 49920 Pa

Step 4: Add, P_hose = 100000 + 49920 = 149920 Pa, about 1.5x10^5 Pa

Final answer: 1.5x10^5 Pa

Question: What is the maximum weight of an aircraft with a wing area of 50 m^2 flying horizontally, is the velocity of the air over the upper surface of the wing is 150 m/s and that the lower surface 140 m/s? Density of air is 1.29 kg/m^3

Answer: Given: wing area A = 50 m^2, v_upper = 150 m/s, v_lower = 140 m/s, rho = 1.29 kg/m^3

Step 1: Pressure difference from Bernoulli, delta P = (1/2) rho (v_upper^2 - v_lower^2)

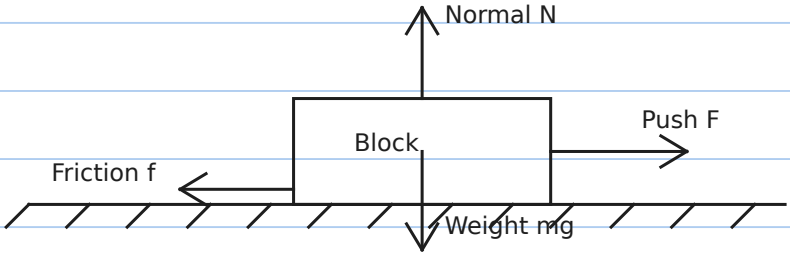
Step 2: Calculate the squares, 150^2 = 22500 and 140^2 = 19600, difference = 2900

Step 3: delta P = 0.5 x 1.29 x 2900 = 1870 Pa

Step 4: Lift force = delta P x A = 1870 x 50 = 9.35x10^4 N, about 9.3x10^4 N

Step 5: For horizontal flight the maximum weight equals the lift

Final answer: 9.3x10^4 N



Question: A liquid flows through a pipe with a diameter of 0.50 m at a speed of 4.20 m/s. What is the rate of flow in L/min?

Answer: Given: d = 0.50 m, v = 4.20 m/s, 1 m^3 = 1000 L

Step 1: Radius r = 0.25 m, area A = pi r^2 = 3.14 x (0.25)^2 = 0.196 m^2

Step 2: Volume flow rate, Q = A v = 0.196 x 4.20 = 0.825 m^3/s

Step 3: Convert to litres per second, 0.825 x 1000 = 825 L/s

Step 4: Convert to per minute, 825 x 60 = 49500 L/min

Final answer: 49500 L/min

Question: Calculate the average speed of blood flow in the major arteries of the body, which have a total cross-sectional area of about 2.1 cm^2. Use the data of example 7.3 (aorta radius 1.2 cm, blood speed in the aorta 40 cm/s).

Answer: Given: aorta radius = 1.2 cm, aorta blood speed = 40 cm/s

(Worked Example 7.3 of this unit), total area of major arteries A2 = 2.1 cm^2

Step 1: Area of the aorta, A1 = pi r^2 = 3.14 x (1.2)^2 = 4.52 cm^2

Step 2: Volume flow rate from the aorta, Q = A1 v1 = 4.52 x 40 = 181 cm^3/s

Step 3: By continuity the same flow rate passes through the arteries, v2 = Q / A2 = 181 / 2.1 = 86 cm/s

Final answer: 86 cm/s (the book's printed 13.6 cm/s does not follow from the data of its own Example 7.3)

Unit 8: Electric Fields

Exercise — Section (A): Multiple Choice Questions (MCQs)

A 2 uC point charge is located a distance "d" away from a 6 uC point charge. What is the ratio of F12/F21?

(a) 1/3

(b) 3

✓ (c) 1

(d) 12

Answer: (c) Working: By Newton's third law the two charges exert equal and opposite forces on each other, so F12 and F21 have the same magnitude whatever the sizes of the charges. The ratio of their magnitudes is therefore 1.

The minimum charge on an object cannot be less than:

✓ (a) 1.6 x 10^-19 C

(b) 3.2 x 10^-19 C

(c) 9.1 x 10^9

(d) No definite value exist

Answer: (a) Working: Charge is quantised as Q = ne, where e = 1.6 x 10^-19 C is the charge of one electron or proton. The smallest free charge an object can carry is one elementary charge, so no charge smaller than this can occur.

Two charges are placed at a certain distance. If the magnitude of each charge is doubled, the force will become

(a) 1/4th of its original value

✓ (b) 4 times of its original value

(c) 1/8th of its original value

(d) 8 times of its original value

Answer: (b) Working: Coulomb's law gives F proportional to q1 x q2. Doubling both charges multiplies the product by 2 x 2 = 4, so the force becomes four times its original value at the same separation.

Which of the following can be deflected while moving in the electric field?

(a) neutron

(b) photon

✓ (c) electron

(d)

(a) and

Answer: (c) Working: An electric field exerts a force F = qE only on a charged particle. The electron carries a negative charge so it is deflected, while the neutron and the photon are electrically neutral and pass through undeflected.

The flux through a flat surface of area "A" in a uniform electric field "E" is maximum when the surface area is:

(a) Parallel to E

✓ (b) perpendicular to E

(c) placed 45 deg to E

(d) placed 60 deg to E

Answer: (b) Working: Flux is phi = EA cos theta, where theta is the angle between E and the normal to the surface. The flux is maximum when the normal is along E, which means the surface itself is perpendicular to the field lines, so the greatest number of lines pass through it.

The product of charge "q" and small separation "d" between two charges

of same magnitude and opposite in nature is known as:

(a) Electric dipole

(b) Moment arm

✓ (c) Electric dipole moment

(d) Flux of electric field

Answer: (c) Working: A pair of equal and opposite charges forms an electric dipole, and its strength is measured by the dipole moment p = qd. It is a vector pointing from the negative charge towards the positive charge.

12 J of work is to be done against an existing electric field to take a charge of 0.01 C from one point A to another point B. The potential difference between B and A is

(a) 120 V

✓ (b) 1200 V

(c) 1.2 V

(d) 12 V

Answer: (b) Working: Potential difference is work done per unit charge, V = W/q = 12 / 0.01 = 1200 V. So point B is at a potential 1200 V higher than point A for this positive charge.

The force between two charges placed in air is F. If air is replaced by a medium of relative permittivity epsilon_r, then force is reduced to:

(a) F epsilon_r

✓ (b) F / epsilon_r

(c) epsilon_r / F

(d) epsilon epsilon_r

Answer: (b) Working: In a dielectric medium Coulomb's force becomes F_medium = F / epsilon_r, because the permittivity in the denominator increases by the factor epsilon_r. A medium with larger relative

permittivity therefore weakens the force more strongly.

The negative gradient of the potential is:

(a) potential energy

(b) voltage

✓ (c) electric field intensity

(d) electric flux

Answer: (c) Working: The field and potential are related by E = -dV/dr, so the electric field intensity is the negative rate of change of potential with distance. The field therefore points in the direction of decreasing potential.

The electric flux through a plane area will be half of its maximum value when area is held at angle of _____ with electric field

(a) 30 deg

✓ (b) 60 deg

(c) 45 deg

(d) 90 deg

Answer: (b) Working: Here the angle is measured between the area (its normal) convention such that phi = phi_max cos theta, and cos 60 deg = 0.5. So at 60 deg the flux falls to exactly one half of its maximum value.

Exercise — Section (B): Structured Questions — CRQs

Question: Why do most objects tend to contain nearly equal numbers of positive and negative charges?

Answer: Ordinary matter is made of atoms, and each neutral atom contains equal numbers of protons and electrons, so bulk objects are normally electrically neutral. Any large imbalance creates very strong electric forces that quickly attract opposite charges from the surroundings

and neutralise the excess. That is why everyday objects contain almost equal numbers of positive and negative charges and show no net charge.

Question: When measuring an electric field, could we use a negative rather than a positive test charge?

Answer: Yes, a negative test charge can be used, because the field is defined as force per unit charge, E = F/q, and this ratio is independent of the test charge used. The only difference is that the force on a negative charge acts opposite to the field direction, so the direction must be reversed when finding E. By convention a small positive test charge is preferred simply because the force on it points along the field.

Question: During fair weather, the electric field due to the net charge on Earth points downward. Is Earth charged positively or negatively?

Answer: Earth is charged negatively. Electric field lines point towards a negative charge and away from a positive charge, so a downward field near the surface means the field is directed into the Earth. This shows that the Earth's surface carries a net negative charge, balanced by positive charge in the upper atmosphere.

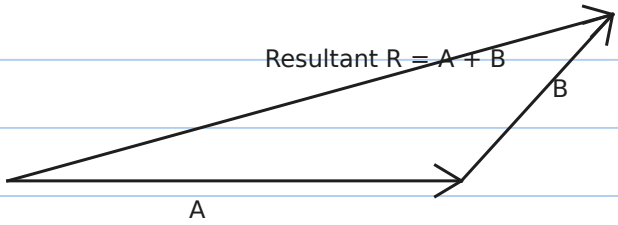
Question: How is the electric flux through a closed surface independent of the size or shape of the surface enclosed the charge?

Answer: By Gauss's law the total flux through a closed surface is phi = Q_enclosed / epsilon_0, which depends only on the net charge inside the surface. Enlarging the surface spreads the same field lines over a larger area while the field itself becomes weaker, and the two effects cancel exactly. Changing the shape has the same result, because every field line from the enclosed charge still crosses the surface once.

Question: What is electric dipole and electric dipole moment?

Answer: An electric dipole is a pair of equal and opposite charges, +q and -q, separated by a small distance d. Its dipole moment is p = qd, a vector

directed from the negative charge to the positive charge along the line joining them. The dipole moment measures the strength of the dipole and decides the field it produces and the torque it experiences in an external field.



Question: A charged particle is seen to be moving in an electric field along a straight line. How it effects the path of motion of the particle?

Answer: Straight line motion occurs when the electric force acts along, or opposite to, the direction of motion, so there is no sideways component to curve the path. The field then only changes the speed of the particle, accelerating it if the force is along the motion and slowing it if it is against it. If the velocity had a component perpendicular to the field, the path would bend into a parabola instead.

Question: An electron has a speed of 10^6 m/s. Find its energy in electron volt.

Answer: Given: v = 10^6 m/s, mass of electron m = 9.1 x 10^-31 kg, 1 eV = 1.6 x 10^-19 J

Step 1: Kinetic energy KE = (1/2) m v^2 = 0.5 x 9.1 x 10^-31 x (10^6)^2

Step 2: KE = 0.5 x 9.1 x 10^-31 x 10^12 = 4.55 x 10^-19 J

Step 3: Convert to electron volts, KE = 4.55 x 10^-19 / 1.6 x 10^-19 = 2.84 eV

Final answer: 2.84 eV

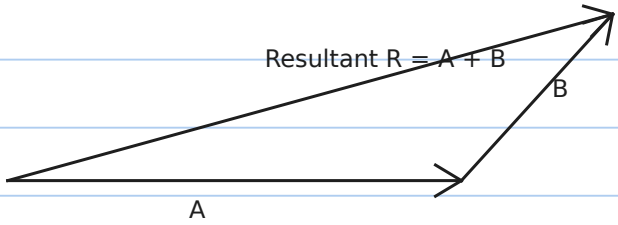
Exercise — Section (B): Structured Questions — ERQs

Question: Define electric charge. Discuss the fundamental properties of electric charge, including the principles of conservation of charge and quantization of charge.

Answer: Electric charge is the basic property of matter due to which it experiences electric forces, and it exists in two kinds, positive and negative, with like charges repelling and unlike charges attracting. Charge is conserved: in any isolated system the total charge remains constant, so charge can be transferred but never created or destroyed. Charge is also quantised, meaning it always occurs in whole multiples of the elementary charge, Q = ne where e = 1.6 x 10^-19 C. In addition, charge is additive and is invariant, having the same value in all frames of reference.

Question: State and explain Coulomb's law. Apply it to calculate the electric field due to an isolated point charge.

Answer: Coulomb's law states that the force between two point charges is directly proportional to the product of the charges and inversely proportional to the square of the distance between them, F = (1 / 4 pi epsilon_0) q1 q2 / r^2, acting along the line joining them. To find the field of an isolated charge Q, place a small test charge q0 at distance r; the force on it is F = (1 / 4 pi epsilon_0) Q q0 / r^2. Dividing by q0 gives E = F/q0 = (1 / 4 pi epsilon_0) Q / r^2, directed away from Q if Q is positive and towards Q if Q is negative.



Question: Define electric flux and explain its significance in the study of electromagnetism. How does it relate to electric fields and charged particles?

Answer: Electric flux is a measure of the number of electric field lines passing through a given area, defined by phi = EA cos theta for a uniform field, with unit N.m^2/C. Its great significance is Gauss's law: the total flux through any closed surface equals the enclosed charge divided by epsilon_0, whatever the shape of the surface. This lets us calculate fields of symmetric charge distributions very easily. In this way flux connects the field spread out in space directly to the charges that produce it.

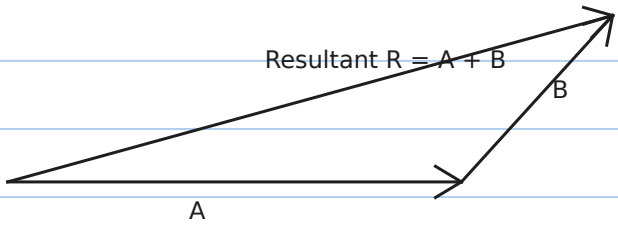
Question: What is electric potential? Derive expression for potential due to an isolated point charge.

Answer: Electric potential at a point is the work done per unit positive charge in bringing a test charge from infinity to that point, V = W/q0, measured in volts. For a point charge Q, the field at distance r is E = (1 / 4 pi epsilon_0) Q / r^2, and the work in moving q0 from infinity to r against this field is W = (1 / 4 pi epsilon_0) Q q0 / r. Dividing by q0 gives V = (1 / 4 pi epsilon_0) Q / r. The potential is a scalar, positive for a positive charge and negative for a negative charge, and it falls as 1/r.

Question: Define an electric dipole. Derive formula for the electric field due to an electric dipole at a point "P" placed on its axial line.

Answer: An electric dipole is a pair of equal and opposite charges +q and -q separated by a small distance 2a, with dipole moment p = q x 2a. For a

point P on the axis at distance r from the centre, the field due to +q is (1 / 4 pi epsilon_0) q / (r - a)^2 away from the dipole, and the field due to -q is (1 / 4 pi epsilon_0) q / (r + a)^2 towards it. Subtracting, E = (1 / 4 pi epsilon_0) x (2pr) / (r^2 - a^2)^2, and for r much greater than a this reduces to E = (1 / 4 pi epsilon_0) x 2p / r^3, along the direction of p.



Exercise — Numericals

Question: (a) Calculate the value of two equal charges if they repel one another with a force of 0.1 N when situated 50 cm apart in a vacuum. (b) What would be the size of the charges if they were situated in an insulating liquid whose permittivity was ten times that of a vacuum?

Answer: Given: F = 0.1 N, r = 50 cm = 0.50 m, q1 = q2 = q, k = 9 x 10^9 N.m^2/C^2

Step 1: In vacuum F = k q^2 / r^2, so q^2 = F r^2 / k

Step 2: q^2 = 0.1 x (0.50)^2 / (9 x 10^9) = 0.025 / 9 x 10^9 = 2.78 x 10^-12 C^2

Step 3: q = sqrt(2.78 x 10^-12) = 1.7 x 10^-6 C = 1.7 uC

Step 4: In the liquid the force is reduced by epsilon_r = 10, so to keep the same force q^2 must increase 10 times, q_liquid = q x sqrt(10) = 1.7 x 3.16 = 5.3 uC

Final answer: (a) 1.7 uC each, (b) 5.3 uC each

Question: How far apart must two protons be if the magnitude of the electrostatic force acting on either one due to the other is equal to the magnitude of the gravitational force on a proton at Earth's surface? (mass of proton = 1.67 x 10^-27 kg, charge of proton = 1.6 x 10^-19 C)

Answer: Given: m = 1.67 x 10^-27 kg, e = 1.6 x 10^-19 C, g = 9.8 m/s^2

Step 1: Set electrostatic force equal to weight, k e^2 / r^2 = mg

Step 2: r^2 = k e^2 / (mg) = 9 x 10^9 x (1.6 x 10^-19)^2 / (1.67 x 10^-27 x 9.8)

Step 3: r^2 = 2.30 x 10^-28 / 1.64 x 10^-26 = 1.41 x 10^-2 m^2

Step 4: r = sqrt(1.41 x 10^-2) = 0.119 m

Final answer: 0.119 m apart

Question: An electron of charge 1.6 x 10^-19 C is situated in a uniform electric field of intensity 1200 volt cm^-1. Find the force on it, its acceleration, and the time it takes to travel 2 cm from rest (electronic mass, m_e = 9.1 x 10^-31 kg).

Answer: Given: q = 1.6 x 10^-19 C, E = 1200 V/cm = 1200 x 100 = 1.2 x 10^5 V/m, s = 2 cm = 0.02 m

Step 1: Force F = qE = 1.6 x 10^-19 x 1.2 x 10^5 = 1.92 x 10^-14 N

Step 2: Acceleration a = F/m = 1.92 x 10^-14 / 9.1 x 10^-31 = 2.12 x 10^16 m/s^2

Step 3: From s = (1/2) a t^2, t = sqrt(2s/a) = sqrt(2 x 0.02 / 2.12 x 10^16)

Step 4: t = sqrt(1.89 x 10^-18) = 1.37 x 10^-9 s

Final answer: F = 1.92 x 10^-14 N, a = 2.12 x 10^16 m/s^2, t = 1.37 x

10^-9 s

Question: An alpha particle (the nucleus of a helium atom) has a mass of 6.64 x 10^-27 kg and a charge of 2e. What are the (a) magnitude and (b) direction of the electric field that will balance the gravitational force on the particle?

Answer: Given: m = 6.64 x 10^-27 kg, q = 2e = 2 x 1.6 x 10^-19 = 3.2 x 10^-19 C, g = 9.8 m/s^2

Step 1: For balance the electric force must equal the weight, qE = mg

Step 2: E = mg / q = 6.64 x 10^-27 x 9.8 / 3.2 x 10^-19

Step 3: E = 6.51 x 10^-26 / 3.2 x 10^-19 = 2.03 x 10^-7 N/C

Step 4: The alpha particle is positively charged, so the field must push it upward against gravity

Final answer: (a) 2.03 x 10^-7 N/C, (b) the electric field is directed upwards

Question: A proton and an electron form two corners of an equilateral triangle of side length 2.0 x 10^-6 m. What is the magnitude of the net electric field these two particles produce at the third corner?

Answer: Given: r = 2.0 x 10^-6 m, e = 1.6 x 10^-19 C for both particles

Step 1: Field of each particle at the third corner, E = k e / r^2 = 9 x 10^9 x 1.6 x 10^-19 / (2.0 x 10^-6)^2 = 3.6 x 10^2 N/C

Step 2: The proton's field points away from the proton and the electron's field points towards the electron, and in the equilateral geometry their components perpendicular to the base cancel

Step 3: The components along the base add, giving E_net = 2 x E x cos 60 deg = 2 x 360 x 0.5 = 360 N/C

Final answer: 3.6 x 10^2 N/C

Question: Figure shows two charged particles on an x-axis: -q = -3.20 x 10^-19 C at x = -3.00 m and q = 3.20 x 10^-19 C at x = 3.00 m. What are the (a) magnitude and (b) direction (relative to the positive direction of the x-axis) of the net electric field produced at point P at y = 4.00 m?

Answer: Given: q = 3.20 x 10^-19 C, charges at x = -3.00 m and x = +3.00 m, P at y = 4.00 m

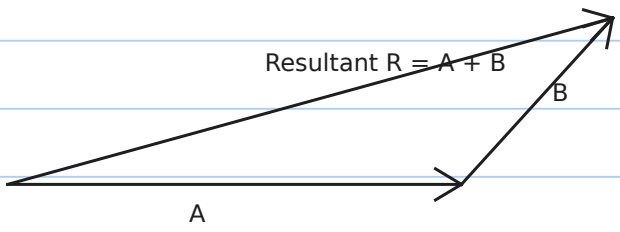
Step 1: Distance of each charge from P, r = sqrt(3^2 + 4^2) = 5.00 m

Step 2: Field of each charge, E = k q / r^2 = 9 x 10^9 x 3.20 x 10^-19 / 25 = 1.15 x 10^-10 N/C

Step 3: The y-components of the two fields cancel by symmetry, while the x-components both point in the -x direction and add: E_net = 2 x E x (3/5) = 2 x 1.15 x 10^-10 x 0.6

Step 4: E_net = 1.38 x 10^-10 N/C

Final answer: (a) 1.38 x 10^-10 N/C, (b) the net electric field points in the -x direction



Question: A proton and an electron form two corners of an equilateral triangle of side length 5 um. What is the magnitude of the net electric field these two particles produce at the third corner?

Answer: Given: r = 5 um = 5 x 10^-6 m, e = 1.6 x 10^-19 C

Step 1: Field of each particle, E = k e / r^2 = 9 x 10^9 x 1.6 x 10^-19 / (5 x 10^-6)^2 = 57.6 N/C

Step 2: As in the equilateral triangle geometry, the perpendicular components cancel and the components along the base add

Step 3: E_net = 2 x E x cos 60 deg = 2 x 57.6 x 0.5 = 57.6 N/C

Final answer: 57.6 N/C (about 5.8 x 10^1 N/C)

Question: The square surface shown in figure measures 3.2 mm on each side. It is immersed in a uniform electric field with magnitude E = 1800 N/C and with field lines at an angle of theta = 35 deg with a normal to the surface, as shown. Take that normal to be directed "outward," as though the surface were one face of a box. Calculate the electric flux through the surface.

Answer: Given: side = 3.2 mm = 3.2 x 10^-3 m, E = 1800 N/C, field lines enter the face so the angle between E and the outward normal is 180 deg - 35 deg = 145 deg

Step 1: Area A = (3.2 x 10^-3)^2 = 1.024 x 10^-5 m^2

Step 2: Flux phi = E A cos theta = 1800 x 1.024 x 10^-5 x cos 145 deg

Step 3: phi = 1.843 x 10^-2 x (-0.819) = -1.5 x 10^-2 N.m^2/C

Final answer: -1.5 x 10^-2 N.m^2/C, the negative sign showing net flux enters the surface

Question: An electron is liberated from the lower of two large parallel metal plates separated by a distance h = 2 cm. The upper plate has a potential of 2400 volts relative to the lower. How long does the electron take to reach it?

Answer: Given: h = 2 cm = 0.02 m, V = 2400 V, e = 1.6 x 10^-19 C, m =

9.1 x 10^-31 kg

Step 1: Field between the plates E = V/h = 2400 / 0.02 = 1.2 x 10^5 V/m

Step 2: Acceleration a = eE/m = 1.6 x 10^-19 x 1.2 x 10^5 / 9.1 x 10^-31 = 2.1 x 10^16 m/s^2

Step 3: From h = (1/2) a t^2, t = sqrt(2h/a) = sqrt(2 x 0.02 / 2.1 x 10^16)

Step 4: t = sqrt(1.9 x 10^-18) = 1.4 x 10^-9 s

Final answer: about 1.4 x 10^-9 s (field between plates is 1.2 x 10^5 V/m)

Question: Two large parallel metal plates are 1.5 cm apart and have charges of equal magnitudes but opposite signs on their facing surfaces. Take the potential of the negative plate to be zero. If the potential halfway between the plates is then 5.0 V, what is the electric field in the region between the plates?

Answer: Given: separation d = 1.5 cm = 0.015 m, potential at halfway point = 5.0 V, negative plate at 0 V

Step 1: The field between large parallel plates is uniform, so potential increases linearly from the negative plate

Step 2: Halfway potential is half the total potential difference, so total V = 2 x 5.0 = 10 V

Step 3: E = V/d = 10 / 0.015 = 6.7 x 10^2 V/m

Final answer: 6.7 x 10^2 V/m, directed from the positive plate to the negative plate

Unit 9: Capacitors

Exercise — Section (A): Multiple Choice Questions (MCQs)

The capacitance of a capacitor is NOT influenced by:

✓ (a) Plate thickness

(b) Plate area

(c) Plate separation

(d) Nature of the dielectric

Answer: (a) Working: Capacitance is C = epsilon_0 epsilon_r A / d, so it depends on plate area A, separation d and the dielectric between the plates. The thickness of the metal plates themselves does not appear in the formula and has no effect.

What is the value of capacitance of a capacitor which has a voltage of 4 V and has 16 C of charge?

(a) 2 F

✓ (b) 4 F

(c) 6 F

(d) 8 F

Answer: (b) Working: Capacitance is charge per unit voltage, C = Q/V = 16 / 4 = 4 F. A larger charge stored at the same voltage would mean a larger capacitance.

Capacitors are used in electric power supply system to:

✓ (a) Improve power factor

(b) Reduce line current

(c) Provide voltage stability

(d) switching

Answer: (a) Working: Capacitor banks supply reactive power locally, bringing the current and voltage more nearly in phase. This improves the power factor of the supply system, which also reduces losses, but power factor improvement is their primary purpose in power systems.

In a variable capacitor, capacitance can be varied by:

✓ (a) Turning the rotatable plates in or out

(b) Changing the plates

(c) Sliding the rotatable plates

(d) Changing the material of plates

Answer: (a) Working: In an air variable capacitor one set of plates rotates inside another fixed set. Turning them changes the overlapping area A, and since C is proportional to A, the capacitance changes smoothly.

Energy stored in the capacitor is:

(a) E = 1/4 C V

✓ (b) E = 1/2 C V^2

(c) E = C V^2

(d) E = 1/2 C V

Answer: (b) Working: The work done in charging a capacitor against the growing potential difference gives stored energy E = (1/2) Q V, and using Q = C V this becomes E = (1/2) C V^2. The energy resides in the electric field between the plates.

The time constant of a series RC circuit consisting of 100 uF capacitor in series with a 100 ohm resistor is:

(a) 0.1 s

(b) 0.1 ms

✓ (c) 0.01 s

(d) 0.01 ms

Answer: (c) Working: Time constant tau = RC = 100 x 100 x 10^-6 = 10^4 x 10^-6 = 0.01 s. In one time constant the charge on the capacitor rises to about 63 percent of its final value.

The charging of a capacitor through a resistance follows:

(a) linear law

(b) square law

✓ (c) exponential law

(d) none of the above

Answer: (c) Working: During charging the charge grows as q = Q_0 (1 - exp(-t/RC)), because the charging current falls as the capacitor voltage rises towards the supply voltage. The growth is therefore exponential, not linear.

When the total charge in a capacitor is doubled, the energy stored

(a) remains the same

(b) is doubled

(c) is halved

✓ (d) is quadrupled

Answer: (d) Working: Stored energy can be written E = Q^2 / (2C), so energy is proportional to the square of the charge. Doubling Q multiplies the energy by 2^2 = 4, provided the capacitance stays the same.

The capacitance C is charged through a resistor R. The time constant of the charging circuit is given by:

(a) C/R

(b) 1/RC

✓ (c) RC

(d) R/C

Answer: (c) Working: The time constant tau = RC has the unit of time and

sets the speed of charging and discharging. A larger resistance or a larger capacitance makes the circuit respond more slowly.

Capacitor blocks:

(a) alternating current

(b) direct current

(c) both alternating and direct current

(d) neither alternating nor direct current

Answer: (b) Working: Once a capacitor is charged to the supply voltage, no steady direct current can flow through its insulating dielectric, so it blocks DC. Alternating current appears to pass because the capacitor repeatedly charges and discharges on each half cycle.

Exercise — Section (B): Structured Questions — CRQs

Question: State the factors on which the capacitance of a parallel plate capacitor depends.

Answer: The capacitance is C = epsilon_0 epsilon_r A / d, so it increases with the area A of overlap of the plates and with the relative permittivity epsilon_r of the dielectric between them. It decreases as the separation d between the plates increases. It does not depend on the charge stored, the voltage applied, or the thickness and material of the metal plates themselves.

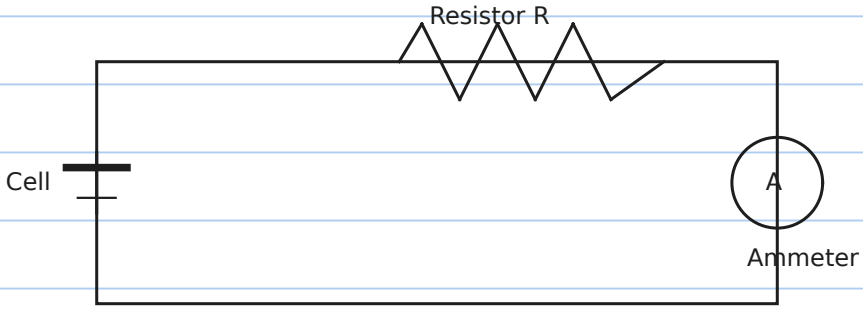
Question: Explain what is meant by dielectric constant (relative permittivity). State two physical properties desirable in a material to be used as the dielectric in a capacitor.

Answer: The dielectric constant, or relative permittivity epsilon_r, is the ratio of the capacitance with the dielectric filling the space between the

plates to the capacitance with vacuum (or air) in the same space, epsilon_r = C / C_0. It tells how many times the material increases the capacitance. A good dielectric should have a high dielectric strength so it can withstand a large voltage without breaking down, and a high relative permittivity so a large capacitance is obtained in a small size; low leakage and low energy loss are also desirable.

Question: Derive expressions for the combined capacitance of two capacitors (a) connected in series, (b) connected in parallel.

Answer: In series the same charge Q sits on each capacitor, and the voltages add: V = V1 + V2 = Q/C1 + Q/C2, so 1/C = 1/C1 + 1/C2 and C = C1 C2 / (C1 + C2), always smaller than either capacitor. In parallel both capacitors have the same voltage V and the charges add: Q = Q1 + Q2 = C1 V + C2 V, so C = C1 + C2. Series connection therefore reduces capacitance while parallel connection increases it.



Question: Derive an expression for the energy stored in a capacitor C when there is a potential difference V between the plates.

Answer: Suppose the capacitor is charged step by step; when its charge is q, its potential difference is v = q/C, and adding a small charge dq needs work dW = v dq = q dq / C. Integrating from q = 0 to the final charge Q, W = Q^2 / (2C). Using Q = C V this stored energy is E = (1/2) C V^2, and it is held in the electric field between the plates until the capacitor discharges.

Question: A capacitor gets a charge of 50 C when it is connected to a battery of emf 5 V. Calculate the capacity of the capacitor.

Answer: Given: Q = 50 C, V = 5 V

Step 1: Capacitance is defined by C = Q / V

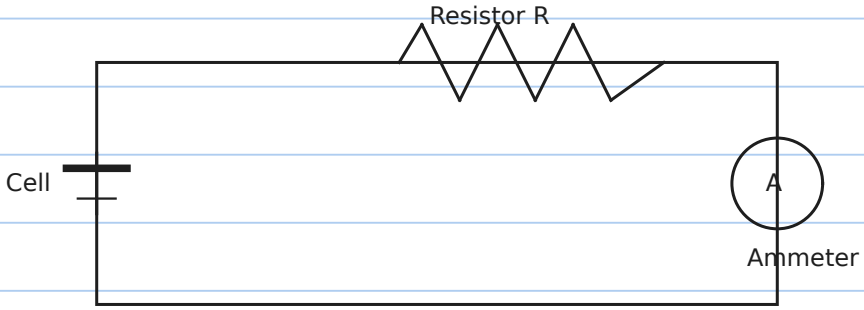
Step 2: C = 50 / 5 = 10

Final answer: 10 F

Exercise — Section (B): Structured Questions — ERQs

Question: Calculate combined capacitance of capacitors in series and in parallel.

Answer: For capacitors in series the charge on each is the same and the total voltage is the sum of the separate voltages, which leads to 1/C = 1/C1 + 1/C2 + 1/C3 + ... , so the combined capacitance is less than the smallest individual capacitance. For capacitors in parallel the voltage across each is the same and the total charge is the sum of the separate charges, giving C = C1 + C2 + C3 + ... As a check, 10 uF and 15 uF give 6 uF in series and 25 uF in parallel.



Question: Prove that energy stored in a capacitor is E = (1/2) C V^2.

Answer: While charging, work must be done against the repulsion of the charge already on the plates. At an intermediate stage with charge q the potential difference is q/C, so the work to add dq is dW = (q/C) dq. Integrating from 0 to Q gives total work W = Q^2/(2C), and this work is

stored as potential energy in the field. Substituting Q = CV gives E = (1/2) C V^2, which can also be written as (1/2) Q V.

Question: Describe the factors affecting the capacitance of a parallel plate capacitor.

Answer: Capacitance increases in direct proportion to the overlapping area of the plates, because a larger area can hold more charge at the same voltage. It increases when a dielectric of high relative permittivity is inserted, since the dielectric polarises and reduces the effective field for the same charge. It decreases in inverse proportion to the plate separation, because a wider gap gives a smaller field and less induced charge on the opposite plate. These factors are summarised in C = epsilon_0 epsilon_r A / d.

Exercise — Numericals

Question: The capacitance of air-filled parallel-plate capacitor is 1.3 pF. If the separation of the plates is doubled and wax is inserted between them, the new capacitance is 2.6 pF. Find the dielectric constant of the wax.

Answer: Given: C_0 = 1.3 pF, separation doubled, C_new = 2.6 pF

Step 1: Doubling the separation alone would halve the capacitance to C_0/2, and inserting wax multiplies it by epsilon_r, so C_new = epsilon_r x C_0 / 2

Step 2: epsilon_r = 2 x C_new / C_0 = 2 x 2.6 / 1.3

Step 3: epsilon_r = 4.0

Final answer: dielectric constant of wax = 4.0

Question: Three capacitors have capacitances 10, 50 and 25 uF

respectively as shown in figure. Calculate (i) charge on each when connected in parallel to a 250 V supply (ii) total capacitance and (iii) potential difference across each when connected in series.

Answer: Given: C1 = 10 uF, C2 = 50 uF, C3 = 25 uF, V = 250 V

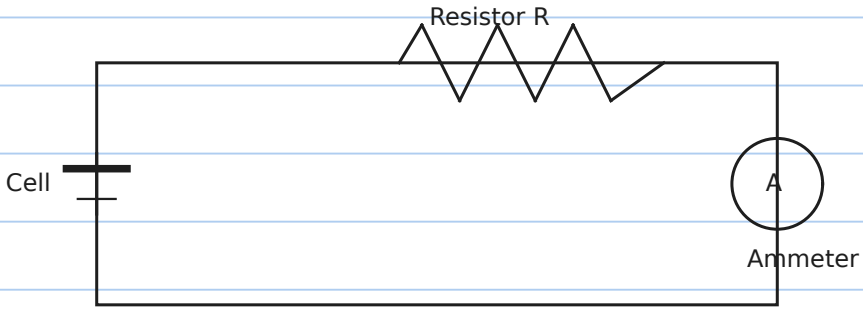
Step 1: In parallel each capacitor has 250 V, so Q1 = C1 V = 10 x 10^-6 x 250 = 2.5 mC, Q2 = 50 x 10^-6 x 250 = 12.5 mC, Q3 = 25 x 10^-6 x 250 = 6.25 mC

Step 2: Total parallel capacitance C = 10 + 50 + 25 = 85 uF

Step 3: In series, 1/C_s = 1/10 + 1/50 + 1/25 = 0.16, so C_s = 6.25 uF and the common charge is Q = C_s V = 6.25 x 10^-6 x 250 = 1.5625 mC

Step 4: Voltages in series: V1 = Q/C1 = 156.25 V, V2 = Q/C2 = 31.25 V, V3 = Q/C3 = 62.5 V, which add to 250 V

Final answer: (i) 2.5 mC, 12.5 mC, 6.25 mC (ii) 85 uF in parallel (6.25 uF in series) (iii) 156.25 V, 31.25 V, 62.5 V



Question: Three capacitors are connected through a potential difference of 100 volts as shown in figure (Figure for 9.10: C1 = 2 uF in series with the parallel combination of C2 = 3 uF and C3 = 5 uF). Find the charges and potential difference across each capacitor.

Answer: Given: C1 = 2 uF, C2 = 3 uF, C3 = 5 uF, total V = 100 V

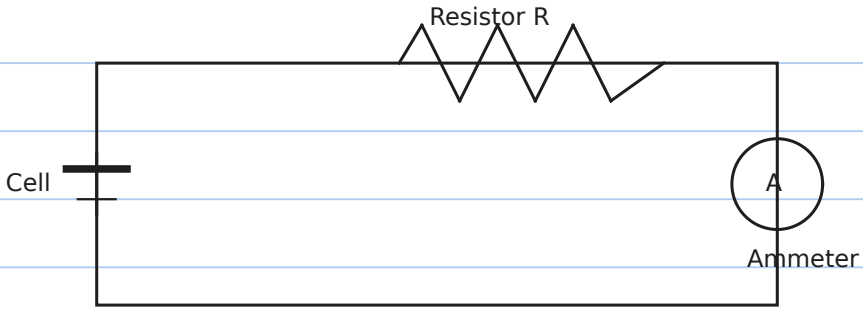
Step 1: C2 and C3 in parallel give C_23 = 3 + 5 = 8 uF

Step 2: C_23 in series with C1 gives 1/C = 1/2 + 1/8 = 0.625, so C = 1.6 uF

Step 3: Charge on C1 equals the total charge, Q1 = C V = 1.6 x 100 = 160 uC, so V1 = Q1/C1 = 160/2 = 80 V

Step 4: Voltage across the parallel pair V_23 = 100 - 80 = 20 V, so Q2 = 3 x 20 = 60 uC and Q3 = 5 x 20 = 100 uC

Final answer: Q1 = 160 uC, Q2 = 60 uC, Q3 = 100 uC, V1 = 80 V, V2 = V3 = 20 V



Question: Capacitor is charged through a large non-reactive resistance by a battery of constant voltage V. For this arrangement, if the capacitor has a capacitance of 10 uF and the resistance is 1 M ohm, calculate the time taken for the capacitor to receive 90% of its final charge.

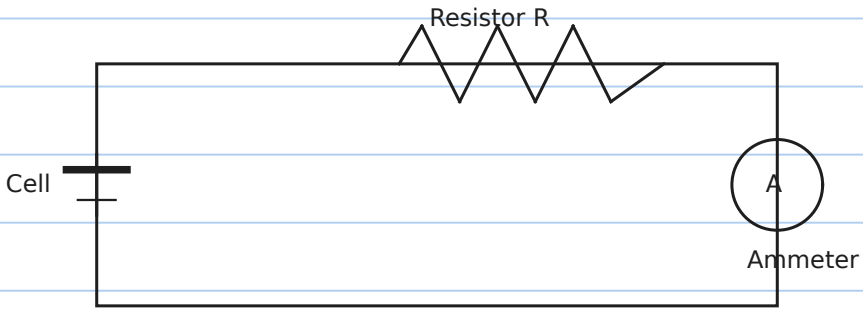
Answer: Given: C = 10 uF = 10 x 10^-6 F, R = 1 M ohm = 10^6 ohm, q = 0.90 Q_0

Step 1: Time constant tau = RC = 10^6 x 10 x 10^-6 = 10 s

Step 2: Charging follows q = Q_0 (1 - exp(-t/tau)), so 0.90 = 1 - exp(-t/10) and exp(-t/10) = 0.10

Step 3: -t/10 = ln 0.10 = -2.303, so t = 10 x 2.303 = 23.03 s

Final answer: about 23 s



Question: What capacitance is required to store energy of 10 kWh at a potential difference of 1000 V?

Answer: Given: E = 10 kWh = 10 x 3.6 x 10^6 = 3.6 x 10^7 J, V = 1000 V

Step 1: Stored energy E = (1/2) C V^2, so C = 2E / V^2

Step 2: C = 2 x 3.6 x 10^7 / (1000)^2 = 7.2 x 10^7 / 10^6

Step 3: C = 72 F

Final answer: 72 F

Question: A 2.0 uF capacitor and a 4.0 uF capacitor are connected in parallel across a 300 V potential difference. Calculate the total energy stored in the capacitors.

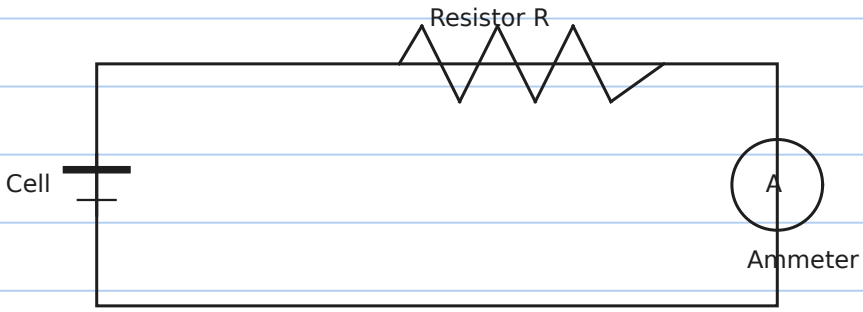
Answer: Given: C1 = 2.0 uF, C2 = 4.0 uF, V = 300 V

Step 1: Parallel combination C = 2.0 + 4.0 = 6.0 uF

Step 2: Total energy E = (1/2) C V^2 = 0.5 x 6.0 x 10^-6 x (300)^2

Step 3: E = 0.5 x 6.0 x 10^-6 x 9 x 10^4 = 0.27 J

Final answer: 0.27 J



Question: A 12.0-V battery is connected to a capacitor, resulting in 54.0 uC of charge stored on the capacitor. How much energy is stored in the capacitor?

Answer: Given: V = 12.0 V, Q = 54.0 uC = 54.0 x 10^-6 C

Step 1: Stored energy E = (1/2) Q V

Step 2: E = 0.5 x 54.0 x 10^-6 x 12.0 = 3.24 x 10^-4 J

Final answer: 3.24 x 10^-4 J

Unit 10: D.C Circuits

Exercise — Section (A): Multiple Choice Questions (MCQs)

Kirchhoff's laws are useful in determining:

✓ (a) current flowing in a circuit

(b) emf and voltage drops in a circuit

(c) power in a circuit

(d) only emf in a circuit

Answer: (a) Working: Kirchhoff's current law (junction rule) and voltage law (loop rule) give enough equations to solve for the unknown branch currents in a network. Once the currents are known, voltage drops and powers can also be found, but determining the currents is their main use.

The resistance of a superconductor is:

(a) finite

(b) infinite

(c) changes with every conductor

✓ (d) zero

Answer: (d) Working: Below its critical temperature a superconductor loses all electrical resistance, so current can flow through it without any loss of energy. Its resistivity also falls to zero in the superconducting state.

Reciprocal of resistance is called:

✓ (a) conductance

(b) resistivity

(c) resonance

(d) capacitance

Answer: (a) Working: Conductance G = 1/R measures how easily a conductor allows current to pass, and its unit is the siemens. A small resistance therefore means a large conductance.

The graphical representation of Ohm's law is:

(a) parabola

(b) hyperbola

(c) ellipse

✓ (d) straight line

Answer: (d) Working: Ohm's law states V = IR, so for a fixed resistance the graph of V against I is a straight line through the origin. The slope of this line gives the resistance R.

A potential difference is applied across the ends of a wire. If the potential difference is doubled, then the drift velocity of free electrons will:

(a) be quadrupled

✓ (b) be doubled

(c) be halved

(d) remain unchanged

Answer: (b) Working: Drift velocity is proportional to the electric field in the wire, and doubling the potential difference doubles the field for the same length. The average drift speed of the electrons therefore doubles, and the current also doubles.

Internal resistance is the resistance offered by:

(a) Capacitor

(b) resistor

(c) Conductor

✓ (d) source of emf

Answer: (d) Working: Every real source, such as a cell or battery, has

materials inside it that oppose the flow of charge. This opposition is its internal resistance r, which causes the terminal voltage to fall below the emf when current is drawn.

Question: Power dissipation in a resistor can be calculated using which formula: (a) P = V^2 / R (b) P = I^2 x R (c) P = V x I (d) P = R / (V x I)

Answer: (b) P = I^2 x R. Working: Electrical power is P = V I, and using Ohm's law V = I R this becomes P = I^2 R, the Joule heating formula. Note options (a) and (c) are also correct forms of power, but I^2 R is the direct expression for power dissipated in a resistor in terms of its own current and resistance.

What is a potentiometer primarily used for?

(a) Measuring electric current

(b) Measuring electric charge

(d) Measuring electric resistance

Answer: (c) Measuring potential difference (voltage). Working: A potentiometer compares an unknown potential difference with a known potential gradient along a uniform wire at a balance point where no current is drawn. It is therefore used to measure pd and emf very accurately.

A heat-sensitive device whose resistivity changes with the change in temperature is called:

(a) conductor

(b) resistor



(c) thermistor

(d) thermometer

Answer: (c) Working: A thermistor is a semiconductor device whose resistance changes rapidly and in a known way with temperature. This property makes it useful in temperature sensors, thermostats and fire

alarms.

A wire of uniform area of cross-section A, length L and resistance R is cut into two parts. The resistivity of each part:

(a) Becomes zero

(b) is halved

(c) Is doubled

(d) remains same

Answer: (d) Working: Resistivity is a property of the material and temperature only, not of the size or shape of the piece. Cutting the wire halves the resistance of each part, but the resistivity of each part stays exactly the same.

Exercise — CRQs

Question: Why is the terminal voltage of a cell less than its emf?

Answer: The cell has internal resistance r, so when current I flows, a voltage I r is dropped inside the cell itself. The terminal voltage available to the external circuit is therefore V = E - I r, which is less than the emf E. On open circuit no current flows, I r becomes zero, and the terminal voltage then equals the emf.

Question: Why is a potentiometer preferred over a voltmeter for determining the emf of a cell?

Answer: At the balance point a potentiometer draws no current from the cell being measured, so there is no voltage drop across the cell's internal resistance and the full emf is measured. A voltmeter always draws some current, so it measures the terminal voltage V = E - I r, which is smaller than the true emf. The potentiometer is therefore both more accurate and

effectively has infinite resistance at balance.

Question: Nichrome and copper wires of same length and same radius are connected in series. Current I is passed through them. Which wire gets heated up more? Justify your answer.

Answer: The nichrome wire gets heated more. In series the same current passes through both wires, and heat produced per second is P = I^2 R, so the wire with the larger resistance heats more. Nichrome has a much higher resistivity than copper, and with equal length and radius its resistance R = rho L / A is much larger, so it produces much more Joule heat.

Question: Explain why the terminal potential of a battery decreases when the current drawn from it is increased?

Answer: The terminal voltage is V = E - I r, where r is the internal resistance of the battery. As the current I drawn increases, the voltage lost inside the battery, I r, increases in proportion. A larger share of the emf is then used up inside the battery itself, so the potential available at the terminals falls.

Question: What are thermistors? Write their importance.

Answer: A thermistor is a heat-sensitive semiconductor resistor whose resistance changes markedly with temperature; in the common NTC type the resistance falls rapidly as temperature rises. Because of this sharp, predictable change, thermistors are used as temperature sensors and controllers. They are important in fire alarms, thermostats, electronic thermometers and in protecting circuits from overheating.

Question: State Kirchhoff's Laws.

Answer: Kirchhoff's current law states that the algebraic sum of currents at a junction is zero, meaning the total current entering a junction equals the total current leaving it; it is based on conservation of charge. Kirchhoff's

voltage law states that the algebraic sum of potential changes around any closed loop is zero, so the sum of emfs in a loop equals the sum of the voltage drops. This second law is based on conservation of energy.

Question: If Copper and Aluminum wires of the same length have same resistance, which has the larger diameter? And why?

Answer: The aluminium wire has the larger diameter. Resistance is R = rho L / A, so for the same R and L a material of larger resistivity needs a larger cross-sectional area A. Aluminium has a higher resistivity than copper (about 2.8 x 10^-8 ohm m against 1.7 x 10^-8 ohm m), so the aluminium wire must be thicker to keep the same resistance.

Question: What is the difference between potential difference and emf?

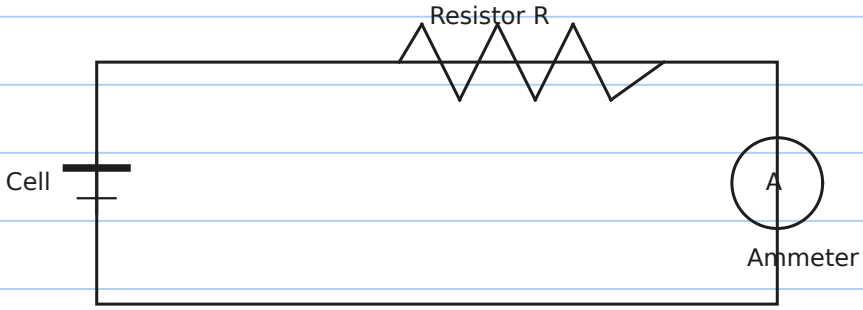
Answer: Emf is the total energy supplied by a source per unit charge, E = W/Q, including the energy used against the internal resistance of the source. Potential difference is the energy transferred per unit charge between two points in a circuit, for example across a resistor, V = W/Q for that part of the circuit. Emf is a property of the source and exists even on open circuit, while potential difference appears across components when the circuit is working; for a source supplying current the terminal pd is less than its emf.

Exercise — ERQs

Question: State the principle of working of a meter bridge. Explain how it is used to find an unknown resistance.

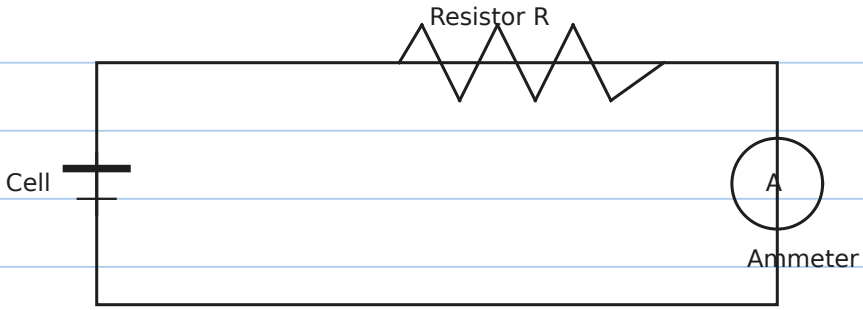
Answer: A meter bridge works on the principle of the Wheatstone bridge: at balance, the ratio of resistances in the two arms equals the ratio of the lengths of the bridge wire on the two sides of the balance point. The

unknown resistance X and a known resistance box are placed in the two gaps, and the sliding contact is moved until the galvanometer shows zero deflection at length l from one end. Then X / R = l / (100 - l), so X = R x l / (100 - l), giving the unknown resistance from a length measurement.



Question: Define thermoelectricity. Explain the working of a Fire Alarm system using thermistor.

Answer: Thermoelectricity refers to effects in which heat and electricity are directly linked, such as resistance changing with temperature in a thermistor and the Seebeck effect in thermocouples. In a thermistor fire alarm, the thermistor forms part of a potential divider in a transistor or comparator circuit at normal temperature, keeping the alarm off. When a fire raises the temperature, the resistance of an NTC thermistor falls sharply, the divider voltage crosses the trigger level, and the circuit switches on a bell or buzzer to warn of the fire.



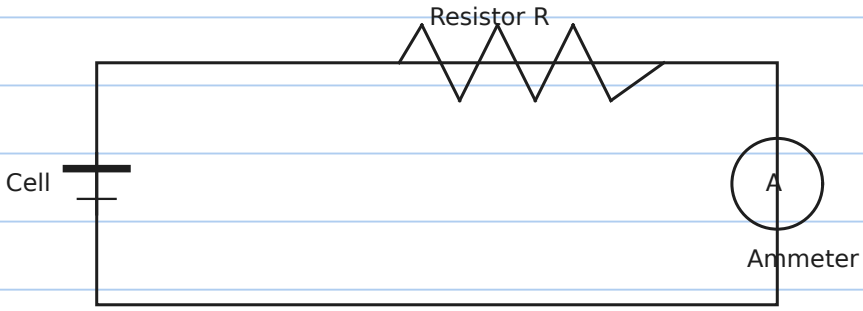
Question: Define resistivity. Explain dependence of resistivity on temperature.

Answer: Resistivity rho is the resistance of a specimen of unit length and

unit cross-sectional area, R = rho L / A, with unit ohm m; it depends only on the material and its temperature. For metals, resistivity increases almost linearly with temperature, rho_t = rho_0 (1 + alpha theta), because hotter ions vibrate more and scatter the electrons more often. For semiconductors and insulators resistivity decreases with temperature, since heating frees many more charge carriers.

Question: What is a rheostat? How can we use a rheostat as a potential divider?

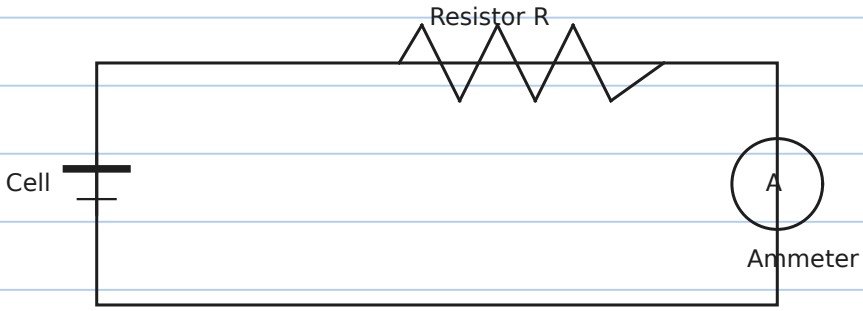
Answer: A rheostat is a variable resistor, usually a coil of resistance wire with a sliding contact, used to control current in a circuit. To use it as a potential divider, the supply is connected across the two fixed ends of the resistance, and the output is taken between one end and the sliding contact. Moving the slider changes the fraction of the resistance in the output part, so the output voltage can be varied smoothly from zero up to the full supply voltage.



Question: State the underlying principle of a potentiometer. Describe briefly, giving the necessary circuit diagram, how a potentiometer is used to measure the internal resistance of a given cell?

Answer: The principle is that the potential drop along a uniform wire carrying a constant current is directly proportional to the length, V proportional to l. To find internal resistance, first balance the cell on open circuit at length l1, so E = k l1, where k is the potential gradient. Then connect a known resistance R across the cell, draw current, and balance

again at length l2, giving the terminal voltage V = k l2. Since V = E R / (R + r), dividing gives E/V = l1/l2 = (R + r)/R, so r = R (l1/l2 - 1).



Exercise — Numericals

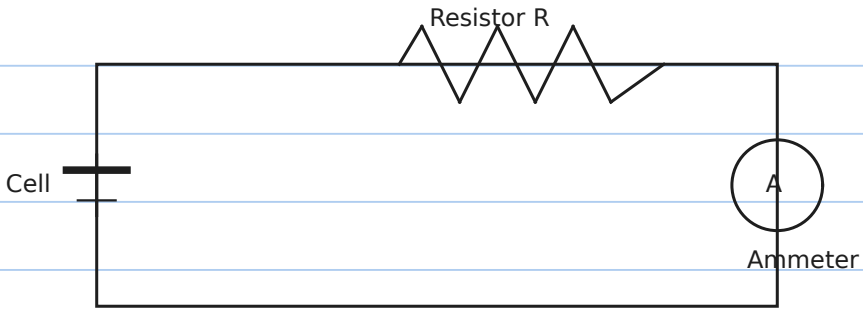
Question: The storage battery of a car has an emf of 12 V. If the internal resistance of the battery is 0.5 ohm, what is the maximum current that can be drawn from the battery?

Answer: Given: E = 12 V, r = 0.5 ohm

Step 1: Current is maximum when the external resistance is zero (short circuit), so I_max = E / r

Step 2: I_max = 12 / 0.5 = 24 A

Final answer: 24 A



Question: A negligibly small current is passed through a wire of length 12 m and uniform cross-section 4.0 x 10^-7 m^2, and its resistance is

measured to be 6.0 ohm. What is the resistivity of the material at the temperature of the experiment?

Answer: Given: L = 12 m, A = 4.0 x 10^-7 m^2, R = 6.0 ohm

Step 1: R = rho L / A, so rho = R A / L

Step 2: rho = 6.0 x 4.0 x 10^-7 / 12

Step 3: rho = 24 x 10^-7 / 12 = 2 x 10^-7 ohm m

Final answer: 2 x 10^-7 ohm m

Question: In a potentiometer arrangement, a cell of emf 1.20 V gives a balance point at 40.0 cm length of the wire. If the cell is replaced by another cell and the balance point shifts to 74.0 cm, what is the emf of the second cell?

Answer: Given: E1 = 1.20 V, l1 = 40.0 cm, l2 = 74.0 cm

Step 1: For a potentiometer, emf is proportional to balance length, E2 / E1 = l2 / l1

Step 2: E2 = E1 x l2 / l1 = 1.20 x 74.0 / 40.0

Step 3: E2 = 1.20 x 1.85 = 2.22 V

Final answer: 2.22 V

Question: (a) Three resistors 1 ohm, 2 ohm, and 3 ohm are combined in series. What is the total resistance of the combination? (b) If the combination is connected to a battery of emf 24 V and negligible internal resistance, obtain the potential drop across each resistor.

Answer: Given: R1 = 1 ohm, R2 = 2 ohm, R3 = 3 ohm in series, E = 24 V

Step 1: Total resistance R = 1 + 2 + 3 = 6 ohm

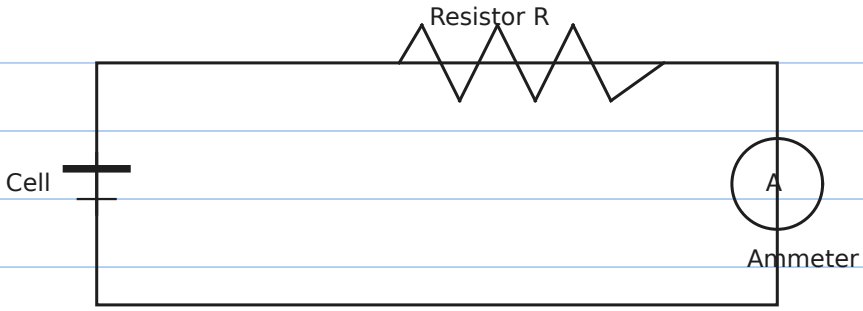
Step 2: Circuit current I = E / R = 24 / 6 = 4 A, the same through each

resistor in series

Step 3: Potential drops V1 = I R1 = 4 x 1 = 4 V, V2 = 4 x 2 = 8 V, V3 = 4 x 3 = 12 V

Step 4: Check: 4 + 8 + 12 = 24 V, equal to the emf

Final answer: (a) 6 ohm (b) V1 = 4 V, V2 = 8 V, V3 = 12 V



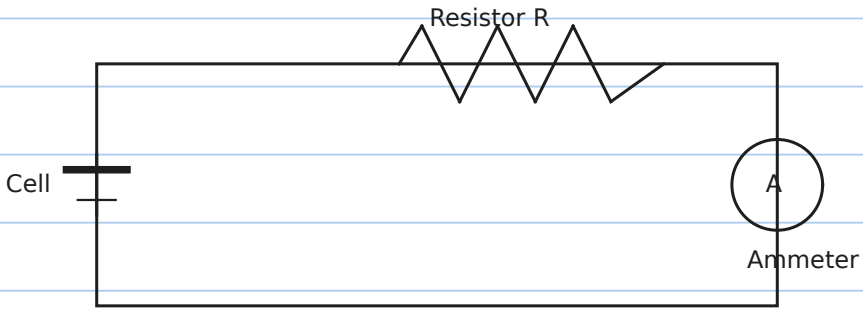
Question: From the given circuit find the value of I. (Junction figure: currents 0.4 A, 0.2 A, 0.6 A, 0.5 A and 0.7 A are shown in the other branches meeting at one junction, with current I in the remaining branch.)

Answer: Given: branch currents at the junction are 0.4 A, 0.2 A, 0.6 A, 0.5 A, 0.7 A and the unknown current I

Step 1: Apply Kirchhoff's current law: sum of currents entering the junction = sum of currents leaving the junction

Step 2: Adding the entering currents and subtracting those leaving, as shown by the arrows in the figure, the balance gives I = 0.6 A

Final answer: I = 0.6 A, in the direction shown in the figure



Question: In a meter bridge with a standard resistance of 15 ohm in the right gap, the ratio of balancing length is 5:3. Find the value of the other resistance.

Answer: Given: standard resistance Q = 15 ohm in the right gap, ratio of balancing lengths P side : Q side = 5 : 3

Step 1: Meter bridge balance condition: P / Q = l1 / l2

Step 2: P = Q x (l1 / l2) = 15 x (5 / 3)

Step 3: P = 15 x 1.667 = 25 ohm

Final answer: P = 25 ohm

Question: By using KVL find Current flowing through 10 ohm resistance. (Circuit figure: 4 V source with 5 ohm and 15 ohm in the left mesh, 10 ohm and 8 ohm in the middle mesh, and 12 ohm with a 6 V source in the right mesh, with mesh currents I1, I2 and I3 marked clockwise.)

Answer: Given: left mesh contains 4 V source, 5 ohm and shared 15 ohm; middle mesh contains 10 ohm and shared 8 ohm; right mesh contains 12 ohm and 6 V source; mesh currents I1, I2, I3 are clockwise

Step 1: Write KVL for mesh 1: 4 = 5 I1 + 15 (I1 - I2)

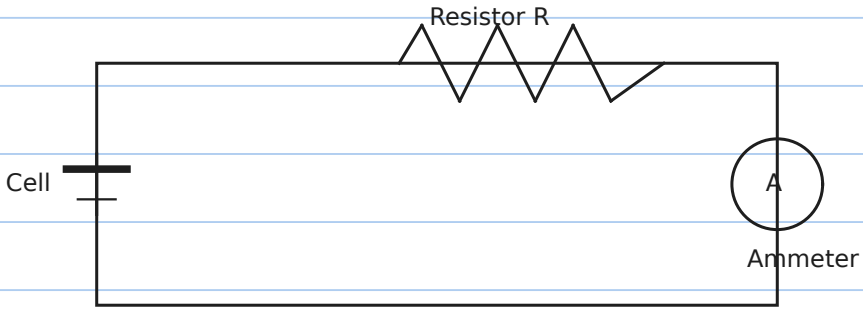
Step 2: Write KVL for mesh 2: 0 = 15 (I2 - I1) + 10 I2 + 8 (I2 - I3)

Step 3: Write KVL for mesh 3, taking care of the sign of the 6 V source: 0 = 8 (I3 - I2) + 12 I3 - 6, with the source opposing the left source

Step 4: Solving the three simultaneous equations gives I1 = 0.22 A, I2 = 0.034 A, I3 = -0.29 A (signs show the right mesh current is opposite to the assumed sense)

Step 5: The 10 ohm resistor lies only in mesh 2, so it carries I2

Final answer: I_10 ohm = 0.0341 A (clockwise)



Unit 11: Oscillations

Exercise — Section (A): Multiple Choice Questions (MCQs)

Two simple pendulums A and B with same lengths, and equal amplitude of vibrations, but the mass of A is twice the mass of B, their period are T_A and T_B and energies are E_A and E_B respectively. Choose the correct statement.

✓ (a) T_A = T_B and E_A > E_B

(b) T_A < T_B and E_A > E_B

(c) T_A > T_B and E_A < E_B

(d) T_A = T_B and E_A < E_B

Answer: (a) Working: The period T = 2 pi sqrt(L/g) does not depend on mass, so both pendulums with the same length have equal periods. The energy at the same amplitude is proportional to the mass (E proportional to m omega^2 A^2), so the heavier pendulum A stores more energy.

In order to double the period of a simple pendulum:

(a) Its length should be doubled

✓ (b) Its length should be quadrupled

(c) The mass should be doubled

(d) The mass should be quadrupled

Answer: (b) Working: Since T = 2 pi sqrt(L/g), the period is proportional to sqrt(L). To make T twice as large, sqrt(L) must double, so the length must become four times its original value; changing the mass has no effect.

A simple harmonic oscillator has amplitude A and time period t. Its

maximum speed is:

(a) 4A/t

(b) 2A/t

(c) 4 pi A/t

✓ (d) 2 pi A/t

Answer: (d) Working: Maximum speed in SHM is v_max = omega A, and omega = 2 pi / T. With the period written as t, v_max = 2 pi A / t, reached as the oscillator passes through its equilibrium position.

A spring attached by a load of weight W is vibrating with a period T. If the spring is divided in four equal parts and the same load is suspended from one of these parts, the new time period is:

(a) T/4

(b) 2T

✓ (c) T/2

(d) 4T

Answer: (c) Working: Cutting a spring into four equal parts makes each part four times stiffer, so the new spring constant is 4k. Since T = 2 pi sqrt(m/k), the new period is T' = T / sqrt(4) = T/2 for the same load.

The total energy of a particle executing simple harmonic motion is proportional to:

(a) frequency of oscillation

(b) maximum velocity of motion

(c) amplitude of motion

✓ (d) square of amplitude of motion

Answer: (d) Working: The total energy is E = (1/2) m omega^2 A^2, so for a given oscillator it depends on the square of the amplitude. Doubling the amplitude therefore makes the total energy four times larger.

A child swinging on a swing in sitting position, stands up, then the time

period of the swing will:

(a) Increase

✓ (b) decrease

(c) remains the same

(d) increases if the child is long and decreases if the child is short

Answer: (b) Working: A swing behaves like a pendulum whose effective length is the distance from the pivot to the centre of mass of the child. On standing up, the centre of mass rises, the effective length decreases, and since T is proportional to sqrt(L), the period decreases and the swing moves faster.

If a body oscillates at the angular frequency omega_d of the driving force, then the oscillations are called:

✓ (a) Forced oscillations

(b) Coupled oscillations

(c) Free oscillations

(d) Maintained oscillations

Answer: (a) Working: In forced oscillations an external periodic force drives the system, and after the starting transient dies away the body settles to oscillate at the frequency of the driving force, omega_d. Its own natural frequency no longer decides the frequency of the motion.

A simple harmonic oscillator with a natural frequency omega_N is forced to oscillate with a driving frequency omega_d. The Resonance occurred when:

(a) omega_N < omega_d

(b) omega_N > omega_d

✓ (c) omega_N = omega_d

(d) omega_N approx omega_d

Answer: (c) Working: Resonance is the sharp increase in amplitude that

occurs when the driving frequency equals the natural frequency of the system. At this condition energy is transferred most efficiently from the driver to the oscillator on every cycle.

In vehicles, shock absorber reduced the jerks:

(a) The shock absorber is the application of damped oscillations.

(b) Damping effect is due to the fractional loss of energy

(c) Shock absorbers in vehicles reduced jerk

✓ (d) All of these

Answer: (d) Working: A shock absorber is a practical application of damped oscillations, converting the spring's vibrational energy into heat through friction in oil. This fractional loss of energy on every cycle quickly kills the bouncing, so the vehicle settles smoothly and jerks are reduced.

A heavily damped system has a fairly flat resonance curve in:

(a) An acceleration time graph

✓ (b) An amplitude frequency graph

(c) Velocity time graph

(d) Distance-time graph

Answer: (b) Working: The resonance curve is a plot of amplitude against driving frequency. Heavy damping lowers and broadens the peak, making the amplitude-frequency curve flat, so the system responds with almost equal, small amplitude over a wide range of frequencies.

Exercise — Section (B): Structured Questions — CRQs

Question: Explain the concept of periodic motion to oscillatory motion. Discuss the terms period, frequency, and amplitude.

Answer: Periodic motion is any motion that repeats itself after a fixed

interval of time, such as the hands of a clock or the Earth going round the Sun. Oscillatory motion is a special periodic motion in which a body moves back and forth about a fixed mean position, like a pendulum bob or a mass on a spring. The period T is the time for one complete cycle, the frequency f = 1/T is the number of cycles per second measured in hertz, and the amplitude is the maximum displacement from the mean position.

Question: Explain the concept of phase and phase difference in oscillatory motion. Discuss how phase is related to the position and time in an oscillating system.

Answer: The phase of an oscillating body tells exactly where it is in its cycle at a given instant; for x = A sin(omega t + phi), the quantity (omega t + phi) is the phase and phi is the initial phase. Two oscillators have a phase difference when their phases differ, for example by pi radians when one is at its positive extreme while the other is at its negative extreme. Phase thus links position with time: knowing the phase fixes both the displacement and the direction of motion of the body.

Question: Explain the concept of damping and its effects on oscillatory motion. Discuss the types of damping, such as over-damping, under-damping, and critical-damping.

Answer: Damping is the gradual loss of energy of an oscillating system, usually to friction or air resistance, which causes the amplitude to decrease with time. In under-damping the system still oscillates, but with an amplitude that decays exponentially. In over-damping the resistance is so large that the system creeps back to equilibrium without oscillating at all, while in critical damping it returns to equilibrium in the shortest possible time without overshooting, as used in shock absorbers and galvanometers.

Question: Discuss the concept of resonance frequency and its relationship

to the natural frequency of an oscillating system.

Answer: Every system that can oscillate has a natural frequency at which it vibrates freely once disturbed. When an external periodic force drives the system at exactly this frequency, resonance occurs and the amplitude grows to a large value because energy is fed in at just the right moments. This special driving frequency is called the resonance frequency, and for light damping it is practically equal to the natural frequency of the system.

Question: Discuss the factors that affect the period of a simple pendulum. Explain how the length of the pendulum, the acceleration due to gravity, mass and the amplitude of oscillation influence the period.

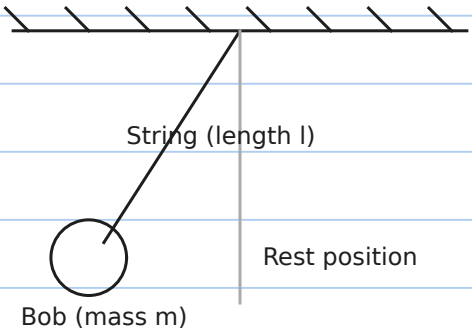
Answer: The period of a simple pendulum is T = 2 pi sqrt(L/g) for small oscillations. It increases as the square root of the length L, so a longer pendulum swings more slowly, and it decreases where g is larger, for example at the poles compared with a mountain top. The mass of the bob has no effect, because both the restoring force and the inertia increase in the same proportion. The amplitude also has no effect provided the swings are small, so the oscillations are isochronous.

Exercise — Section (B): Structured Questions — ERQs

Question: Define simple harmonic motion (SHM). Discuss the key characteristics of a system undergoing SHM.

Answer: Simple harmonic motion is an oscillatory motion in which the acceleration is directly proportional to the displacement from the mean position and is always directed towards that position, a = -omega^2 x. Its key characteristics are a fixed period and frequency independent of amplitude, a sinusoidal variation of displacement with time, maximum

speed at the mean position and zero speed at the extremes. The acceleration is zero at the mean position and maximum at the extremes, and the total mechanical energy remains constant if no damping acts.

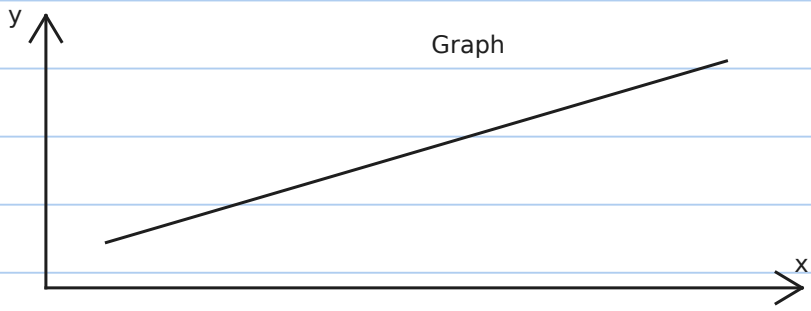


Question: Derive the equation of motion for a mass-spring system in SHM, illustrating each step of the derivation.

Answer: Let a mass m hang from a spring of constant k and be displaced by x from equilibrium. Step 1: by Hooke's law the spring exerts a restoring force F = -k x, the negative sign showing it opposes the displacement. Step 2: by Newton's second law F = m a = m d^2x/dt^2. Step 3: equating them, m d^2x/dt^2 = -k x, that is d^2x/dt^2 = -(k/m) x. This is the SHM equation with omega^2 = k/m, giving period T = 2 pi sqrt(m/k) and general solution x = A sin(omega t + phi).

Question: Discuss the concept of energy in SHM. Explain how kinetic energy and potential energy vary throughout the motion of a particle in SHM and how the total mechanical energy is conserved.

Answer: In SHM energy continually changes between kinetic and potential forms. At the mean position the displacement is zero, so potential energy is minimum (taken as zero) and kinetic energy is maximum, (1/2) m omega^2 A^2. At the extremes the particle is momentarily at rest, kinetic energy is zero and potential energy is maximum with the same value. At intermediate positions both forms are present, and their sum E = (1/2) m omega^2 A^2 stays constant throughout, provided friction and damping are absent.



Question: Discuss the concept of resonance in simple harmonic motion. Explain how resonance occurs and its effects on the amplitude and energy transfer in a driven oscillating system.

Answer: Resonance occurs in a driven system when the frequency of the driving force equals the natural frequency of the system. The driver then pushes in step with the motion every cycle, so energy is transferred to the system at the maximum possible rate. As a result the amplitude builds up to a large value, limited only by damping. Resonance is useful in tuning circuits and musical instruments, but it can be destructive in bridges and buildings if it is not controlled.

Question: Discuss the factors that affect the sharpness of resonance in an oscillatory system. Explain how damping and quality factor influence the width and peak of the resonance curve.

Answer: The sharpness of resonance is decided mainly by the amount of damping in the system. Light damping gives a tall, narrow resonance peak, because energy stays in the system for many cycles; heavy damping gives a low, broad and flat peak. The quality factor Q measures this sharpness: a high Q means a sharp, high peak and a small fractional energy loss per cycle, while a low Q means a broad response spread over many frequencies.

Question: Explain how the period of a mass-spring system can be independent of amplitude, even though the distance travelled during each cycle is proportional to amplitude.

Answer: For a larger amplitude the mass has farther to travel in each cycle, but Hooke's law gives a proportionally larger restoring force and hence a larger acceleration and speed, because a = -(k/m) x grows with x. The extra distance is therefore covered at a proportionally higher average speed, and the two effects cancel exactly. The period T = 2 pi sqrt(m/k) depends only on mass and spring constant, not on amplitude, which is the defining property of SHM.

Question: A mass hanging vertically from a spring and a simple pendulum both have a period of oscillation of 1 s on Earth. The two devices are sent to another planet, where gravitational field is stronger than that of Earth. For each of the two systems, state whether the period is now longer than 1 s, shorter than 1 s, or equal to 1 s. Explain your reasoning.

Answer: The mass-spring period T = 2 pi sqrt(m/k) does not contain g at all, because the spring force depends on extension, not on gravity; gravity only shifts the equilibrium position. Its period therefore stays equal to 1 s on the new planet. The pendulum period T = 2 pi sqrt(L/g) decreases when g increases, so on the planet with stronger gravity the pendulum swings faster and its period becomes shorter than 1 s.

Exercise — Numericals

Question: The period of oscillation of an object in an ideal spring and mass system is 0.50 s and the amplitude is 5.0 cm. What is the speed at the equilibrium point and the acceleration at the point of maximum extension of the spring?

Answer: Given: T = 0.50 s, A = 5.0 cm = 0.050 m

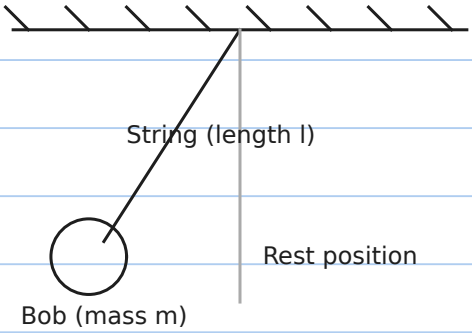
Step 1: Angular frequency omega = 2 pi / T = 2 pi / 0.50 = 12.57 rad/s

Step 2: Speed at equilibrium is the maximum speed, v_max = omega A = 12.57 x 0.050 = 0.628 m/s = 62.8 cm/s

Step 3: Acceleration at maximum extension is the maximum acceleration, a_max = omega^2 A = (12.57)^2 x 0.050

Step 4: a_max = 158 x 0.050 = 7.9 m/s^2

Final answer: speed at equilibrium = 62.8 cm/s, acceleration at maximum extension = 7.9 m/s^2



Question: A sewing machine needle moves with a rapid vibratory motion, like SHM, as it sews a seam. Suppose the needle moves 8.4 mm from its highest to its lowest position and it makes 24 stitches in 9.0 s. What is the maximum needle speed?

Answer: Given: total travel highest to lowest = 8.4 mm, 24 stitches in 9.0 s

Step 1: Amplitude A = 8.4 / 2 = 4.2 mm = 0.42 cm

Step 2: Frequency f = 24 / 9.0 = 2.67 Hz, so period T = 1/f = 0.375 s

Step 3: Maximum speed v_max = 2 pi f A = 2 pi x 2.67 x 0.42

Step 4: v_max = 7.0 cm/s

Final answer: 7.0 cm/s

Question: An ideal spring with a spring constant of 15 N/m is suspended vertically. A body of mass 0.60 kg is attached to the upstretched spring

and released. (a) What is the extension of the spring when the speed is a maximum? (b) What is the maximum speed?

Answer: Given: k = 15 N/m, m = 0.60 kg, g = 9.8 m/s^2

Step 1: Speed is maximum at the equilibrium position, where the spring force balances the weight, k x_0 = mg

Step 2: x_0 = mg / k = 0.60 x 9.8 / 15 = 0.39 m

Step 3: Released from the unstretched position, the body oscillates with amplitude A = x_0 = 0.39 m about this equilibrium

Step 4: Angular frequency omega = sqrt(k/m) = sqrt(15 / 0.60) = 5.0 rad/s, so v_max = omega A = 5.0 x 0.39 = 2.0 m/s

Final answer: (a) 0.39 m (b) 2.0 m/s

Question: A body is suspended vertically from an ideal spring of spring constant 2.5 N/m, the spring is initially in its relaxed position. The body is then released and oscillates about its equilibrium position. The motion is described by y = (4.0 cm) sin[(0.70 rad/s) t]. What is the maximum kinetic energy of the body?

Answer: Given: k = 2.5 N/m, amplitude A = 4.0 cm = 0.040 m (from the given equation of motion)

Step 1: Maximum kinetic energy equals the total mechanical energy of the oscillator

Step 2: E = (1/2) k A^2 = 0.5 x 2.5 x (0.040)^2

Step 3: E = 0.5 x 2.5 x 1.6 x 10^-3 = 2.0 x 10^-3 J

Final answer: 2.0 mJ

Question: The period of oscillation of a simple pendulum does not depend

on the mass of the bob. By contrast the period of a mass-spring system does depend on mass. Explain the apparent condition.

Answer: For the pendulum the restoring force comes from gravity and is proportional to the mass (component mg sin theta), while the inertia that resists acceleration is also proportional to mass, so mass cancels in a = F/m and T = 2 pi sqrt(L/g) has no mass term. For the spring, the restoring force F = -k x is fixed by the spring and the displacement and does not grow when a heavier mass is attached. The heavier mass therefore accelerates more slowly, the motion takes longer, and T = 2 pi sqrt(m/k) increases with mass. There is no contradiction; the difference lies in whether the restoring force itself depends on mass.

Question: What is the period of a simple pendulum of a 6.0 kg mass oscillating on a 4.0 m long string?

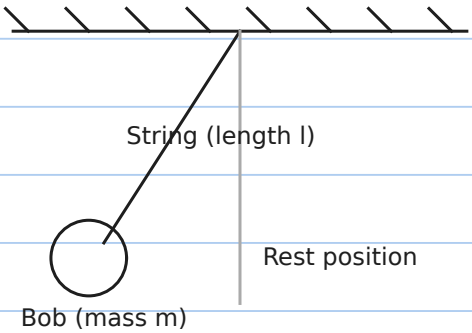
Answer: Given: L = 4.0 m, m = 6.0 kg (mass does not affect the period), g = 9.8 m/s^2

Step 1: T = 2 pi sqrt(L/g)

Step 2: T = 2 pi x sqrt(4.0 / 9.8) = 2 pi x sqrt(0.408)

Step 3: T = 2 pi x 0.639 = 4.01 s

Final answer: 4.01 s



Question: A pendulum of length 75 cm and mass 2.5 kg swings with a mechanical energy of 0.015 J. What is its amplitude?

Answer: Given: L = 75 cm = 0.75 m, m = 2.5 kg, E = 0.015 J

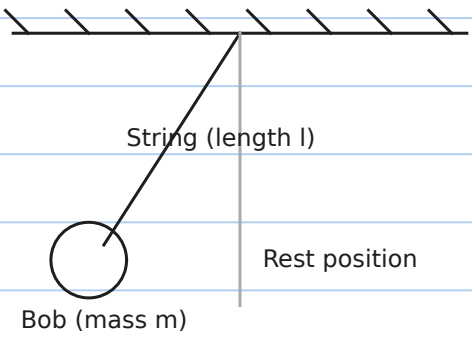
Step 1: For small oscillations the total energy is E = (1/2) m (g/L) A^2, where A is the arc amplitude

Step 2: Rearranging, A = sqrt(2 E L / (m g))

Step 3: A = sqrt(2 x 0.015 x 0.75 / (2.5 x 9.8)) = sqrt(0.0225 / 24.5) = sqrt(9.18 x 10^-4)

Step 4: A = 0.030 m = 3.0 cm

Final answer: 3.0 cm



Question: A pendulum of length L1 has a period of T1 = 0.950 s, the length of the pendulum is adjusted to a new value L2 such that T2 = 1.00 s. What is the ratio L2/L1?

Answer: Given: T1 = 0.950 s, T2 = 1.00 s

Step 1: Since T is proportional to sqrt(L), L2 / L1 = (T2 / T1)^2

Step 2: L2 / L1 = (1.00 / 0.950)^2 = (1.0526)^2

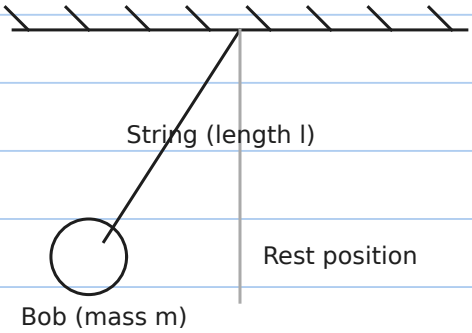
Step 3: L2 / L1 = 1.11

Final answer: 1.11

Question: A wire is hanging from the top of a tower such that the top is not visible due to darkness. How do you calculate the height of tower?

Answer: Attach a small heavy bob to the lower end of the wire so that the

wire and bob form a pendulum whose length is the height of the tower. Set it swinging with a small amplitude and time many complete oscillations to find the period T accurately. Then use T = 2 pi sqrt(L/g), giving L = g T^2 / (4 pi^2), and this length L is the height of the tower. Timing, say, 20 oscillations and dividing reduces the error in T.



Question: The amplitude of oscillation of a pendulum decays by a factor of 20.0 in 120 s. By what factor has its energy decayed in that time?

Answer: Given: amplitude falls by a factor of 20.0 in 120 s, so A_final = A_initial / 20

Step 1: Energy of an oscillator is proportional to the square of its amplitude, E proportional to A^2

Step 2: E_final / E_initial = (A_final / A_initial)^2 = (1/20)^2 = 1/400

Final answer: The energy has decreased by a factor of 400

Unit 12: Acoustics

Exercise — Section (A): Multiple Choice Questions (MCQs)

The speed v of a wave represented by y = A sin(omega t - kx) is:

(a) k/omega

✓ (b) omega/k

(c) omega k

(d) 1/omega k

Answer: (b) Working: for a wave y = A sin(omega t - kx), a point of constant phase satisfies omega t - kx = constant, so dx/dt = omega/k. Hence the wave speed is v = omega/k = f lambda.

Two sound waves are y = A sin(omega t - kx) and y = A cos(omega t - kx). The phase difference between the two waves is:

✓ (a) pi/2

(b) pi/4

(c) pi

(d) 0

Answer: (a) Working: cos(theta) can be written as sin(theta + pi/2), so the cosine wave leads the sine wave by pi/2 radians. The amplitudes, frequencies and wave numbers are the same, so pi/2 is the only phase difference between them.

If va, vh and vm are the speeds of sound in air, hydrogen gas, and a metal at the same temperature, then:

(a) va > vh > vm

(b) vm > vh > va

(c) vh > vm > va

(d) vh > va > vm

Answer: (b) Sound travels fastest in a metal because solids have a very large elasticity, even though their density is also high. Among gases, hydrogen has a much lower density than air at the same pressure, so sound travels faster in hydrogen than in air, giving vm > vh > va.

The speed of sound in air at STP is 332 m/s. If the air pressure becomes double at the same temperature, the speed of sound becomes:

(a) 1382 m/s

(b) 664 m/s

✓ (c) 332 m/s

(d) 166 m/s

Answer: (c) The speed of sound in a gas is v = sqrt(gamma P/rho). When pressure is doubled at constant temperature, the density rho also doubles, so the ratio P/rho stays unchanged and the speed remains 332 m/s.

How does the speed of sound v in air depend on the atmospheric pressure P:

✓ (a) v proportional to P^0

(b) v proportional to P^-1

(c) v proportional to P^2

(d) v proportional to P^1

Answer: (a) The exponent zero means the speed does not depend on pressure at all. This is because pressure and density change in the same proportion at constant temperature, so their ratio, which fixes the speed, is constant.

The speed of sound in a gas is proportional to:

(a) square root of isothermal elasticity ✓

(b) square root of adiabatic elasticity

(c) isothermal elasticity

(d) adiabatic elasticity

Answer: (b) Sound compressions and rarefactions happen so quickly that no heat is exchanged, so the process is adiabatic, not isothermal. Laplace corrected Newton's formula to v = sqrt(gamma P/rho), where gamma P is the adiabatic elasticity, so v is proportional to the square root of the adiabatic elasticity divided by density.

The length of a pipe closed at one end is L. In the standing wave whose frequency is 7 times the fundamental frequency, what is the closest distance between nodes?

Answer: (c) (2/7)L. Working: for a closed pipe the fundamental wavelength is 4L, and at 7 times the fundamental frequency the wavelength becomes 4L/7. In any standing wave, neighbouring nodes are half a wavelength apart, so the node spacing is (1/2)(4L/7) = (2/7)L.

A 620 Hz frequency song of an ice cream trolley approaches with speed v a boy standing at the door of his house; it is heard with frequency f1. If the trolley is stopped and the boy approaches the ice cream trolley with the same speed v, the boy now hears the sound with frequency f2. Choose the correct statement:

(a) f1 = f2; both are greater than 620 Hz

(b) f2 > f1 > 620 Hz

(c) f1 = f2; both are lesser than 620 Hz

(d) f1 > f2 > 620 Hz

Answer: (d) When the source moves, the wavelength itself is shortened, f1 = f v/(v - vs), which gives a larger increase. When only the observer moves, the wavelength is unchanged and f2 = f (v + vo)/v, which gives a smaller increase for the same speed. Both are above 620 Hz because the

distance is closing in both cases, so f1 > f2 > 620 Hz.

The speed of sound in a gas in which two waves of wavelength 50 cm and 50.4 cm produce 6 beats per second is:

(a) 338 m/s

(b) 350 m/s

✓ (c) 378 m/s

(d) 400 m/s

Answer: (c) Working: the beat frequency is |f1 - f2| = v(1/0.50 - 1/0.504) = v(0.004/0.252) = 0.01587 v. Setting this equal to 6 beats per second gives v = 6/0.01587 = about 378 m/s.

The speed of a wave in a medium is 760 m/s. If 3600 waves are passing through a point in the medium in 2 minutes, then its wavelength is:

(a) 13.85 m

✓ (b) 25.3 m

(c) 41.5 m

(d) 57.2 m

Answer: (b) Working: the frequency is f = 3600 waves / 120 s = 30 Hz. Using lambda = v/f = 760/30 = 25.3 m, so the wavelength is 25.3 m.

Exercise — CRQs

Question: Why can sound not travel through vacuum?

Answer: Sound is a mechanical wave, so it needs a material medium whose particles can vibrate and pass the disturbance on by compressions and rarefactions. In a vacuum there are no particles at all, so there is nothing to carry the vibrations from the source to the listener. This is why an electric bell ringing inside an evacuated jar cannot be heard, while

light, which is electromagnetic, crosses vacuum easily.

Question: On what factors does the speed of sound depend?

Answer: The speed of sound depends on the elasticity and the density of the medium, through v = sqrt(elastic modulus/rho), so it is much greater in solids than in gases. In a gas it also depends on temperature, rising by about 0.61 m/s for every 1 deg C rise, and on humidity, because moist air is less dense. It does not depend on pressure at constant temperature, nor on the frequency or amplitude of the sound.

Question: What are the conditions for the interference of waves?

Answer: The two waves must be coherent, meaning they have the same frequency and a constant phase difference between them. They should travel in the same or nearly the same direction and have equal or comparable amplitudes so that the maxima and minima are clearly visible. For light, the sources must also be monochromatic and derived from a single source so that the phase relation stays fixed.

Question: Is the energy of a wave maximum or minimum at nodes?

Answer: At a node the displacement of the particles is always zero, so the particles there never move and the kinetic energy at a node is minimum (in fact zero). The energy of a standing wave is concentrated at the antinodes, where the amplitude and particle speed are greatest. Energy in a standing wave in fact stays trapped between nodes instead of being transported along the medium.

Question: Why is it possible to understand the words spoken by two people at the same time?

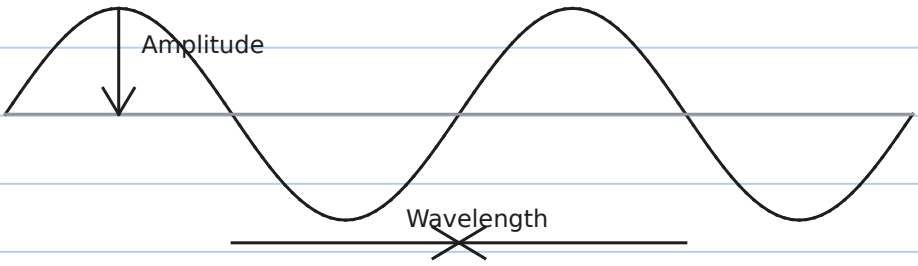
Answer: Sound waves obey the principle of superposition, so when two people speak at once their waves cross the same region of air without permanently disturbing or destroying each other. After crossing, each wave continues with its own frequency, pitch and quality unchanged, so

each voice keeps its own identity. Our ear and brain can then separate the two voices because their pitches and timbres are different.

Exercise — ERQs

Question: Define a sound wave and explain its nature as a longitudinal wave. Discuss the key properties of sound waves, such as frequency, wavelength, amplitude, and speed of propagation.

Answer: A sound wave is a mechanical disturbance produced by a vibrating body and carried through a material medium as a series of compressions and rarefactions. It is longitudinal because the particles of the medium vibrate back and forth parallel to the direction in which the wave travels, unlike transverse waves where the vibration is perpendicular. Its frequency is the number of complete vibrations per second and fixes the pitch, while its wavelength is the distance between two consecutive compressions. Its amplitude fixes the loudness or intensity of the sound, and its speed is given by v = f lambda and depends on the elasticity and density of the medium.

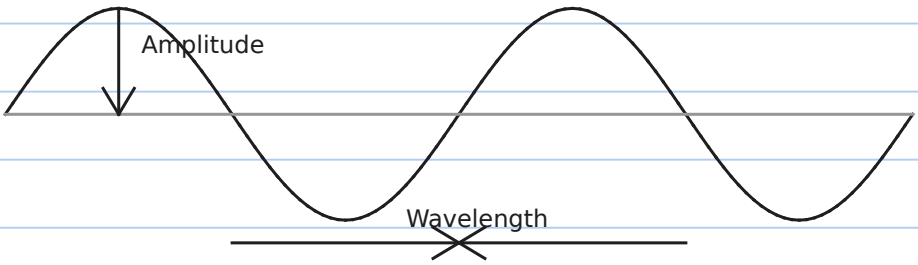


Question: Discuss the concept of the Doppler Effect in sound waves. Explain how the motion of a source or observer affects the perceived frequency and pitch of a sound wave. Provide examples to illustrate the Doppler Effect in daily life.

Answer: The Doppler Effect is the apparent change in the frequency of a sound when there is relative motion between the source and the observer. When the source or observer moves so that the distance between them decreases, the wavefronts are received closer together, so the observed frequency and pitch rise; when the distance increases, the observed frequency and pitch fall. A moving source actually changes the wavelength in the medium, whereas a moving observer only changes the rate at which wavefronts are met. A familiar example is the whistle of a passing train or the siren of an ambulance, which sounds higher in pitch while approaching and suddenly lower after it passes.

Question: What is a standing wave? How are they produced? Also elaborate the concept of nodes and antinodes in standing waves. Explain their locations and the relationship between node spacing and wavelength.

Answer: A standing wave is a wave pattern that does not travel forward but stays in a fixed region, formed when two waves of the same frequency, wavelength and amplitude travel in opposite directions and superpose. It is commonly produced when a wave reflects from a fixed end, a wall or an obstacle and overlaps its own reflection, as on a stretched string or in an air column. Points that always remain at rest are called nodes, while points that vibrate with maximum amplitude are called antinodes. Nodes and antinodes alternate along the medium, neighbouring nodes are lambda/2 apart, neighbouring antinodes are also lambda/2 apart, and a node and the nearest antinode are lambda/4 apart.



Question: Define the Doppler Effect and derive the Doppler Effect equation for sound waves in terms of the relative velocity between the source, observer, and the speed of sound.

Answer: The Doppler Effect is the change in the observed frequency of a wave caused by relative motion between the source and the observer. Let a source emit frequency fo in a medium where sound travels at speed v. If the observer moves towards the source with speed vo, the wave speed relative to the observer is v + vo, so the observed frequency is f = fo (v + vo)/v. If the source moves towards the observer with speed vs, the wavelength is shortened to lambda = (v - vs)/fo, giving f = fo v/(v - vs). Combining both motions gives the general equation f = fo (v + vo)/(v - vs), with signs reversed when the motion increases the separation.

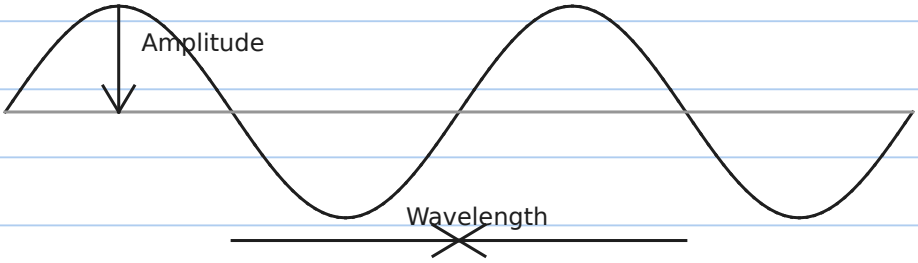
Question: Define standing waves and explain how they are formed.

Answer: A standing wave is a stationary wave pattern in which every point of the medium vibrates with a fixed amplitude that depends on its position, and no energy is carried forward. It is formed by the superposition of two identical waves travelling in opposite directions through the same medium at the same time. In practice this happens when a wave is reflected back on itself, for example at the fixed ends of a guitar string or at the closed end of a pipe. The result is alternate nodes of zero displacement and antinodes of maximum displacement spaced regularly along the medium.

Question: Discuss the concept of harmonics in standing waves. Explain

how harmonics are formed and their relationship with the fundamental frequency.

Answer: Harmonics are the higher modes of vibration of a system whose frequencies are whole-number multiples of the lowest possible frequency, called the fundamental or first harmonic. They are formed when the length of a string or air column fits a whole number of half-wavelengths (or odd numbers of quarter-wavelengths in a closed pipe), so standing waves of shorter wavelength can also be sustained. For a string fixed at both ends the harmonics are f, 2f, 3f and so on, while a pipe closed at one end gives only odd harmonics f, 3f, 5f. The mixture of harmonics present gives each instrument its characteristic quality or timbre.



Exercise — Numericals

Question: The equation of a wave is y(x, t) = 3.5 sin{ (pi/3.0) x - 66 t } cm where t is in seconds and x and y both are in cm. Find (a) the amplitude and (b) the wavelength of this wave.

Answer: Given: y(x, t) = 3.5 sin{ (pi/3.0) x - 66 t } cm

Step 1: Compare with the standard wave equation y = A sin(kx - omega t), so the amplitude is A = 3.5 cm and the wave number is k = pi/3.0 per cm

Step 2: Use lambda = 2 pi / k

Step 3: Substitute lambda = 2 pi / (pi/3.0) = 2 x 3.0 = 6.0 cm

Final answer: (a) amplitude = 3.5 cm, (b) wavelength = 6.0 cm

Question: Why is it that your own voice sounds strange to you when you hear it played back on a tape recorder, but your friends all agree that it is just what your voice sounds like?

Answer: When you speak, you hear your own voice by two routes at once: through the air to your ears, and through the bones of your skull directly to the inner ear. The bone-conduction route carries the lower frequencies strongly, so your own voice sounds deeper and fuller to you than it really is. A tape recorder records only the air-carried sound, which is exactly what your friends always hear. That is why the recording sounds correct to them but strange and thinner to you.

Question: Why is the speed of sound in solids much faster than the speed of sound in air?

Answer: The speed of sound is v = sqrt(elastic modulus/rho), so it grows with the stiffness of the medium. Solids have a very large elastic modulus because their particles are held close together by strong forces and pass a push on to their neighbours almost instantly. Although solids are also denser than air, their elasticity is larger by a far greater factor, so the ratio is much bigger. As a result sound travels at about 5000 m/s in steel compared with only about 343 m/s in air.

Question: An increase in pressure of 100 kPa causes a certain volume of water to decrease by 5 x 10^-3 percent of its original volume. Find (a) the bulk modulus of water. (b) What is the speed of sound in water?

Answer: Given: increase in pressure delta P = 100 kPa = 1.0 x 10^5 Pa, fractional decrease in volume delta V/V = 5 x 10^-3 percent = 5 x 10^-5, density of water rho = 1000 kg/m^3

Step 1: Bulk modulus B = delta P / (delta V/V)

Step 2: Substitute B = (1.0 x 10^5) / (5 x 10^-5) = 2.0 x 10^9 Pa = 2000 MPa

Step 3: Speed of sound in a liquid is v = sqrt(B/rho)

Step 4: Substitute v = sqrt(2.0 x 10^9 / 1000) = sqrt(2.0 x 10^6) = about 1414 m/s, i.e. about 1400 m/s

Final answer: (a) B = 2000 MPa, (b) speed of sound in water = about 1400 m/s

Question: A uniform string of length 10.0 m and weight 0.25 N is attached to the ceiling. A weight of 1.00 kN hangs from its lower end. The lower end of the string is suddenly displaced horizontally. How long does it take the resulting wave pulse to travel to the upper end? Neglect the weight of the string in comparison to the hanging mass.

Answer: Given: length L = 10.0 m, weight of string = 0.25 N, tension T = hanging weight = 1.00 kN = 1000 N

Step 1: Mass of string m = 0.25 / 9.8 = 0.0255 kg, so linear mass density mu = m/L = 0.0255/10.0 = 2.55 x 10^-3 kg/m

Step 2: Wave speed on a string is v = sqrt(T/mu)

Step 3: Substitute v = sqrt(1000 / 2.55 x 10^-3) = sqrt(3.92 x 10^5) = about 626 m/s

Step 4: Time = distance / speed = 10.0 / 626 = 0.016 s

Final answer: about 16 ms (approx)

Question: A travelling sine wave is the result of the superposition of two other sine waves with equal amplitudes, wavelengths, and frequencies.

The two component waves each have amplitude 5.00 cm. If the resultant wave has amplitude 6.69 cm, what is the phase difference phi between the component waves?

Answer: Given: amplitude of each component A = 5.00 cm, resultant amplitude AR = 6.69 cm

Step 1: For two equal waves with phase difference phi, AR = 2 A cos(phi/2)

Step 2: Substitute 6.69 = 2 x 5.00 x cos(phi/2), so cos(phi/2) = 6.69/10.00 = 0.669

Step 3: Take inverse cosine, phi/2 = cos^-1 (0.669) = 48 deg

Step 4: Therefore phi = 2 x 48 deg = 96 deg

Final answer: phase difference phi = 96 deg

Question: In order to decrease the fundamental frequency of a guitar string by 4.0%, by what percentage should you reduce the tension?

Answer: Given: new frequency f2 = f1 - 4.0 percent of f1 = 0.96 f1, string length and mass per unit length unchanged

Step 1: Fundamental frequency of a string is f = (1/2L) sqrt(T/mu), so f is proportional to sqrt(T)

Step 2: Therefore T2/T1 = (f2/f1)^2 = (0.96)^2 = 0.9216

Step 3: New tension T2 = 0.9216 T1, so the reduction is T1 - T2 = 0.0784 T1

Step 4: Percentage reduction = 0.0784 x 100 = 7.84 percent (note: the value 76.96 percent printed in the book is a misprint)

Final answer: reduce the tension by 7.84%

Question: A string 2.0 m long is held fixed at both ends. If a sharp blow is

applied to the string at its centre, it takes 0.050 s for the pulse to travel to both ends of the string and return to the middle. What is the fundamental frequency of oscillation for this string?

Answer: Given: string length L = 2.0 m, round-trip time from centre to the ends and back t = 0.050 s

Step 1: The pulse travels from the centre to an end and back, a distance of 2 x 1.0 m = 2.0 m

Step 2: Wave speed v = distance / time = 2.0 / 0.050 = 40 m/s

Step 3: For the fundamental mode of a string fixed at both ends, wavelength lambda = 2L = 4.0 m

Step 4: Fundamental frequency f = v / lambda = 40 / 4.0 = 10 Hz

Final answer: fundamental frequency = 10 Hz

Question: A sound source of frequency fo and an observer are located at a fixed distance apart. Both the source and observer are at rest. However, the propagation medium (through which the sound waves travel at speed v) is moving at a uniform velocity vm in an arbitrary direction. Find the frequency detected by the observer giving physical explanation.

Answer: Given: source frequency fo, source and observer both at rest, medium moving with velocity vm making an angle theta with the line joining source to observer

Step 1: The component of the medium velocity along the source-observer line is vm cos theta, so relative to the ground the wavefronts travel towards the observer at speed v + vm cos theta

Step 2: The source is at rest, so it still emits fo wavefronts per second, and in the moving medium these are spaced lambda = (v + vm cos theta)/fo

apart

Step 3: The stationary observer receives wavefronts at rate f = (speed of wavefronts)/(their spacing) = (v + vm cos theta) / lambda

Step 4: Substituting lambda gives f = (v + vm cos theta) x fo / (v + vm cos theta) = fo

Final answer: the detected frequency is fo, unchanged, because the moving medium carries the wavefronts faster (or slower) but spaces them out (or squeezes them) by exactly the same factor, so the rate at which they arrive at a fixed observer is the same as the rate at which they are emitted

Question: A train sounds its whistle while passing by a railroad crossing. An observer at the crossing measures a frequency of 219 Hz as the train approaches the crossing and a frequency of 184 Hz as the train leaves. The speed of the sound is 340 m/s. Find the speed of the train and frequency of its whistle.

Answer: Given: approach frequency fa = 219 Hz, receding frequency fr = 184 Hz, speed of sound v = 340 m/s, observer stationary

Step 1: For approach, fa = fo v/(v - vT), and for recession, fr = fo v/(v + vT), where vT is the train speed

Step 2: Divide the two equations: fa/fr = (v + vT)/(v - vT) = 219/184

Step 3: Cross-multiply: 219(340 - vT) = 184(340 + vT), so 219 x 340 - 184 x 340 = (219 + 184) vT

Step 4: Solve vT = 340 x (219 - 184)/(219 + 184) = 340 x 35/403 = 29.5 m/s

Step 5: Use fa = fo v/(v - vT) to get fo = fa (v - vT)/v = 219 x (340 -

29.5)/340 = 219 x 310.5/340 = 200 Hz

Final answer: speed of train vT = 29.5 m/s and whistle frequency fo = 200 Hz

Unit 13: Physical Optics

Exercise — Section (A): Multiple Choice Questions (MCQs)

If the wavelength of an electromagnetic wave is about the diameter of a cricket ball, what type of radiation is it?

(a) X-ray

(b) Ultraviolet

✓ (c) Radio waves

(d) Visible light

Answer: (c) A cricket ball has a diameter of about 7 cm, which is far longer than the wavelengths of X-rays, ultraviolet and visible light, which are all measured in nanometres. A wavelength of several centimetres lies in the short radio wave (microwave) region of the electromagnetic spectrum, so among the given choices it is a radio wave.

Electromagnetic waves from an unknown source in space are found to be diffracted when passing through gaps of the order of 10^-5 m, which type of wave are they most likely to be?

(a) microwaves

(b) Ultraviolet

(c) Radio waves

✓ (d) infra-red waves

Answer: (d) Noticeable diffraction occurs only when the gap is comparable with the wavelength, so the wavelength here is about 10^-5 m, that is about 10 micrometres. This wavelength lies in the infra-red region,

between visible light and microwaves, so the waves are infra-red waves.

Huygens's conception of secondary waves:

(a) helps us to find the focal length of a thick lens

✓ (b) is a geometrical method to find a wavefront

(c) is used to determine the velocity of light

(d) is used to explain the polarization of light

Answer: (b) According to Huygens, every point on a wavefront acts as a source of secondary wavelets, and the envelope of these wavelets after a short time gives the new position of the wavefront. It is therefore a geometrical construction for tracing how a wavefront advances, reflects and refracts.

Interference fringes are produced using monochromatic light of same intensity from a double slit screen. If the intensity of light emerging from one of the slit is reduced, the effect on interference pattern will be:

(a) All the dark and bright fringes become brighter

(b) All the dark and bright fringes become darker

(c) Bright fringes become brighter and dark fringes become darker

✓ (d) Bright fringes become darker and dark fringes become brighter

Answer: (d) With unequal amplitudes the two waves no longer cancel completely at the minima, so the dark fringes are no longer fully dark and appear brighter. At the maxima the two amplitudes add to a smaller total than before, so the bright fringes become less bright, and overall the contrast of the pattern decreases.

In Young's experiment when the distance between slits and screen is doubled, while separation of slits is halved, then fringe width will be:

✓ (a) 4 times

(b) 1/4 times

(c) doubled

(d) unchanged

Answer: (a) Fringe width is beta = lambda D/d, where D is the slit-to-screen distance and d is the slit separation. Doubling D doubles beta, and halving d doubles it again, so the new fringe width is 2 x 2 = 4 times the original.

A ray of light passes from air into water, striking the surface of the water with an angle of incidence 45 deg. Which of these quantities change as the light enters the water? i) Wavelength ii) frequency iii) speed of propagation iv) direction of propagation

(a) i and ii only

(b) iii and iv only

✓ (c) i, iii, and iv only

(d) all of them

Answer: (c) On entering the denser medium the light slows down, so its speed changes, and because the ray meets the surface at an angle it bends towards the normal, so its direction changes. Since v = f lambda and the frequency is fixed by the source and never changes at a boundary, the wavelength must decrease in the same ratio as the speed.

A hill separates a television (TV) transmitter from a house. The transmitter cannot be seen from the house but still the TV in the house has good reception. What wave phenomena make it possible?

(a) Coherence of waves

✓ (b) Diffraction of waves

(c) Interference of waves

(d) Refraction of waves

Answer: (b) TV waves have wavelengths of the order of metres, comparable with the size of obstacles, so they bend appreciably around the edges of the hill into the shadow region behind it. This spreading of

waves around an obstacle is diffraction, and it allows the signal to reach a house that has no straight line of sight to the transmitter.

Monochromatic light is incident on a diffraction grating and a diffraction pattern is observed. Which effect is observed by replacing the grating with one that has more lines per millimeter?

(a) Number of maxima decreases with decrease in angle between first and second order maxima

✓ (b) Number of maxima decreases with increase in angle between first and second order maxima

(c) Number of maxima increases with decrease in angle between first and second order maxima

(d) Number of maxima increases with increase in angle between first and second order maxima

Answer: (b) More lines per millimetre means a smaller grating spacing d, so from d sin theta = m lambda the diffraction angles theta for each order become larger and the first and second order maxima move further apart. Since sin theta cannot exceed 1, the maximum possible order m = d/lambda becomes smaller, so fewer maxima can be observed.

Optically active substances are those substances which:

(a) produce polarized light

✓ (b) rotate the plane of polarization of polarized light

(c) produce double refraction

(d) convert a plane polarized light into circularly polarized light

Answer: (b) When plane polarized light passes through an optically active substance such as a sugar solution or a quartz crystal, the plane in which the light vibrates is turned through an angle. The amount of rotation depends on the length of the substance and, for a solution, on its concentration, which is the basis of the polarimeter.

Plane polarized light is passed through a Polaroid. On viewing through the Polaroid we find that when Polaroid is given one complete rotation about the direction of light:

(a) The intensity of light gradually decreases to zero and remains at zero

(b) The intensity of light gradually increases to maximum and remains at maximum

(c) There is no change in the intensity of light

✓ (d) The intensity of light varies such that it is twice maximum and twice zero

Answer: (d) By Malus law the transmitted intensity is I = Io cos^2 theta, where theta is the angle between the plane of polarization and the Polaroid axis. In one full rotation theta passes twice through 0 deg, giving maximum intensity, and twice through 90 deg, giving zero intensity, so there are two maxima and two minima in each complete turn.

Exercise — CRQs

Question: In every day experience, visible light seems to travel in straight lines while radio waves do not. Explain.

Answer: Diffraction, the bending of waves around obstacles, is only noticeable when the size of the obstacle or opening is comparable with the wavelength of the wave. Visible light has a very tiny wavelength, about 5 x 10^-7 m, so its bending around everyday objects is far too small to notice and it appears to travel in straight lines, casting sharp shadows. Radio waves have wavelengths of metres or more, similar to the sizes of buildings and hills, so they diffract strongly around them and do not seem

to follow straight lines.

Question: Explain why two waves of significantly different frequencies cannot be coherent.

Answer: Coherent waves must maintain a constant phase difference with time, which is only possible if they have exactly the same frequency. If the frequencies differ, even slightly, the phase of one wave continually drifts relative to the other, so their phase difference keeps changing. Any interference pattern then shifts rapidly and averages out, so no steady pattern of maxima and minima can be observed.

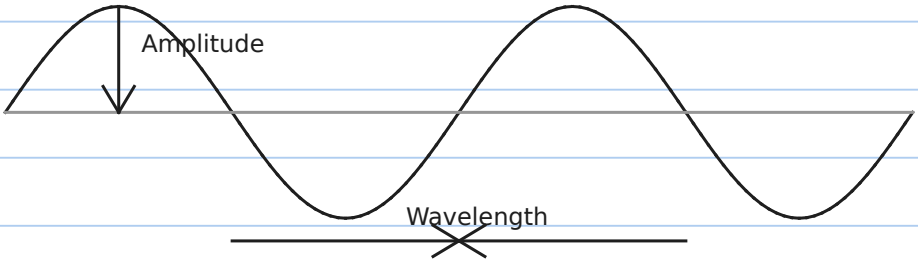
Question: In a Young's double slit experiment, how the interference phenomenon is affected by changing the slits separation and the distance between the slits and screen?

Answer: The fringe width is beta = lambda D/d, where d is the separation of the slits and D is the distance from the slits to the screen. Increasing the slit separation d makes the fringes narrower and more closely packed, while decreasing d spreads the fringes further apart. Increasing the slit-to-screen distance D makes the fringes wider, and decreasing D makes them narrower, while the brightness of the pattern also falls as the screen is moved away.

Question: Explain how the double-slit experiment provides evidence for the wave nature of light.

Answer: In Young's double-slit experiment a single light source illuminates two narrow slits, and instead of just two bright lines the screen shows a series of equal bright and dark bands called fringes. Such a pattern can only be produced by two overlapping waves that reinforce where they arrive in phase and cancel where they arrive out of phase, which is interference, a purely wave property. Two streams of particles could never cancel each other to produce dark bands, so the experiment is strong

evidence that light travels as a wave.



Question: Discuss the concept of monochromatic light in the context of Newton's rings.

Answer: Monochromatic light is light of a single wavelength or colour. In the Newton's rings experiment, monochromatic light falling on the air film between a convex lens and a glass plate produces clear, concentric bright and dark rings, because for a given film thickness the interference condition is satisfied at one definite radius for that one wavelength. If white light were used, each colour would form rings of a different size, the rings of different colours would overlap, and only a few coloured rings near the centre would be visible before the pattern washed out.

Question: Explain why monochromatic light is preferred for obtaining clear and well-defined interference patterns in the experiment.

Answer: Each wavelength of light produces its own interference pattern with its own fringe width, since beta depends on lambda. With white or multi-wavelength light, the patterns of all the colours fall on top of each other and overlap, so away from the centre the bright fringe of one colour falls on the dark fringe of another and the pattern becomes blurred and disappears. Monochromatic light gives a single pattern of sharp, high-contrast bright and dark fringes that can be observed and measured accurately.

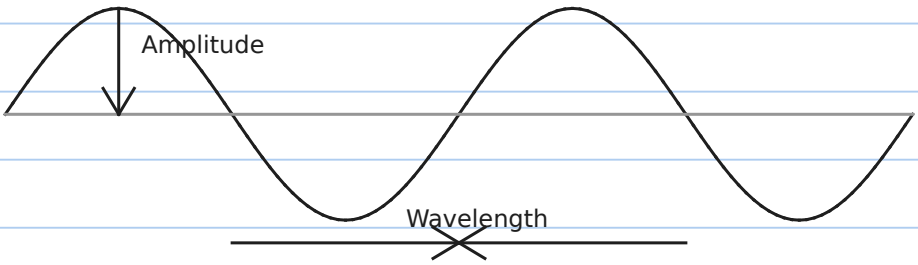
Question: Discuss how the interference of light waves leads to the formation of bright and dark rings in the experiment.

Answer: In Newton's rings, light reflects from the top and bottom surfaces of the thin air film between the lens and the plate, and these two reflected beams interfere. Where the film thickness makes the path difference an odd multiple of lambda/2 (allowing for the phase change of pi at the denser surface), the beams cancel and a dark ring forms; where the path difference is a whole multiple of lambda, they reinforce and a bright ring forms. Because the film thickness increases uniformly with distance from the point of contact, these conditions are met on circles, producing alternate bright and dark concentric rings with a dark centre.

Exercise — ERQs

Question: Explain the concept of interference in physical optics. Discuss constructive and destructive interference.

Answer: Interference is the phenomenon in which two coherent light waves superpose and produce a steady pattern in which the resultant intensity is not uniform but varies from place to place. Where the waves arrive in phase, with a path difference of a whole number of wavelengths, their amplitudes add up and the resultant intensity is maximum; this is constructive interference and produces a bright fringe. Where the waves arrive out of phase, with a path difference of an odd number of half-wavelengths, their amplitudes cancel and the intensity is minimum or zero; this is destructive interference and produces a dark fringe. Energy is not destroyed at the dark fringes but is redistributed, appearing in extra measure at the bright fringes.



Question: Discuss the conditions required for interference and provide examples of interference in daily life.

Answer: For observable interference the two sources must be coherent, that is of the same frequency with a constant phase difference, which in practice means they must be derived from a single source. The waves should have nearly equal amplitudes for good contrast, the sources must be close together so the fringes are wide enough to see, and for light the source should be monochromatic and narrow. Everyday examples include the brilliant colours seen on soap bubbles and thin oil films on wet roads, and the coloured rings seen when a lens rests on a glass plate. Anti-reflection coatings on camera lenses and spectacles also work by destructive interference of reflected light.

Question: Describe the setup and working principle of the Michelson interferometer. Also explain how the Michelson interferometer can be used to measure the wavelength of monochromatic light.

Answer: In the Michelson interferometer, light from a source falls on a half-silvered glass plate that splits it into two beams of equal intensity. One beam travels to a fixed mirror and the other to a movable mirror; both are reflected back, recombine at the plate, and interfere to form circular fringes in a telescope. Because each beam makes a double journey, moving the movable mirror by a distance d changes the path difference by 2d. Counting the number of fringes N that cross the field of view while the mirror moves a measured distance d gives the wavelength from lambda =

2d/N. This makes the instrument an extremely precise tool for measuring wavelengths and tiny lengths.

Question: Describe the setup and procedure of the diffraction of X-rays through a crystal experiment.

Answer: In this experiment a narrow beam of X-rays from an X-ray tube is collimated and directed onto the face of a single crystal, such as rock salt, mounted on a spectrometer table that can be rotated through measured angles. The crystal planes of atoms act as a three-dimensional grating, and the reflected X-ray beam is detected by an ionization chamber or photographic detector mounted on an arm that turns through twice the crystal angle. The angle of the crystal and the intensity of the diffracted beam are recorded as the crystal is slowly rotated. Sharp peaks of intensity appear only at angles satisfying Bragg law, 2d sin theta = m lambda, from which the plane spacing d or the X-ray wavelength can be calculated.

Exercise — Numerical

Question: A monochromatic light of wavelength 6900 A deg is used to illuminate two parallel slits. On a screen that is 3.30 m away from the slits, interference fringes are observed. The distance between adjacent bright fringes in the centre of the pattern is 1.80 cm. What is the distance between the slits?

Answer: Given: wavelength lambda = 6900 A deg = 6900 x 10^-10 m = 6.90 x 10^-7 m, screen distance D = 3.30 m, fringe width beta = 1.80 cm = 0.018 m

Step 1: Fringe width beta = lambda D / d, so slit separation d = lambda D /

beta

Step 2: Substitute d = (6.90 x 10^-7 x 3.30) / 0.018

Step 3: Numerator = 2.277 x 10^-6, so d = 2.277 x 10^-6 / 0.018 = 1.265 x 10^-4 m

Final answer: distance between the slits d = 1.265 x 10^-4 m

Question: A Michelson interferometer is adjusted so that a bright fringe appears on the screen. As one of the mirrors is moved 25.8 micrometer, 92 bright fringes are counted on the screen. What is the wavelength of light used in the interferometer?

Answer: Given: mirror displacement d = 25.8 micrometer = 25.8 x 10^-6 m, number of fringes shifted N = 92

Step 1: When the mirror moves by d, the path difference changes by 2d because the light makes a return journey

Step 2: Each fringe shift corresponds to a path change of one wavelength, so 2d = N lambda

Step 3: Therefore lambda = 2d / N = (2 x 25.8 x 10^-6) / 92

Step 4: lambda = 51.6 x 10^-6 / 92 = 5.60 x 10^-7 m = 560 nm

Final answer: wavelength of light = 560 nm

Question: In section 13.4 we studied interference due to thin films. Why must the film be thin? Why do we not see the interference effect when looking through a window or at a poster covered by a plate of glass, even if the glass is optically flat?

Answer: Interference in a film occurs between light reflected from its two surfaces, and the path difference between the two beams is about 2 n t. For clear fringes this path difference must be small, only a few

wavelengths, so that the two beams remain coherent with each other, which requires the film thickness t to be of the order of the wavelength of light, that is, very thin. In a window pane or a glass plate covering a poster, the thickness is millions of wavelengths, so the order of interference is extremely high and countless overlapping patterns of all colours average out to uniform illumination. In addition, ordinary light has a short coherence length, so beams with such a huge path difference cannot interfere at all, and no colours or fringes are seen.

Question: Newton's rings are formed by the light of 400 nm wavelength. Determine the change in air film thickness between the third and sixth bright fringe. If the radius of curvature of the curved surface is 5.0 m what is the radius of third bright fringe?

Answer: Given: wavelength lambda = 400 nm = 400 x 10^-9 m, radius of curvature R = 5.0 m

Step 1: For a bright ring the film thickness is t = (m + 1/2) lambda/2, so the thickness increases by lambda/2 from one bright fringe to the next

Step 2: From the third to the sixth bright fringe there are 3 steps, so change in thickness delta t = 3 x lambda/2 = 3 x 400/2 = 600 nm

Step 3: Radius of the m-th bright ring is r = sqrt((m + 1/2) lambda R); for the third bright fringe m = 2

Step 4: Substitute r = sqrt(2.5 x 400 x 10^-9 x 5.0) = sqrt(5.0 x 10^-6) = 2.236 x 10^-3 m

Final answer: change in film thickness = 600 nm and radius of the third bright fringe = 2.236 mm

Question: A soap film has an index of refraction n = 1.50. The film is viewed in reflected light. (a) At a spot where the film thickness is 910.0

nm, which wavelengths are missing in the reflected light? (b) Which wavelengths are strongest in the visible light?

Answer: Given: refractive index n = 1.50, film thickness t = 910.0 nm, visible range about 400 nm to 700 nm

Step 1: Compute 2 n t = 2 x 1.50 x 910.0 = 2730 nm

Step 2: Because there is one phase change on reflection, wavelengths are missing (destructive) when 2 n t = m lambda, so lambda = 2730/m; for m = 4, 5, 6 this gives 683 nm, 546 nm and 455 nm, all in the visible range

Step 3: Wavelengths are strongest (constructive) when 2 n t = (m + 1/2) lambda, so lambda = 2730/(m + 1/2); for m = 4, 5, 6 this gives 2730/4.5 = 607 nm, 2730/5.5 = 496 nm and 2730/6.5 = 420 nm

Final answer: (a) missing wavelengths are 683 nm, 546 nm and 455 nm, (b) strongest wavelengths are 607 nm, 496 nm and 420 nm

Question: What is the difference between deviation and diffraction? What do diffraction and interference have in common?

Answer: Deviation is the change in the direction of a light ray when it refracts through a medium such as a prism, where the ray bends because its speed changes across a boundary. Diffraction is the spreading or bending of waves around the edges of an obstacle or through a narrow opening, which happens because each point of the wavefront acts as a source of secondary wavelets. Diffraction and interference are alike because both are wave phenomena produced by the superposition of coherent waves: interference combines waves from a small number of separate sources, while a diffraction pattern can be regarded as interference between the very large number of wavelets coming from different parts of a single wavefront.

Question: Is it possible to increase the orders of maxima for a given energy spectrum from a diffraction grating?

Answer: The highest order that a grating can produce is fixed by d sin theta = m lambda with sin theta at most 1, so m_max is the whole number just below d/lambda. For a given wavelength, the number of orders can therefore be increased only by increasing the grating spacing d, that is by using a grating with fewer lines per unit length. This spreads fewer orders over the available angles, although the spectra then overlap more and the dispersion and resolving power fall. With the same grating and the same light it is not possible to get more orders, because orders beyond m_max would require sin theta greater than 1.

Question: Describe what happens to a single slit diffraction pattern as the width of the slit is slowly decreased.

Answer: The width of the central maximum is 2 lambda D/a, so as the slit width a decreases, the central maximum becomes steadily broader and the whole pattern spreads out over a wider region of the screen. At the same time the total amount of light passing through decreases, so the pattern becomes fainter, and the secondary maxima get weaker and also move further out. When the slit width becomes comparable with the wavelength, the spreading is so great that the central maximum fills most of the screen and the fringes practically disappear.

Question: The diffraction pattern from a single slit of width 0.020 mm is viewed on a screen. If the screen is 1.20 m from the slit and light of wavelength 430 nm is used, what is the width of the central maximum?

Answer: Given: slit width a = 0.020 mm = 0.020 x 10^-3 m = 2.0 x 10^-5 m, screen distance D = 1.20 m, wavelength lambda = 430 nm = 430 x 10^-9 m

Step 1: The first minimum lies at sin theta = lambda/a, so its distance

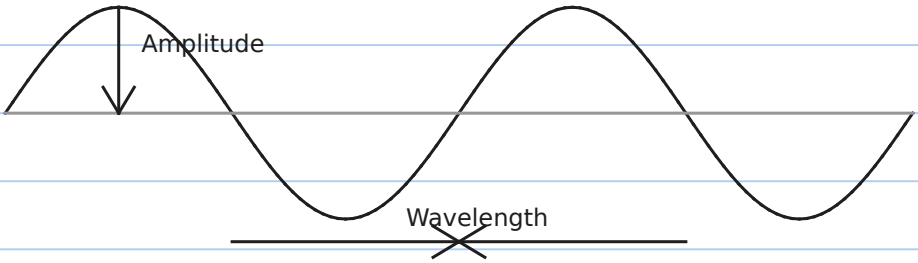
from the centre on the screen is y = D lambda / a

Step 2: The central maximum extends between the first minima on both sides, so its width is W = 2 D lambda / a

Step 3: Substitute W = (2 x 1.20 x 430 x 10^-9) / (2.0 x 10^-5) = (1.032 x 10^-6) / (2.0 x 10^-5)

Step 4: W = 0.0516 m = 5.16 cm

Final answer: width of the central maximum = 5.16 cm



Question: A grating has exactly 8000 lines uniformly spaced over 2.54 cm and is illuminated by light from a mercury vapor discharge lamp. What is the expected angle for the third order maximum of the light of wavelength 546 nm?

Answer: Given: number of lines N = 8000 over width 2.54 cm = 0.0254 m, order m = 3, wavelength lambda = 546 nm = 546 x 10^-9 m

Step 1: Grating spacing d = 0.0254 / 8000 = 3.175 x 10^-6 m

Step 2: Grating equation d sin theta = m lambda, so sin theta = m lambda / d

Step 3: Substitute sin theta = (3 x 546 x 10^-9) / (3.175 x 10^-6) = 1638 x 10^-9 / 3.175 x 10^-6 = 0.5159

Step 4: theta = sin^-1 (0.5159) = 31.06 deg, which is 31 deg 3 min 29 sec

Final answer: angle of the third order maximum = about 31 deg (31 deg 3 min 29 sec)

Question: How many lines per centimeter are there in a grating which gives 1st order spectra at an angle of 30 deg when the wavelength of light is 6 x 10^-5 cm?

Answer: Given: order m = 1, angle theta = 30 deg, wavelength lambda = 6 x 10^-5 cm

Step 1: Grating equation d sin theta = m lambda, so d = m lambda / sin theta

Step 2: Substitute d = (1 x 6 x 10^-5) / sin 30 deg = (6 x 10^-5) / 0.5 = 1.2 x 10^-4 cm

Step 3: Number of lines per cm = 1/d = 1 / (1.2 x 10^-4) = 8333

Final answer: number of lines per cm = 8333

Question: Light of wavelength 450 nm is incident on a diffraction grating on which 5000 lines/cm have been ruled. Determine (i) how many orders of spectra can be observed on either side of spectra? (ii) Determine the angle corresponding to each order.

Answer: Given: wavelength lambda = 450 nm = 450 x 10^-9 m, lines per cm = 5000, so grating spacing d = 1/5000 cm = 2.0 x 10^-6 m

Step 1: From d sin theta = m lambda, the maximum order is m_max = d/lambda (for sin theta = 1) = (2.0 x 10^-6)/(450 x 10^-9) = 4.44, so 4 complete orders can be observed on either side

Step 2: For each order sin theta = m lambda / d = m x 450/2000 = 0.225 m

Step 3: m = 1 gives sin theta = 0.225, theta = 13 deg; m = 2 gives 0.450,

theta = 26.7 deg; m = 3 gives 0.675, theta = 42.5 deg; m = 4 gives 0.900, theta = 64.2 deg

Final answer: (i) 4 orders on either side, (ii) angles are 13 deg, 26.7 deg, 42.5 deg and 64.2 deg

Question: Why does a crystal act as a three dimensional grating for X-rays but not for visible light?

Answer: A grating can diffract a wave strongly only when its spacing is comparable with the wavelength of the wave. In a crystal the atoms are arranged in regular planes with spacings of about 0.1 nm, which is comparable with the wavelength of X-rays, so the crystal planes produce clear Bragg diffraction of X-rays in three dimensions. Visible light has a wavelength of 400 nm to 700 nm, thousands of times larger than the atomic spacing, so no diffraction orders are possible for it and the crystal behaves towards visible light simply as a continuous transparent medium that only refracts it.

Question: A beam of X-rays of wavelength 0.071 nm is diffracted by a diffracting plane of rock salt with distance between the atomic planes are 1.98 A deg. Find the glancing angle for the second-order diffraction.

Answer: Given: wavelength lambda = 0.071 nm = 0.71 A deg, plane spacing d = 1.98 A deg, order m = 2

Step 1: Bragg law is 2 d sin theta = m lambda, where theta is the glancing angle

Step 2: Rearrange sin theta = m lambda / (2 d)

Step 3: Substitute sin theta = (2 x 0.71) / (2 x 1.98) = 1.42 / 3.96 = 0.3586

Step 4: theta = sin^-1 (0.3586) = 21 deg

Final answer: glancing angle for second-order diffraction = 21 deg

Unit 14: Communication

Exercise — Section (A): Multiple Choice Questions (MCQs)

In Radio and Television broadcast, the information signal is in the form of:

(a) analog signal

(b) digital signal

(c) Both analog & digital signals

(d) neither analog nor digital signal

Answer: (a) In radio and television broadcasting the sound and picture information varies continuously with time, like the voice and brightness it represents. Such a continuously varying signal is an analog signal, and it is this analog information signal that is modulated onto the carrier wave for broadcast.

A communication channel consists of:

(a) transmission line only

(b) optical fibre only

(c) free space only

(d) All of them

Answer: (d) A communication channel is simply the medium through which the signal travels from the transmitter to the receiver. That medium may be a transmission line such as a twisted pair or coaxial cable, an optical fibre carrying light signals, or free space carrying radio waves, so all of these count as communication channels.

Voltage signal generated by a microphone is:

(a) digital in nature

✓ (b) analog in nature

(c) hybrid in nature

(d) consists of bits & bytes

Answer: (b) A microphone changes sound pressure into a voltage that rises and falls continuously in step with the sound wave. Because this voltage can take any value within a range and is not restricted to discrete levels, it is an analog signal, not a digital one made of bits.

As compared to sound waves frequency of radio waves is:

✓ (a) higher

(b) equal

(c) lower

(d) may be higher or lower

Answer: (a) Sound waves are audio signals with frequencies roughly from 20 Hz to 20 kHz. Radio waves are electromagnetic waves used as carriers, with frequencies from about 100 kHz up to many gigahertz, which is far higher than the frequency of sound waves.

The process of superimposing signal frequency on the carrier wave is known as:

(a) transmission

(b) detection

(c) reception

✓ (d) modulation

Answer: (d) In modulation the low-frequency information signal is superimposed on a high-frequency carrier wave by varying the carrier's amplitude, frequency or phase in step with the signal. This makes long-distance transmission practical, and the reverse process at the receiver is called detection or demodulation.

What is the frequency range of signals that can be transmitted in case of twisted pair of wires?

(a) 10 MHz to 15 MHz

(b) 5 MHz to 10 MHz

✓ (c) 100 Hz to 5 MHz

(d) 10 Hz to 100 Hz

Answer: (c) A twisted pair of wires suffers increasing attenuation and distortion as frequency rises, which limits the signals it can carry usefully to a few megahertz. Within the range of about 100 Hz to 5 MHz it is widely used for telephone and local network connections.

The maintenance of a satellite's orbital position is called:

(a) maintenance keeping

✓ (b) station keeping

(c) station maintenance

(d) attitude control

Answer: (b) A communication satellite is slowly pulled away from its assigned position by the gravity of the Sun and Moon and by other small disturbances. The manoeuvres, using small thrusters, that keep the satellite within its allotted slot in orbit are called station keeping, whereas attitude control only keeps the satellite pointing in the right direction.

Process of mapping the sampled analog voltage values to discrete voltage levels is called:

(a) sampling

(b) sampling frequency

✓ (c) quantizing

(d) encoding

Answer: (c) In analog-to-digital conversion the sampled voltage values are compared with a set of fixed, discrete levels and each sample is assigned

the nearest level. This step of mapping continuous sampled values to discrete levels is called quantizing, and it is followed by encoding the levels into binary numbers.

AM is used for broadcasting because:

(a) it requires less transmitting power compare with other systems

(b) it is more noise immune than other modulation system

(c) No other modulation can provide the necessary bandwidth faithful transmission

✓ (d) its use avoids receiver complexity

Answer: (d) An amplitude modulation receiver needs only a simple detector circuit to recover the audio signal, so AM radio receivers are cheap and simple to build for a mass audience. Although AM is more affected by noise than FM, this simplicity of the receiver is the main reason it was adopted for broadcasting.

Data in compact disc is stored in the form of:

(a) analog signal

✓ (b) digital signal

(c) noise

(d) both

(a) &

Answer: (b) On a compact disc the sound is first converted into binary numbers, and these are stored as a pattern of tiny pits and flat lands along the track. A laser reads this pattern back as a digital signal, which is then converted to analog form for the loudspeaker, so the stored data is digital.

Exercise — Section (B): Structured Questions — CRQs

Question: A male voice after modulation-transmission sounds like that of a female to the receiver. Give reason.

Answer: The pitch of a voice depends on its frequency, and a female voice has a higher fundamental frequency than a male voice. If during modulation and transmission the frequencies of the signal are shifted upward, for example because of a carrier frequency error or an incorrect oscillator setting at the transmitter or receiver, every component of the male voice is reproduced at a higher frequency. The receiver therefore hears a higher-pitched voice that sounds like a female voice.

Question: Why is an AM signal likely to be more noisy than a FM signal upon transmission through a channel?

Answer: In AM the information is carried in the amplitude of the carrier, and most natural and man-made noise, such as lightning and electrical interference, also appears as unwanted changes in amplitude. This noise therefore mixes directly with the signal and cannot be separated at the receiver. In FM the information is carried in the frequency while the amplitude is kept constant, so the receiver can clip off amplitude variations and remove most of the noise, giving a much cleaner output.

Question: Write 5 differences between analog and digital transmission.

Answer: First, an analog signal varies continuously and can take any value, while a digital signal takes only discrete values, usually two levels representing 0 and 1. Second, analog signals are badly degraded by noise and distortion along the channel, whereas digital signals can be regenerated cleanly at repeaters because only the digits need to be recognised. Third, analog transmission has a limited quality that falls with distance, while digital transmission can use error detection and correction to keep quality almost perfect. Fourth, digital signals can be compressed, encrypted and processed by computers easily, which analog signals

cannot. Fifth, digital systems make more efficient use of bandwidth and allow many types of information, voice, data and video, to be carried in one common form.

Question: Define core and cladding. On what principle does optical fibre work?

Answer: The core is the thin central part of an optical fibre, made of glass or plastic of higher refractive index, through which the light signal actually travels. The cladding is the outer layer surrounding the core, made of material of slightly lower refractive index, which keeps the light confined inside the core. An optical fibre works on the principle of total internal reflection: light entering the core strikes the core-cladding boundary at an angle greater than the critical angle and is reflected completely back into the core again and again, so it travels along the fibre even around bends.

Question: What is a Digital Communication System? Why it is advantageous to use digital communication over analog communication.

Answer: A digital communication system is a system in which the information is converted into a stream of binary digits and transmitted in digital form, then converted back at the receiving end. It is advantageous because digital signals resist noise and can be regenerated perfectly at repeaters, so quality does not deteriorate over long distances. Digital systems also allow error correction, data compression, encryption for security, and the mixing of voice, video and computer data on the same network.

Question: Give 5 advantages of satellite communication.

Answer: First, a satellite can connect very distant and remote places, including mountains, deserts, ships and aircraft, where cables cannot be laid. Second, one geostationary satellite covers about one third of the Earth's surface, so a few satellites give worldwide coverage. Third,

satellite links can be set up quickly and are ideal for broadcasting television and radio to a huge number of receivers at once. Fourth, the cost of a satellite link does not depend on the distance between the two stations. Fifth, satellites provide services such as weather monitoring, navigation support, disaster communication and backup links when ground networks fail.

Question: Write advantages and disadvantages of frequency modulation.

Answer: The main advantages of FM are that it is almost free from noise and interference because the receiver responds only to frequency changes, it gives high-fidelity sound reproduction, and all the transmitted power is useful signal power. Its main disadvantages are that it needs a much wider bandwidth than AM, about ten times more, so fewer channels can be fitted in a given band. FM reception is also limited roughly to line-of-sight range, so it is suitable only for local broadcasting, and its transmitters and receivers are more complex and costly than those of AM.

Question: What are limitations of amplitude modulation?

Answer: In AM most of the transmitted power is carried by the carrier wave, which carries no information, so AM transmission is power-inefficient. The audio quality is limited because the bandwidth allowed is small, so high audio frequencies are lost and reproduction is not high fidelity. AM is also highly disturbed by electrical noise and interference, since noise changes the amplitude in which the information is stored. In addition, only a limited number of AM stations can operate in a band without interfering with each other.

Exercise — Section (B): Structured Questions — ERQs

Question: What is modulation? Why it is needed for the transmission of signals?

Answer: Modulation is the process of superimposing a low-frequency information signal on a high-frequency carrier wave by varying the amplitude, frequency or phase of the carrier in accordance with the signal. It is needed because an audio signal has a very long wavelength, so an antenna of practical size, which must be comparable with the wavelength, cannot radiate it efficiently. Raising the signal to a high carrier frequency makes its wavelength short, so reasonably sized antennas can transmit and receive it over long distances. Modulation also allows many stations to use different carrier frequencies without mixing, and it reduces noise and interference during transmission.

Question: How many types of modulation are there? Explain each type of modulation in detail.

Answer: There are three basic types of analog modulation, based on which property of the carrier is varied. In amplitude modulation (AM) the amplitude of the carrier is changed in step with the signal while its frequency stays constant; it is simple and used for radio broadcasting, but it suffers from noise. In frequency modulation (FM) the frequency of the carrier is changed in step with the signal while its amplitude stays constant; it gives noise-free, high-quality reception but needs a wide bandwidth. In phase modulation (PM) the phase of the carrier is shifted in step with the signal; it is closely related to FM and is widely used in digital communication systems.

Question: What is A/D conversion? Explain sampling, quantizing and encoding involved in A/D conversion in detail.

Answer: A/D (analog-to-digital) conversion is the process of changing a continuous analog signal into a digital signal made of binary numbers, and

it is done in three steps. In sampling, the value of the analog signal is measured at regular time intervals; according to the sampling theorem the sampling rate must be at least twice the highest frequency in the signal so that it can be recovered faithfully. In quantizing, each measured sample value is rounded to the nearest one of a fixed set of discrete voltage levels, which introduces only a very small quantizing error. In encoding, each quantized level is written as a binary number of a fixed number of bits, and these binary numbers are transmitted or stored as the digital signal.

Question: Define communication channel. How information carried through UTP, STP and Coaxial cable? Write their advantages and disadvantages.

Answer: A communication channel is the physical medium or path through which a signal travels from the transmitter to the receiver. In UTP (unshielded twisted pair) and STP (shielded twisted pair) cables the information travels as electrical signals along pairs of copper wires twisted together to reduce interference; they are cheap and easy to install and are used for telephones and local networks, but they have limited bandwidth, high attenuation and work only over short distances. In a coaxial cable the signal travels along a central copper conductor surrounded by insulation and a braided metal shield; it offers higher bandwidth, lower loss and better protection from interference than twisted pairs and is used for cable TV, but it is more expensive, thicker and harder to install.

Question: What is optical Fibre? How information carried out through optical fibre? Give advantages of using optical fibre over other communication channels.

Answer: An optical fibre is a very thin, flexible strand of glass or plastic consisting of a core surrounded by cladding of lower refractive index. Information is converted into pulses of light by an LED or laser and sent

into the core; the light undergoes total internal reflection at the core-cladding boundary again and again and so travels along the fibre to the far end, where a light detector converts it back into an electrical signal. Optical fibre has enormous bandwidth, very low loss over long distances, complete immunity to electrical interference, and it is light, thin and secure against tapping. These advantages make it superior to copper cables for long-distance and high-speed communication.

Question: What are the main components of a Satellite? How communication takes place in a satellite. Explain three orbits of a satellite.

Answer: The main components of a communication satellite are the transponders, which receive, amplify and retransmit signals, the antennas, the solar panels and batteries that supply power, and the station-keeping and attitude-control systems that hold its position and pointing. In satellite communication an earth station sends the signal up to the satellite on an uplink frequency; the transponder amplifies it, changes its frequency, and beams it back down on a downlink frequency to receiving stations over a wide area. The three commonly used orbits are low Earth orbit (LEO), a few hundred kilometres up, used for mobile and data services; medium Earth orbit (MEO), about 10,000 km up, used for navigation systems such as GPS; and geostationary Earth orbit (GEO), about 35,786 km above the equator, where the satellite appears fixed in the sky and is ideal for television broadcasting and continuous communication.